Vectors

06 November 2025

Revision: 4084

Aziz Manva, PGDM, IIM-L

azizmanva@gmail.com

Table of contents

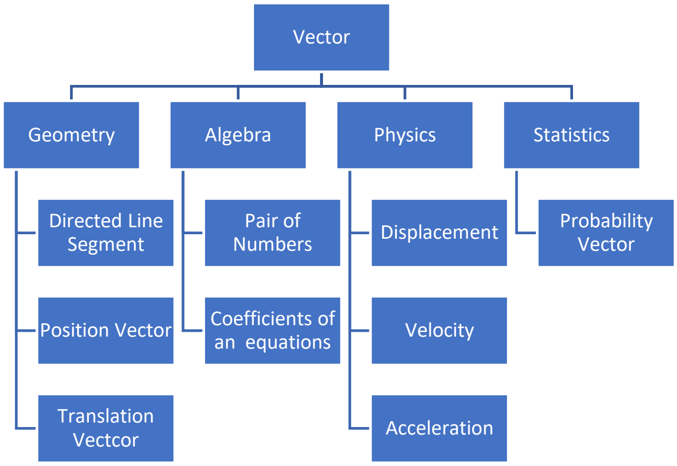
1. Scalar Operations

1.1 Basics

A. Introduction

We introduce vectors as geometrical quantities. However, vectors have many interpretations:

- as a translation from one point to another (Coordinate Geometry)
- as the position of a point (Coordinate Geometry)
- as a pair of numbers (Algebra)
- as the displacement of a quantity (Physics)
- as the velocity of an object (Physics)
- as the acceleration of an object (Physics)
- as the probabilities of the sample space of an event (Statistics)



- Note that we could $define$ vectors to be any one of the above interpretations, and we could still work with them.
- In fact, these would lead to equivalent properties, though we will not show this here.

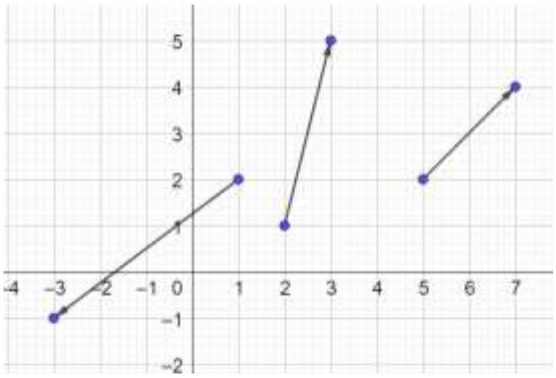
B. Geometrical Definition

We introduce vectors in terms of coordinate geometry. Later on, we show applications in other areas, including algebra, geometry, and physics. Vectors have deep connections to all of these areas.

1.1: Definition

A vector is a *directed* line segment that has a start point and an endpoint.

$$Start\ Point = Tail = (x_1, y_1)$$
$$End\ Point = Tip = (x_2, y_2)$$



Example 1.2

Identify the start points and the endpoints of the vectors drawn alongside.

$$(2,1), (3,5)$$
$$(5,2), (7,4)$$
$$(1,2), (-3,-1)$$

1.3: Notation

Vectors are often indicated in textbooks using boldface notation.

$$Vector\ a = \boldsymbol{a}$$

Vectors when written by hand are usually indicated with an arrow on top of the letter:

$$Vector\ a = \vec{a}$$

We will use the notation interchangeably. We might also combine both notations:

$$Vector\ a = \vec{\boldsymbol{a}}$$

Variables that you have come across so far in Algebra are scalars.

Example 1.4

Decide whether the following quantities are vectors or scalars (based on the notation):

A. x
B. $\boldsymbol{x}$
C. $\vec{x}$

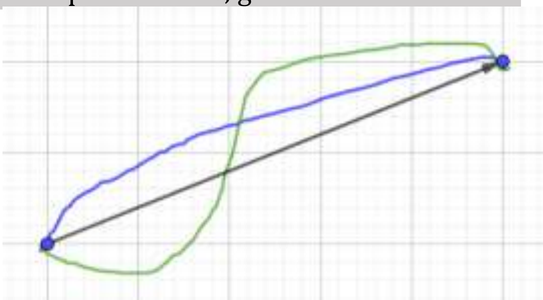
$$Scalar: A$$
$$Vector: B, C$$

C. Translation Vectors

1.5: Vectors as Translation

You can think of a vector as translating a point from its start point to its end point. Hence, given $A = (x_1, y_1), B(x_2, y_2)$:

$$\overrightarrow{\boldsymbol{AB}} = (x_2 - x_1, y_2 - y_1)$$



Example 1.6

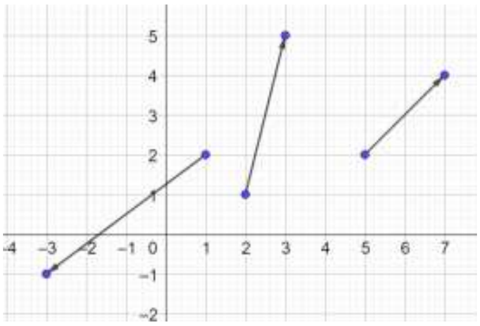
The diagram alongside shows a vector with a start end, and an endpoint. It also shows three different paths from the start point to the end point.
Do these paths represent three different vectors?

$$No.$$

A vector is the translation from its start point to its endpoint.

Example 1.7

Write each vector in the diagram alongside as a translation vector.



$$(3 - 2, 5 - 1) = (1,4)$$
$$(7 - 5, 4 - 2) = (2,2)$$
$$(-3 - 1, -1 - 2) = (-4, -3)$$

Example 1.8

A. Jane was visiting a town that had a rectangular grid layout. She imagined a coordinate system with the town's central square at the origin. Jane is currently at the town fountain at $(3, -2)$. She asked for directions to the mayor's office, and was told she needed to apply a translation vector of $(2, -5)$ to her current position. Identify the coordinates of the mayor's office.
B. Write a translation of 3 units in the x direction, and 2 units in the y direction as a translation vector.
C. Identify the translation in the x direction, and the y direction for the translation vector (π, e).

Part A

$$\underbrace{(3, -2)}_{\substack{Current\\Position}} + \underbrace{(2, -5)}_{\substack{Translation\\Vector}} = (3 + 2, -2 - 5) = \underbrace{(5, -7)}_{\substack{Mayor's\\Office}}$$

Part B

$$(3,2)$$

Part C

$$x - direction\ translation = \pi$$
$$y - direction\ translation = e$$

Example 1.9

Consider the points:

$$A = (1,2), B = (3,5), C = (5,2), D = (7,4), E = (1,2), F = (-3,-1)$$

Plot the following vectors, and find their values

A. $\overrightarrow{\boldsymbol{AB}}$
B. $\overrightarrow{\boldsymbol{CD}}$
C. $\overrightarrow{\boldsymbol{EF}}$

$$\overrightarrow{\boldsymbol{AB}} = (3-1, 5-2) = (2,3)$$
$$\overrightarrow{\boldsymbol{CD}} = (7-5, 4-2) = (2,2)$$
$$\overrightarrow{\boldsymbol{EF}} = (-3-1, -1-2) = (-4,-3)$$

D. Position Vector and Components

1.10: Position Vector

- A vector that starts from the origin is called a position vector.
- It gives the coordinates of a position (along with being a vector)

$$End\ point - Start\ Point$$

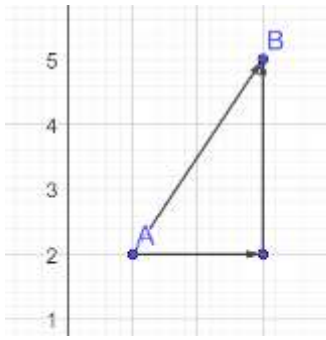
Example 1.11

Consider the points:

$$A = (1,2), B = (3,5), C = (5,2), D = (7,4), E = (1,2), F = (-3,-1)$$

Write the following as position vectors.

A. $\overrightarrow{\boldsymbol{AB}}$
B. $\overrightarrow{\boldsymbol{CD}}$
C. $\overrightarrow{\boldsymbol{EF}}$



The answers are the same as previously calculated:

$$\overrightarrow{\boldsymbol{AB}} = (3-1, 5-2) = (2,3)$$
$$\overrightarrow{\boldsymbol{CD}} = (7-5, 4-2) = (2,2)$$
$$\overrightarrow{\boldsymbol{EF}} = (-3-1, -1-2) = (-4,-3)$$

1.12: Components of a Vector

Given a vector $\vec{\boldsymbol{a}} = (x_1, y_1)$, its components are:

$$x_1 = x\ component$$
$$y_1 = y\ component$$

1.13: $\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k}$ notation

A vector can be written in $\hat{\imath}, \hat{\jmath}, \hat{k}$ notation as

$$(x_1, y_1, z_1) = x_1\hat{\boldsymbol{\imath}} + y_1\hat{\boldsymbol{\jmath}} + z_1\widehat{\boldsymbol{k}}$$

E. Direction

1.14: Order is important

$$A = (x_1, y_1), B(x_2, y_2) \Rightarrow \overrightarrow{\boldsymbol{AB}} = (x_2 - x_1, y_2 - y_1)$$

The vector $\overrightarrow{\boldsymbol{AB}}$ is not equal to the vector $\overrightarrow{\boldsymbol{BA}}$.
The two vectors have the same length but opposite directions.

In fact, we can state that

$$\overrightarrow{\boldsymbol{BA}} = -\overrightarrow{\boldsymbol{AB}} = (x_1 - x_2, y_1 - y_2)$$

Example 1.15

Consider the points

$$A = (1,2), B = (3,5), C = (5,2), D = (7,4), E = (1,2), F = (-3,-1)$$

Plot the following vectors, and find their values:

A. $\overrightarrow{\boldsymbol{BA}}$
B. $\overrightarrow{\boldsymbol{DC}}$
C. $\overrightarrow{\boldsymbol{FE}}$

$$\overrightarrow{\boldsymbol{BA}} = (1 - 3, 2 - 5) = (-2, -3)$$
$$\overrightarrow{\boldsymbol{DC}} = (5 - 7, 2 - 4) = (-2, -2)$$
$$\overrightarrow{\boldsymbol{FE}} = \big(1 - (-3), 2 - (-1)\big) = (4,3)$$

Example 1.16

Consider the points

$$A = (1,2), B = (3,5)$$

A. Are vector $\overrightarrow{\boldsymbol{AB}}$, and vector $\overrightarrow{\boldsymbol{BA}}$ the same or different?
B. If you plot them, will they look the same? If they look different, what will be the difference?
C. What concept in Euclidean Geometry does this connect to?

Part A
The vectors are different.
Part B
The start point and the end point will be the same, but the arrow will point in exactly opposite directions.
Part C
Ray

1.17: Parallel and Anti Parallel Vectors

- Vectors that have the same direction are parallel.
- Vectors that have opposite directions are anti-parallel.

F. Algebraic Definition

1.18: Algebraic Definition

An ordered pair of numbers x and y represent a vector.

$$\vec{\boldsymbol{a}} = (x, y)$$

1.19: Row and Column Vectors

- A row vector is a vector of the form (x_1, y_1)
- A column vector is a vector of the form $\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$
- Both forms are equivalent forms of writing the same vector.

$$\underbrace{(x_1, y_1)}_{\substack{\text{Row Vector}\\ \text{Form}}} = \underbrace{\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}}_{\substack{\text{Column}\\ \text{Vector}\\ \text{Form}}}$$

G. Physical Interpretation

A vector can be interpreted as a directed line segment that represents:

- Displacement
- Velocity
- Acceleration
- Force

1.20: Vectors versus Scalars

- Vectors have both magnitude and direction.

- An example of a vector quantity is force.
- To give a complete description of a force, we must give both the magnitude of the force, and the direction in which it is applied.

1.21: Vectors in Kinematics

- Displacement is the straight-line distance between the start point and the end point of an object that is moving.
- Velocity is the rate of change of displacement with respect to time. $\vec{\boldsymbol{v}} = \frac{d\vec{s}}{dt}$
- Acceleration is the rate of change of velocity. $\vec{\boldsymbol{a}} = \frac{d\vec{v}}{dt}$

Kinematics is one of the most important applications of vectors. The above definitions are crucial to working in Kinematics.

Example 1.22

Given position vectors $\overrightarrow{\boldsymbol{OA}} = (2, -5)$, $\overrightarrow{\boldsymbol{OB}} = (-3,4)$, find

A. $\boldsymbol{BA}$ and interpret it.
B. $\boldsymbol{AB}$ and interpret it.

1.23: Scalar Quantities

- Scalars have only magnitude.

Some scalars can never be negative. Examples of such scalars include

- Mass
- Length
- Distance
- Age

1.24: Signed Scalar Quantities

Some scalar quantities can also be negative.

Examples of such scalars include:

- Temperature
- Bank Balance

Example 1.25

Decide whether the following are vectors or scalars:

A. The bank balance in a savings account
B. The pull exerted by gravity between the moon and object on the moon
C. The distance travelled by an asteroid orbiting the Sun.
D. A plane takes off from a runaway.
E. The wind conditions during a hot air balloon event.
F. The quantity of sand stored in a balloon before it takes off.
G. The temperature of a hot plate used to cook.
H. The distance travelled by a car owner travelling from home to office.

$$Vectors: B, D, E$$
$$Scalars: A, C, F, G, H$$

1.26: Displacement Vector

The displacement vector $\binom{x_1}{y_1}$ moves its input x_1 units in the x direction, and y_1 units in the y direction.

Example 1.27

Rory the dog is pulling a cart. He is currently stationed at $(-2,5)$. Rory applies the displacement vector $(2,-5)$ of the dog cart. Determine the final position of the dog cart.

$$(-2,5) + (2,-5) = (-2+2, 5-5) = (0,0)$$

Example 1.28

Convert the following vectors, if given in row form, into column form, and vice versa.

A. $(-2,8)$
B. $\begin{pmatrix} \frac{1}{2} \\ -3 \end{pmatrix}$

$$\begin{pmatrix} -2 \\ 8 \end{pmatrix}$$
$$\left(\frac{1}{2}, -3\right)$$

1.2 Properties

A. Magnitude

1.29: Length or Magnitude of a Vector

The length of a vector is the length of the line segment connecting its start point and its end point. It is given by the distance formula to be:

$$\text{Length of } \overrightarrow{\boldsymbol{AB}} = \left|\overrightarrow{\boldsymbol{AB}}\right| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$
$$A = (x_1, y_1)$$
$$B = (x_2, y_2)$$

Just as absolute values cannot be negative, similarly, length cannot be negative.
Hence, the symbol for length of a vector is similar to the symbol for absolute value.

Example 1.30

Find the magnitude of the following vectors:

$$A = (3,4), B = (4,9)$$

A. $\overrightarrow{\boldsymbol{AB}}$

$$|\boldsymbol{AB}| = \sqrt{(4-3)^2 + (9-4)^2} = \sqrt{1+25} = \sqrt{26}$$

Example 1.31: Magnitude

Consider points A, B, C and D with position vectors $7\hat{\boldsymbol{\imath}} - 4\hat{\boldsymbol{\jmath}} + 7\hat{\boldsymbol{k}}, \boldsymbol{i} - 6\hat{\boldsymbol{\jmath}} + 10\hat{\boldsymbol{k}}, -\hat{\boldsymbol{\imath}} - 3\hat{\boldsymbol{\jmath}} + 4\hat{\boldsymbol{k}}$ and $5\hat{\boldsymbol{\imath}} - \hat{\boldsymbol{\jmath}} + 5\hat{\boldsymbol{k}}$, respectively. Then, $ABCD$ is a

A. Square
B. Rhombus
C. Rectangle
D. None of these **(JEE Main 2003)**

To determine the nature of the quadrilateral, we need the distance between the points.
This will be given to us by the magnitude:

$$\vec{\boldsymbol{a}} = (7, -4, 7)$$
$$\vec{\boldsymbol{b}} = (1, -6, 10)$$
$$\vec{\boldsymbol{c}} = (-1, -3, 4)$$
$$\vec{\boldsymbol{d}} = (5, -1, 5)$$

$$\left|\overrightarrow{\boldsymbol{AB}}\right| = \sqrt{(7-1)^2 + (-4+6)^2 + (7-10)^2} = \sqrt{36+4+9} = \sqrt{49} = 7$$
$$\left|\overrightarrow{\boldsymbol{BC}}\right| = \sqrt{(1+1)^2 + (-6+3)^2 + (10-4)^2} = \sqrt{4+9+36} = \sqrt{49} = 7$$
$$\left|\overrightarrow{\boldsymbol{CD}}\right| = \sqrt{(-1-5)^2 + (-3+1)^2 + (4-5)^2} = \sqrt{36+4+1} = \sqrt{41}$$
$$\left|\overrightarrow{\boldsymbol{DA}}\right| = \sqrt{(5-7)^2 + (-1+4)^2 + (5-7)^2} = \sqrt{4+9+4} = \sqrt{17}$$

Since $\left|\overrightarrow{\boldsymbol{AB}}\right| \neq \left|\overrightarrow{\boldsymbol{CD}}\right|$, opposite sides are not equal:

$$\text{it is not a rectangle} \Rightarrow \text{Options } A, B, C \text{ are not valid}$$
$$\text{Option } D$$

1.32: Magnitude of a Vector is Invariant under Rotation

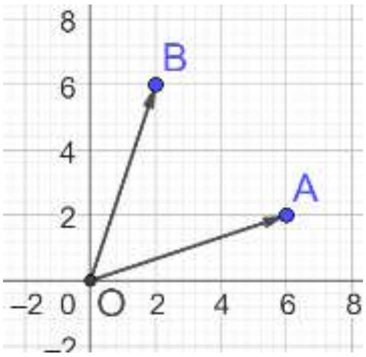
If you rotate a vector, you do not change its magnitude or length.
Hence, we say that magnitude is *invariant* under rotation.

Example 1.33

A vector $\vec{\boldsymbol{a}}$ has components $3p$ and 1 with respect to rectangular cartesian system. This system is rotated through a certain angle about the origin in the counter clockwise sense. If, with respect to new system, $\vec{\boldsymbol{a}}$ has components, $p+1$ and $\sqrt{10}$, then the values that p can take are: (JEE Main 2021, 18 March, Shift-I)

The magnitude of the vector is the same before and after rotation.

$$|\overrightarrow{\boldsymbol{OA}}| = |\overrightarrow{\boldsymbol{OB}}|$$
$$|\overrightarrow{\boldsymbol{OA}}|^2 = |\overrightarrow{\boldsymbol{OB}}|^2$$
$$(3p)^2 + 1 = (p+1)^2 + \left(\sqrt{10}\right)^2$$
$$9p^2 + 1 = p^2 + 2p + 1 + 10$$
$$8p^2 - 2p - 10 = 0$$
$$4p^2 - p - 5 = 0$$
$$(p+1)(4p-5) = 0$$
$$p \in \left\{-1, \frac{5}{4}\right\}$$

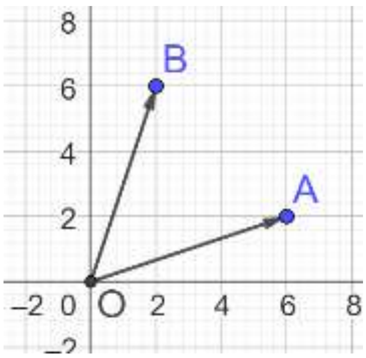


Example 1.34: Rotation Invariance

The vector $\hat{\boldsymbol{\imath}} + x\hat{\boldsymbol{\jmath}} + 3\hat{\boldsymbol{k}}$ is rotated through an angle θ and doubled in magnitude. It becomes $4\hat{\boldsymbol{\imath}} - (4x-2)\hat{\boldsymbol{\jmath}} + 2\hat{\boldsymbol{k}}$. The values of x are: (JEE Main 2002)

The magnitude of the vector is the same before and after rotation.

$$2|\overrightarrow{\boldsymbol{OA}}| = |\overrightarrow{\boldsymbol{OB}}|$$
$$2|(1, x, 3)| = |(4, -4x+2, 2)|$$



Use the definition of magnitude:

$$2\sqrt{1^2 + x^2 + 3^2} = \sqrt{4^2 + (4x-2)^2 + 2^2}$$

Square both sides:

$$4(10 + x^2) = 20 + 16x^2 - 16x + 4$$

Expand:

$$40 + 4x^2 = 24 + 16x^2 - 16x$$

Collate all terms on one side:

$$0 = 12x^2 - 16x - 16$$

Divide both sides by 3:

$$3x^2 - 4x - 4 = 0$$

Factor:

$$(x-2)(3x+2) = 0$$

Use the zero-product property:

$$x \in \left\{-\frac{2}{3}, 2\right\}$$

1.35: Magnitude of a Position Vector

The magnitude of a position vector $\vec{\boldsymbol{a}} = (x, y)$ is the length of the line segment connecting the origin to its end point. It is given by the distance formula to be:

$$\sqrt{x^2 + y^2}$$

$$|\boldsymbol{a}| = \sqrt{(x-0)^2 + (y-0)^2} = \sqrt{x^2 + y^2}$$

Example 1.36
Determine the magnitude of the following position vectors:

A. $\vec{\boldsymbol{a}} = \left(\frac{1}{2}, \frac{3}{4}\right)$

$$|a| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{3}{4}\right)^2} = \sqrt{\frac{1}{4} + \frac{9}{16}} = \sqrt{\frac{13}{16}} = \frac{\sqrt{13}}{4}$$

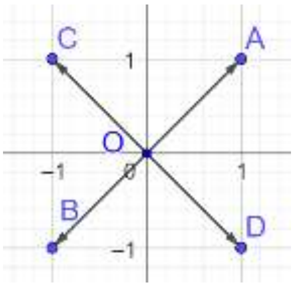
1.37: Zero Vector
- A vector with magnitude zero is called the zero vector.
- The zero vector does not have direction.

$$\textit{Zero vector is written } \mathbf{0}$$

1.38: Changing direction of a vector
If a non-zero vector is rotated, it becomes a different vector.
If the zero vector is rotated, it remains the same.
If a vector remains the same under a rotation which is not 360°, then it is the zero vector.

$$\vec{\boldsymbol{a}} = \overrightarrow{\boldsymbol{OA}} + \overrightarrow{\boldsymbol{OB}} = \vec{\mathbf{0}}$$

Rotate $\overrightarrow{\boldsymbol{OA}} + \overrightarrow{\boldsymbol{OB}}$ to get
$\overrightarrow{\boldsymbol{OC}} + \overrightarrow{\boldsymbol{OD}} = \vec{\mathbf{0}}$



Example 1.39
$\vec{\boldsymbol{a}}$ does not change after being rotated $\theta°, \theta \neq 0$. Identify $\vec{\boldsymbol{a}}$.

$$\vec{\boldsymbol{a}} = \vec{\mathbf{0}}$$

1.40: Zero Vector
The components of the zero vector are:

$$(0,0)$$

Consider a position vector $\vec{\boldsymbol{a}}$ such that

$$|\vec{\boldsymbol{a}}| = \mathbf{0}$$

Square both sides:

$$x^2 + y^2 = 0$$

By the Trivial Inequality $x^2 \geq 0, y^2 \geq 0$:

$$x = y = 0$$

B. Equality of Vectors

1.41: Equality of Two Vectors

- Two vectors are equal $if\ and\ only\ if$ they have the same magnitude, and the same direction.
- Equivalently, two vectors are equal $if\ and\ only\ if$ their individual components are equal.

Example 1.42

Determine the values of the variables in each case

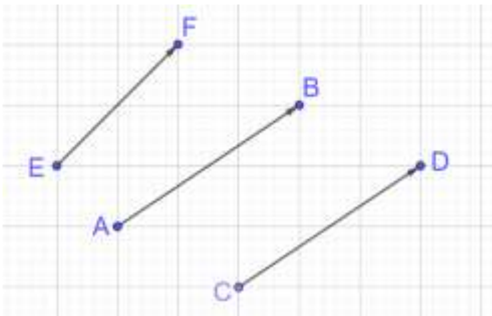
A. $\begin{pmatrix}3\\j\end{pmatrix}+\begin{pmatrix}k\\-4\end{pmatrix}=\mathbf{0}$

Think of the second vector as a translation vector. It translates the first vector back to the origin:

$$\begin{pmatrix}3\\j\end{pmatrix}+\begin{pmatrix}k\\-4\end{pmatrix}=\begin{pmatrix}3+k\\j-4\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}$$

Equating components:

$$x: 3+k=0 \Rightarrow k=-3$$
$$y: j-4=0 \Rightarrow j=4$$



1.43: Position is not important

Changing the position of a vector does not change the vector. The identity of the vector depends only upon its magnitude, and its direction, not its position.

Example 1.44

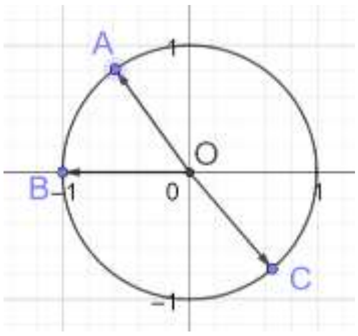
Determine which of the following vectors are equal.

$$\overrightarrow{\boldsymbol{AB}}=\overrightarrow{\boldsymbol{CD}}=(3,2)=\begin{pmatrix}3\\2\end{pmatrix}$$

C. Unit Vectors and Component Form

1.45: Unit Vector

- A vector with magnitude 1 is called a unit vector.
- $(Notation)$ A unit vector is written with a hat on top, instead of an arrow, to focus on the fact that it is a unit vector.



Example 1.46

In the adjoining diagram, we have a circle with its center at the origin, and a radius of 1 unit. Explain why $\widehat{\boldsymbol{OA}},\widehat{\boldsymbol{OB}}\ and\ \widehat{\boldsymbol{OC}}$ are unit vectors.

Each of the vectors mentioned has a magnitude of 1.
Hence, they are unit vectors.

Example 1.47

Checking for Unit Vectors
Determine whether the following are unit vectors.

A. $\vec{\boldsymbol{a}}=(1,0)$
B. $\vec{\boldsymbol{b}}=\left(\frac{1}{2},\frac{1}{2}\right)$

C. $\vec{\boldsymbol{c}} = (0,1)$

D. $\vec{\boldsymbol{d}} = \left(\frac{3}{4}, \frac{\sqrt{7}}{4}\right)$

Back Calculations

Determine the value of the variables such that the vector is a unit vector.

E. $\vec{\boldsymbol{e}} = \left(\frac{1}{2}, e\right)$

F. $\vec{\boldsymbol{f}} = \left(f, \frac{2}{3}\right)$

G. $\vec{\boldsymbol{g}} = (2, g)$

Checking for Unit Vectors

$$|\vec{\boldsymbol{a}}| = \sqrt{1^2 + 0} = 1$$

$$|\vec{\boldsymbol{b}}| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \sqrt{\frac{2}{4}} \neq 1$$

$$|\vec{\boldsymbol{c}}| = \sqrt{0^2 + 1^2} = 1$$

$$|\vec{\boldsymbol{d}}| = \sqrt{\left(\frac{3}{4}\right)^2 + \left(\frac{\sqrt{7}}{4}\right)^2} = \sqrt{\frac{9}{16} + \frac{7}{16}} = \sqrt{\frac{16}{16}} = 1$$

$$\textit{Units vectors are } \vec{\boldsymbol{a}}, \vec{\boldsymbol{c}}, \vec{\boldsymbol{d}}$$

Back Calculations

Part E

$$|\vec{\boldsymbol{e}}| = 1$$

$$\sqrt{\left(\frac{1}{2}\right)^2 + e^2} = 1$$

$$\frac{1}{4} + e^2 = 1$$

$$e^2 = \frac{3}{4}$$

$$e = \pm\sqrt{\frac{3}{4}} = \pm\frac{\sqrt{3}}{2}$$

Part F

$$|\vec{\boldsymbol{f}}| = 1$$

$$\sqrt{f^2 + \left(\frac{2}{3}\right)^2} = 1$$

$$f^2 + \frac{4}{9} = 1$$

$$f^2 = 1 - \frac{4}{9} = \frac{5}{9}$$

$$f = \pm\sqrt{\frac{5}{9}} = \pm\frac{\sqrt{5}}{3}$$

Part G

$$|\vec{\boldsymbol{g}}| = 1$$

$$\sqrt{2^2 + g^2} = 1$$

$$4 + g^2 = 1$$

$$g^2 = -3$$

$$\textit{No real solutions}$$

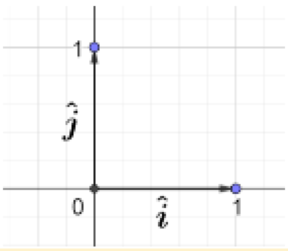
We can also realize that there no solutions since the x component is greater than 1.

1.48: Standard Basis

$$\textit{Unit Vector in the } x \textit{ direction} = \hat{\boldsymbol{\imath}}$$

$$\textit{Unit Vector in the } y \textit{ direction} = \hat{\boldsymbol{\jmath}}$$

$$\textit{Unit Vector in the } z \textit{ direction} = \widehat{\boldsymbol{k}}$$



Example 1.49

Write the following vectors in component form using the standard basis.

A. Position vector $\overline{\boldsymbol{OA}} = (3,4)$

B. Position vector $\overline{\boldsymbol{OB}} = (x, y)$

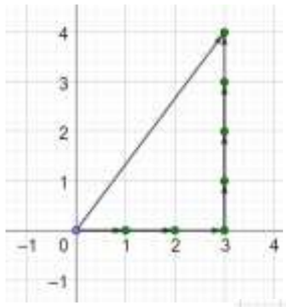
C. Position vector $\overline{\boldsymbol{OB}}$ with a magnitude of 13 units, and which translates its input 5 units in the x direction.

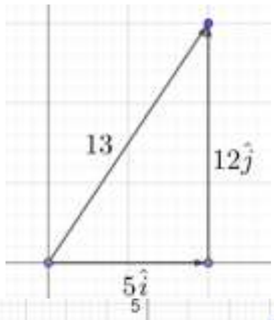
Parts A-B

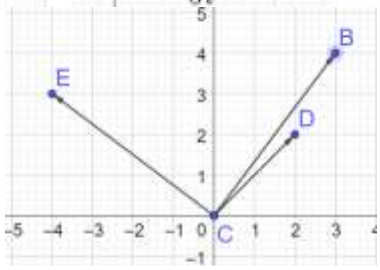
$$\overrightarrow{OA} = 3\hat{\imath} + 4\hat{\jmath}$$
$$\overrightarrow{OB} = x\hat{\imath} + y\hat{\jmath}$$

Parts C

$$\overrightarrow{OC} = 5\hat{\imath} + 12\hat{\jmath}$$







Example 1.50

The adjacent diagram shows vectors $\overrightarrow{CD}, \overrightarrow{CB}$ and $\overrightarrow{CE}$. Write these vectors in component form.

$$\overrightarrow{CD} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$
$$\overrightarrow{CB} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$
$$\overrightarrow{CE} = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$$

Example 1.51

Vector $\vec{a}$ has x −component with magnitude $\left|\frac{1}{2}\vec{a}\right|$, and y-component with magnitude $\left|\frac{\sqrt{3}}{2}\vec{a}\right|$. Determine what kind of triangle is formed by the vector and its components.

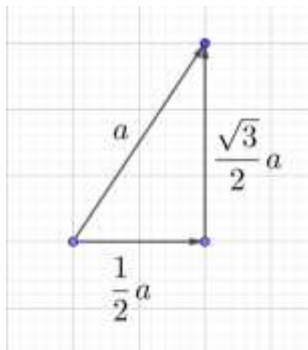
The x component of the vector has magnitude

$$Half\ of\ the\ original\ vector$$

The y component of the vector has magnitude

$$\frac{\sqrt{3}}{2}\ of\ the\ original\ vector$$

Draw a diagram that meets these conditions.



The triangle that is formed is a:

$$30 - 60 - 90\ Triangle$$

D. Slope of a Vector

1.52: Slope of a Vector

The slope of a vector is the slope that the line segment of the vector has, and it is given by:

$$\frac{y_2 - y_1}{x_2 - x_1}$$

The slope of a vector can be used to determine the direction of the vector.

Example 1.53

Consider the points

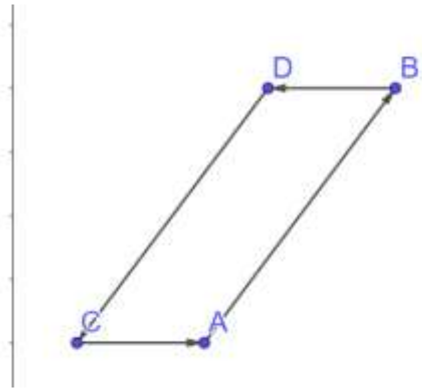
$$A(3,4), B(6,8), C(1,4), D(4,8)$$

A. Show that $\overrightarrow{\boldsymbol{AB}} = \overrightarrow{\boldsymbol{CD}}$
B. Show that $\overrightarrow{\boldsymbol{AB}} \neq \overrightarrow{\boldsymbol{DC}}$

Part A

Length:

$$|\overrightarrow{\boldsymbol{AB}}| = \sqrt{(6-3)^2 + (8-4)^2} = \sqrt{25} = 5$$
$$|\overrightarrow{\boldsymbol{CD}}| = \sqrt{(4-1)^2 + (8-4)^2} = \sqrt{25} = 5$$
$$|\overrightarrow{AB}| = |\overrightarrow{CD}|$$



Direction/Slope

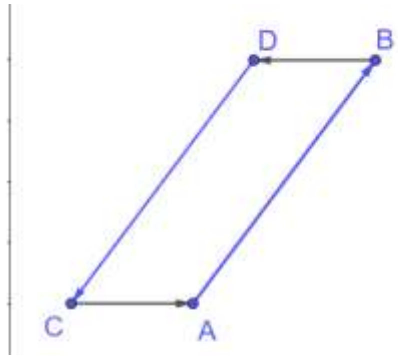
$Slope\ of\ \overrightarrow{AB} = \frac{y_2-y_1}{x_2-x_1} = \frac{8-4}{6-3} = \frac{4}{3}$

$$Slope\ of\ \overrightarrow{CD} = \frac{y_2-y_1}{x_2-x_1} = \frac{8-4}{4-1} = \frac{4}{3}$$

$$Slope\ of\ \overrightarrow{\boldsymbol{AB}} = Slope\ of\ \overrightarrow{\boldsymbol{CD}}$$

The length and the slope are the same.
The direction is also the same.

Part B



The length and the slope of the two vectors are the same.
But $\overrightarrow{\boldsymbol{DC}}$ is oriented in 180° from $\overrightarrow{\boldsymbol{CD}}$. Hence, it is going in the opposite direction.

E. Angle between two vectors

1.54: Angle between two Vectors

The angle between two vectors is the *smaller angle* formed when the vectors are arranged:

- Tail to Tail
- OR Head-to-Head

- The angle between $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$ can be denoted $(\vec{\boldsymbol{a}}, \vec{\boldsymbol{b}})$
- The angle between two vector is always in the range [0,180°]

Example 1.55

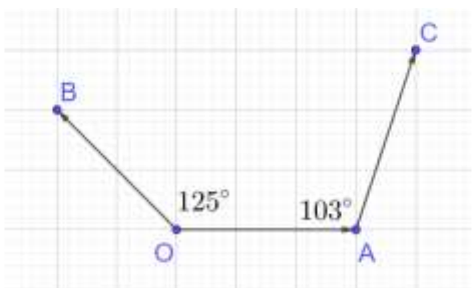
θ is the angle between two vectors. Find the range of θ.

$$0 \le \theta \le 180°$$

Example 1.56

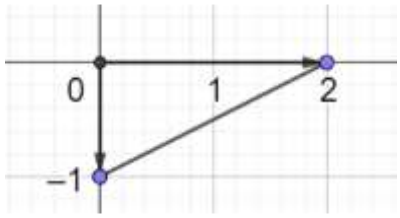
Find the angle between the vectors in each case:

A. The position vectors given by (2,0) and $(0,-1)$ on the coordinate plane.
B. $\overrightarrow{\boldsymbol{OB}}$ and $\overrightarrow{\boldsymbol{OA}}$
C. $\overrightarrow{\boldsymbol{OA}}$ and $\overrightarrow{\boldsymbol{AC}}$
D. $\overrightarrow{\boldsymbol{OB}}$ and $\overrightarrow{\boldsymbol{AC}}$



Part A
Draw the vectors, and note that we get a right-angled triangle:

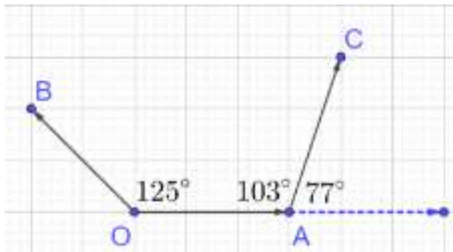
$$Angle\ between\ Vectors = 90°$$



Part B
The vectors are already arranged tail to tail. Hence, we can directly get the answer from the diagram:

$$Angle\ between\ \overrightarrow{OB}\ and\ \overrightarrow{OA} = 125°$$

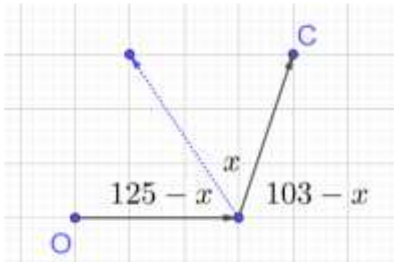
Part C
$\overrightarrow{\boldsymbol{OA}}$ and $\overrightarrow{\boldsymbol{AC}}$ are currently arranged tip to tail. To determine the angle between them, we arrange them tail to tail.
We do this by moving $\overrightarrow{\boldsymbol{OA}}$ to the right until its tip coincides with the tip of $\overrightarrow{\boldsymbol{AC}}$.



Then, the angle between them

$$= 180 - 103 = 77°$$

Part D
Move $\overrightarrow{\boldsymbol{OB}}$ to the right until its tail coincides with the tail of $\overrightarrow{\boldsymbol{AC}}$.
Call the angle between the two vectors



$$(125 - x) + x + (103 - x) = 180$$
$$x = 48$$

Shortcut
If you recognize that the angle between the vectors is the overlap of the two angles, then you can directly calculate:

$$Angle\ between\ Vectors = 125 + 103 - 180 = 48$$

F. n-Dimensional Vectors

So far, the vectors we have been working with have been two dimensional. However, the framework that we have created generalizes easily to more dimensions.

1.57: Vectors in 3 Dimensions

A vector in three dimensions consists of an ordered triple of numbers:

$$(x_1, y_1, z_1)$$

Example 1.58

Write the row vector (x_1, y_1, z_1) as a column vector

$$\underbrace{(x_1, y_1, z_1)}_{Row\ Vector} = \underbrace{\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}}_{\substack{Column \\ Vector}}$$

1.59: Vectors in n Dimensions

A vector in n dimensions consists of n ordered numbers:

$$(x_1, x_2, \dots, x_n)$$

1.60: Magnitude

Given the position vector $\vec{\boldsymbol{b}} = (x_1, x_2, \dots, x_n)$, its magnitude of the vector is by:

$$|\vec{\boldsymbol{b}}| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$

The formula for the magnitude of an n dimensional vector is a straightforward generalization of the formula in two dimensions:

$$\vec{\boldsymbol{a}} = (x_1, x_2) \Rightarrow |\vec{\boldsymbol{a}}| = \sqrt{x_1^2 + x_2^2}$$

Example 1.61

Find the distance from the origin of the vector in infinite dimensions given below:

$$\left(\frac{1}{\sqrt{2}}, \frac{1}{2}, \frac{1}{\sqrt{8}}, \dots, \frac{1}{2^{\frac{n}{2}}}, \dots\right)$$

The distance is given by:

$$\sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{\sqrt{8}}\right)^2 + \cdots + \left(\frac{1}{2^{\frac{n}{2}}}\right) + \cdots}$$

$$\sqrt{\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^n} + \cdots}$$

The expression inside the square root is a geometric series with $a = \frac{1}{2}, r = \frac{1}{2}$.

Substitute $a = \frac{1}{2}, r = \frac{1}{2}$ in the formula for the sum of a geometric series:

$$\frac{a}{1-r} = \frac{\frac{1}{2}}{1 - \frac{1}{2}} = \frac{\frac{1}{2}}{\frac{1}{2}} = 1$$

G. Probability Vectors

1.62: Probability Vectors

A vector $(x_1, x_2, \dots, x_n)$ is a probability vector if:

$$0 \le x_i \le 1$$
$$x_1 + x_2 + \cdots + x_n = 1$$

Consider events $e_1, e_2, \dots, e_n$ which are mutually exclusive and exhaustive.
Then

$$0 \le P(e_i) \le 1$$
$$e_1 + e_2 + \cdots + e_n = 1$$

Example 1.63

A.

Identification

State, with reasons, whether the vectors below are probability vectors.

A. $\left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right)$

B. $\left(\frac{2}{5}, \frac{2}{7}\right)$

Back Calculations

Can the vectors given below be probability vectors? If so, find the values of the missing variables.

C. $\left(\frac{1}{4}, p\right)$
D. $(2, c)$
E. $\left(\frac{3}{5}, \frac{4}{5}, x\right)$

Part A
Consider the first condition. Each component of the vector is between 0 and 1. That is:

$$0 \le \frac{1}{2} \le 1$$
$$0 \le \frac{1}{3} \le 1$$
$$0 \le \frac{1}{6} \le 1$$

Consider the second condition. The sum of the components is:

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{6} = \frac{3}{6} + \frac{2}{6} + \frac{1}{6} = 1$$

Both the conditions are satisfied. Hence, the vector is a probability vector.

Part B

$$\frac{2}{5} + \frac{2}{7} = \frac{14}{35} + \frac{10}{35} = \frac{24}{35} \neq 1$$

Not a probability vector

Part C

$$0 \le \frac{1}{4} \le 1$$

Hence, the given vector can be a probability vector.

$$\frac{1}{4} + p = 1 \Rightarrow p = \frac{3}{4}$$

Part D

$$2 > 1$$

Not a probability vector

Part E

$$\frac{3}{5} + \frac{4}{5} = \frac{7}{5} > 1$$

Not a probability vector

Example 1.64

An urn that has three green balls, two blue balls, four yellow balls and five purple balls.

A. (*Warmup*) I draw a single ball from the urn. Represent the sample space of the possibilities using a probability vector.
B. I draw a ball from the urn, replace it, and then draw another ball from the urn. What is the probability that the two balls have distinct colors?

Part A

$$\vec{\boldsymbol{p}} = (Green, Blue, Yellow, Purple) = \left(\underbrace{\frac{3}{14}}_{\substack{Green\\Ball}}, \underbrace{\frac{2}{14}}_{\substack{Blue\\Ball}}, \underbrace{\frac{4}{14}}_{\substack{Yellow\\Ball}}, \underbrace{\frac{5}{14}}_{\substack{Purple\\Ball}}\right)$$

Part B

$$1 - P(Same\ Color)$$

Without using vectors

$$1 - P(Same\ Color)$$

$$= 1 - \left[\underbrace{\left(\frac{3}{14}\right)^2}_{\substack{Two\\Green\ Balls}} + \underbrace{\left(\frac{2}{14}\right)^2}_{\substack{Two\\Blue\ Balls}} + \underbrace{\left(\frac{4}{14}\right)^2}_{\substack{Two\\Yellow\ Balls}} + \underbrace{\left(\frac{5}{14}\right)^2}_{\substack{Two\\Purple\ Balls}}\right]$$

$$= 1 - \left[\frac{9}{196} + \frac{4}{196} + \frac{16}{196} + \frac{25}{196}\right] = 1 - \frac{54}{196} = 1 - \frac{27}{98} = \frac{71}{98}$$

Vectors

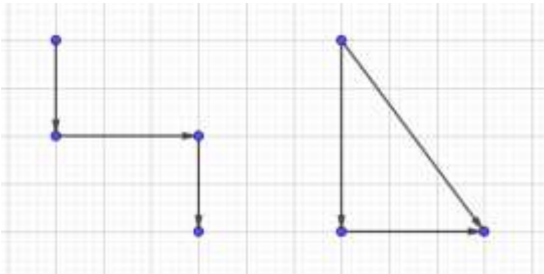
$$1 - |\boldsymbol{p}|^2 = 1 - \left[\underbrace{\left(\frac{3}{14}\right)^2}_{\substack{Two\\Green\ Balls}} + \underbrace{\left(\frac{2}{14}\right)^2}_{\substack{Two\\Blue\ Balls}} + \underbrace{\left(\frac{4}{14}\right)^2}_{\substack{Two\\Yellow\ Balls}} + \underbrace{\left(\frac{5}{14}\right)^2}_{\substack{Two\\Purple\ Balls}}\right] = \frac{71}{98}$$

1.3 Addition and Subtraction

A. Basics

Example 1.65

Bill walks $\frac{1}{2}$ mile south, then $\frac{3}{4}$ mile east, and finally $\frac{1}{2}$ mile south. How many miles is he, in a direct line, from his starting point? (AMC 8 2005/7)



Bill has the following movements:

- $\frac{1}{2}$ mile south($\boldsymbol{a}$)
- $\frac{3}{4}$ mile east($\boldsymbol{b}$)
- $\frac{1}{2}$ mile south($\boldsymbol{c}$)

The diagram shows three vectors (one for each movement). Note that

- the starting point of $\boldsymbol{b}$ is placed precisely where $\boldsymbol{a}$ ends.
- the starting point of $\boldsymbol{c}$ is placed precisely where $\boldsymbol{b}$ ends.

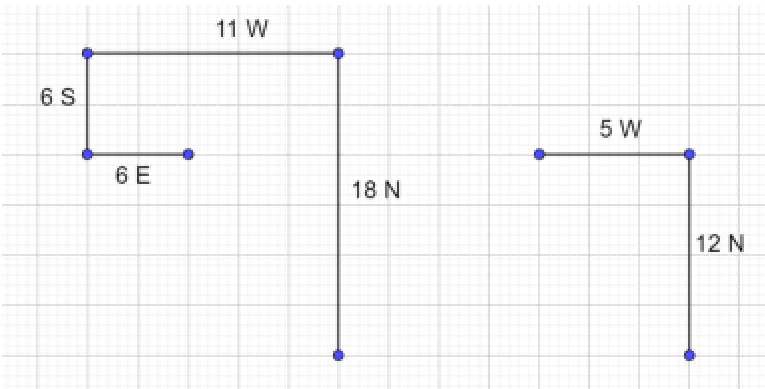
(This approach, which can be done intuitively to get Bill's movement is exactly the definition of vector addition.)

Using the Pythagorean Theorem, we get:

$$d = \sqrt{1^2 + \left(\frac{3}{4}\right)^2} = \sqrt{1 + \frac{9}{16}} = \sqrt{\frac{25}{16}} = \frac{5}{4}$$

Example 1.66

While walking on a plane surface, a traveler first headed 18 miles north, then 11 miles west, then 6 miles south and finally 6 miles east. How many miles from the starting point was the traveler after these four legs of the journey? (MathCounts 1996 Warm-Up 8)



The diagram on the left shows the movements of the traveler.

$$(5, 12, x) \Rightarrow x = 13$$

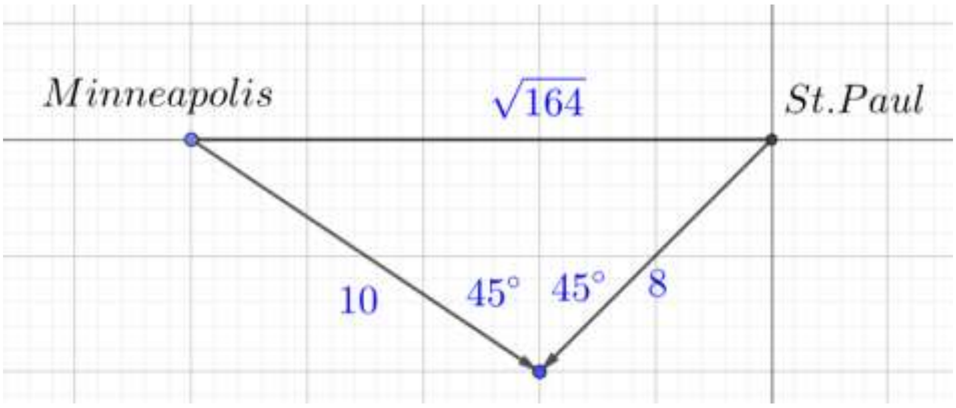
1.67: Orthogonal

An angle of $90°$ is called orthogonal.

Example 1.68

Minneapolis-St. Paul International Airport is 8 miles southwest of downtown St. Paul and 10 miles southeast of downtown Minneapolis. Which integer is closest to the number of miles between downtown St. Paul and downtown Minneapolis? **(AMC 10B 2004/8)**

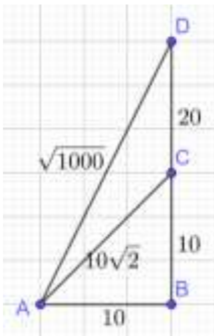
Southwest and southeast are orthogonal. Hence, we get a right triangle.



$$\sqrt{8^2 + 10^2} = \sqrt{64 + 100} = \sqrt{164} \approx \sqrt{169} = 13$$

Example 1.69

Point B is due east of point A. Point C is due north of point B. The distance between points A and C is $10\sqrt{2}$, and $\angle BAC = 45°$. Point D is 20 meters due north of point C. The distance AD is between which two integers? **(AMC 10B 2012/12)**



$$AB = \frac{10\sqrt{2}}{\sqrt{2}} = 10$$
$$AD = \sqrt{10^2 + 30^2} = \sqrt{1000}$$
$$31 < AD < 32$$

B. Algebraic Method

1.70: Addition and Subtraction: Algebraic Method

Given $\vec{\boldsymbol{a}} = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}, \vec{\boldsymbol{b}} = \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$

$$\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}} = \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \\ z_1 + z_2 \end{pmatrix}, \qquad \vec{\boldsymbol{a}} - \vec{\boldsymbol{b}} = \begin{pmatrix} x_1 - x_2 \\ y_1 - y_2 \\ z_1 - z_2 \end{pmatrix}$$

- When vectors are given in component form, they can be added or subtracted by adding/subtracting their individual components.
- This is one of the most useful properties in applications of vectors.

Example 1.71

Add the vectors $\overrightarrow{\boldsymbol{OP}} = \left(1, \frac{1}{2}\right)$ *and* $\overrightarrow{\boldsymbol{OQ}} = \left(\frac{2}{3}, 2\right)$ where O is the origin.

$$\overrightarrow{\boldsymbol{OP}}+\overrightarrow{\boldsymbol{OQ}}=\begin{pmatrix}1+\frac{2}{3}\\ \frac{1}{2}+2\end{pmatrix}=\begin{pmatrix}\frac{5}{3}\\ \frac{5}{2}\end{pmatrix}$$

1.72: Addition and Subtraction in Component Form

Given two vectors

$$\vec{\boldsymbol{X}}=a\hat{\boldsymbol{\imath}}+b\hat{\boldsymbol{\jmath}}+c\hat{\boldsymbol{k}},\qquad \vec{\boldsymbol{Y}}=d\hat{\boldsymbol{\imath}}+e\hat{\boldsymbol{\jmath}}+f\hat{\boldsymbol{k}}$$

their sum is:

$$\vec{\boldsymbol{X}}+\vec{\boldsymbol{Y}}=(a+d)\hat{\boldsymbol{\imath}}+(b+e)\hat{\boldsymbol{\jmath}}+(c+f)\hat{\boldsymbol{k}}$$
$$\vec{\boldsymbol{X}}-\vec{\boldsymbol{Y}}=(a-d)\hat{\boldsymbol{\imath}}+(b-e)\hat{\boldsymbol{\jmath}}+(c-f)\hat{\boldsymbol{k}}$$

Example 1.73

Given that $\vec{\boldsymbol{a}}=2\hat{\boldsymbol{\imath}}+3\hat{\boldsymbol{\jmath}},\vec{\boldsymbol{b}}=5\hat{\boldsymbol{\imath}}-2\hat{\boldsymbol{\jmath}}$ find

A. $\vec{\boldsymbol{a}}+\vec{\boldsymbol{b}}$
B. $\vec{\boldsymbol{a}}-\vec{\boldsymbol{b}}$

$$\vec{\boldsymbol{a}}+\vec{\boldsymbol{b}}=2\hat{\boldsymbol{\imath}}+3\hat{\boldsymbol{\jmath}}+5\hat{\boldsymbol{\imath}}-2\hat{\boldsymbol{\jmath}}=7\hat{\boldsymbol{\imath}}+\hat{\boldsymbol{\jmath}}$$
$$\vec{\boldsymbol{a}}-\vec{\boldsymbol{b}}=2\hat{\boldsymbol{\imath}}+3\hat{\boldsymbol{\jmath}}-5\hat{\boldsymbol{\imath}}+2\hat{\boldsymbol{\jmath}}=-\boldsymbol{3}\hat{\boldsymbol{\imath}}+5\hat{\boldsymbol{\jmath}}$$

1.74: Identity Vector for Addition

$$\vec{\boldsymbol{a}}+\boldsymbol{0}=\vec{\boldsymbol{a}}$$

The zero vector is the identity vector for addition.

1.75: Additive Inverse

Additive inverse of $\vec{\boldsymbol{a}}$ is $-\vec{\boldsymbol{a}}$

$$\vec{\boldsymbol{a}}+(-\vec{\boldsymbol{a}})=\boldsymbol{0}$$

Example 1.76: Vector Equation

Solve for $\vec{\boldsymbol{x}}$:

$$\vec{\boldsymbol{x}}+\vec{\boldsymbol{a}}=\vec{\boldsymbol{b}}$$

$$\vec{\boldsymbol{x}}=\vec{\boldsymbol{b}}-\vec{\boldsymbol{a}}$$

C. Geometric Method: Addition

1.77: Geometrical Addition: Triangle Approach

To add vectors $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$:

- Place the tail of the second vector where the tip of the first vector ends.
- The sum of the two vectors is the vector starting from the tail of the first vector, and ending at the tip of the second vector

This approach is

- Called the triangle approach because it creates a triangle.
- Also called the *tip to tail* approach.

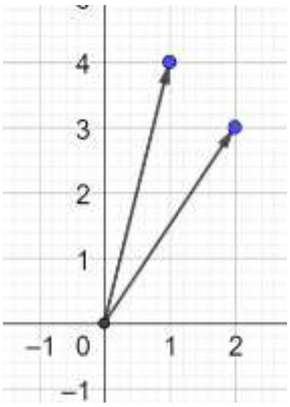
Example 1.78

Add the vectors

$$\overrightarrow{\boldsymbol{OA}} = (2,3)\ and\ \overrightarrow{\boldsymbol{OB}} = (1,4)$$

Where O is the origin.

Method I: Use the tip to tail approach

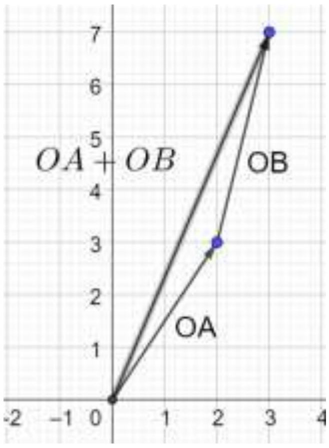


If you draw the vectors, you will get the diagram on the left.

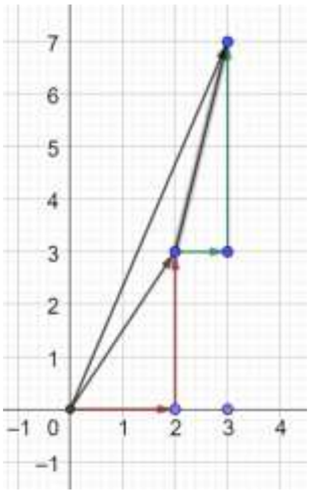
To add the vectors, pick any vector and move it to the tip of the other vector.
Suppose, we choose to move $\overrightarrow{OA}$ and put it at the tip of $\overrightarrow{OB}$. We then get the diagram to the right.

And

$$\overrightarrow{OA} + \overrightarrow{OB} = (3,7)$$



Method II: Add the components



We can work out the components of each of the vectors we want to add up (see the diagram on the left):

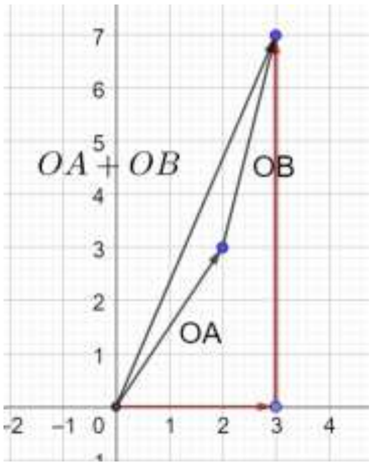
$$\overrightarrow{OA} = (2,3) \Rightarrow 2\ Right, 3\ Up$$
$$\overrightarrow{OB} = (1,4) \Rightarrow 1\ Right, 4\ Up$$

We can add the components in the right direction, and the components in the up direction directly:

$$Right: 2 + 1 = 3$$
$$Up: 3 + 4 = 7$$

And, hence, the final answer is:

$$\overrightarrow{OA} + \overrightarrow{OB} = (2 + 1, 3 + 4) = (3,7)$$



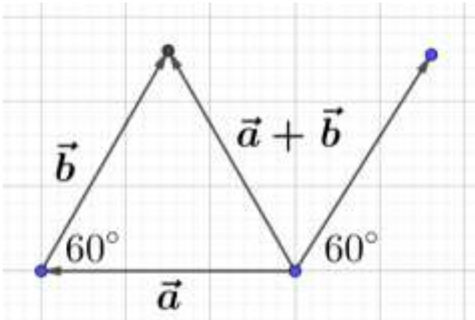
Example 1.79: Important

If $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$ are two unit vectors such that $|\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}}|$ is also a unit vector, then find the angle between $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$. (CBSE 2014)

Geometric Method

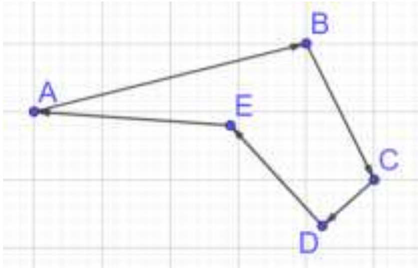
We get an equilateral triangle, as shown alongside. The angle between the line segments is 60°, but this is not the angle between the vectors, because the vectors are arranged *tip to tail*.

To find the angle between the vectors, we move $\vec{\boldsymbol{b}}$ to the right until its tail coincides with the tail of $\vec{\boldsymbol{a}}$.



The vectors are now arranged tail to tail, and the angle is:

$$180 - 60 = 120°$$



1.80: Vector Polygon

A polygon can be thought of as a set of vectors.

Example 1.81

Identify the sum of the following vectors in the vector polygon drawn alongside.

We want to find:

$$\overrightarrow{\boldsymbol{AB}} + \overrightarrow{\boldsymbol{BC}} + \overrightarrow{\boldsymbol{CD}} + \overrightarrow{\boldsymbol{DE}} + \overrightarrow{\boldsymbol{EA}}$$

This looks complicated until we realize we can use the tip to tail approach, and we end up exactly where we began and hence:

$$\overrightarrow{\boldsymbol{AB}} + \overrightarrow{\boldsymbol{BC}} + \overrightarrow{\boldsymbol{CD}} + \overrightarrow{\boldsymbol{DE}} + \overrightarrow{\boldsymbol{EA}} = \overrightarrow{\boldsymbol{AB}} = 0$$

1.82: Sum of a Vector Polygon

The sum of the vectors in a vector polygon is zero.

Example 1.83

Simplify:

A. $\overrightarrow{\boldsymbol{AB}} + \overrightarrow{\boldsymbol{BC}} + \overrightarrow{\boldsymbol{CD}}$

We do not know the details of the vectors involved. But, using the tip to tail approach (for any diagram that we draw), we get:

$$\overrightarrow{\boldsymbol{AB}} + \overrightarrow{\boldsymbol{BC}} + \overrightarrow{\boldsymbol{CD}} = \overrightarrow{\boldsymbol{AC}} + \overrightarrow{\boldsymbol{CD}} = \overrightarrow{\boldsymbol{AD}}$$

Example 1.84

True or False

State whether the following statement is true or false. If it is false, correct it.

The converse of the statement “If three vectors can be arranged to form the sides of a triangle, the sum of three vectors is zero.” is true.

The converse is: “If the sum of three vectors is zero, then the three vectors can be arranged to form the sides of a triangle.”

This is not true.

Exception I: Zero Vectors

$$\vec{\boldsymbol{a}} = \vec{\boldsymbol{b}} = \vec{\boldsymbol{c}} = \mathbf{0} \Rightarrow \vec{\boldsymbol{a}} + \vec{\boldsymbol{b}} + \vec{\boldsymbol{c}} = \mathbf{0}$$

In the above, the three vectors are all zero, and hence their sum is zero. But they do not form the sides of a triangle.

Exception I: Collinear Vectors

$$\vec{\boldsymbol{a}} = \vec{\boldsymbol{b}}, \vec{\boldsymbol{c}} = -2\vec{\boldsymbol{a}} \Rightarrow \vec{\boldsymbol{a}} + \vec{\boldsymbol{b}} + \vec{\boldsymbol{c}} = \vec{\boldsymbol{a}} + \vec{\boldsymbol{a}} - 2\vec{\boldsymbol{a}} = 0$$

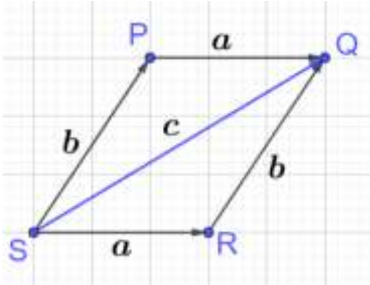
Even though the vectors are non-zero, they are collinear, and hence do not form the sides of a triangle.

We need to incorporate these two exceptions to make the statement true. The corrected version is:
"If the sum of three non-zero, non-collinear vectors is zero, then the three vectors form the sides of a triangle."

1.85: Geometrical Addition: Parallelogram Approach

To add vectors $\vec{a}$ and $\vec{b}$:

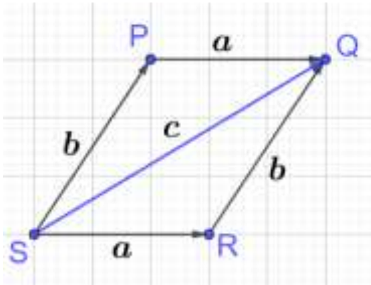
- Place the tails of both the vectors at a common starting point.
- Create a parallelogram by adding two vectors parallel in direction and equal in magnitude to the two vectors to be added.
- The resultant vector is the diagonal of the parallelogram.



- This approach is called the parallelogram approach because it creates a parallelogram
- Alternatively, take the triangle approach, and reflect the triangle across the resultant vector to get a parallelogram.

1.86: Vector Addition is Commutative

$$\vec{a}+\vec{b}=\vec{b}+\vec{a}$$



Consider parallelogram $PQRS$:

$$PQ \parallel SR, \qquad PQ = SR \Rightarrow \overrightarrow{PQ} = \overrightarrow{SR} = \vec{a}$$
$$SP \parallel RQ, \qquad SP = RP \Rightarrow \overrightarrow{SP} = \overrightarrow{RQ} = \vec{b}$$

Using the tip to tail approach:

$$In\ \Delta SPQ: \vec{b}+\vec{a}=\vec{c}$$
$$In\ \Delta SRQ: \vec{a}+\vec{b}=\vec{c}$$
$$\therefore \vec{a}+\vec{b}=\vec{b}+\vec{a}$$

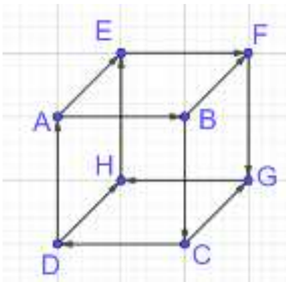
Example 1.87

The cube alongside consists of unit vectors.

A. Simplify $\overrightarrow{EA}+\overrightarrow{AD}+\overrightarrow{DC}+\overrightarrow{CB}$

B. $\sqrt{\left|\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CG}+\overrightarrow{GH}\right|}$

C. $\sqrt[3]{\left|\overrightarrow{AD}+\overrightarrow{DH}+\overrightarrow{EF}+\overrightarrow{HE}+\overrightarrow{FG}\right|}$



Part A

$$\overrightarrow{ED}+\overrightarrow{DC}+\overrightarrow{CB}=\overrightarrow{EC}+\overrightarrow{CB}=\overrightarrow{EB}$$

Part B

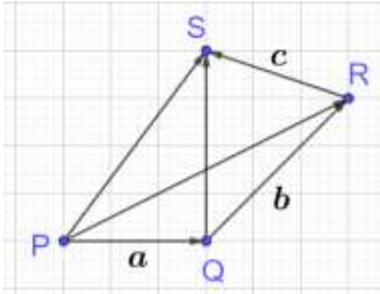
$$\sqrt{\left|\overrightarrow{AC}+\overrightarrow{CG}+\overrightarrow{GH}\right|}=\sqrt{\left|\overrightarrow{AG}+\overrightarrow{GH}\right|}=\sqrt{\left|\overrightarrow{AH}\right|}=\sqrt{\sqrt{2}}=\sqrt[4]{2}$$

Part C

$$\sqrt[3]{\left|\overrightarrow{AH}+\overrightarrow{HE}+\overrightarrow{EF}+\overrightarrow{FG}\right|}=\sqrt[3]{\left|\overrightarrow{AE}+\overrightarrow{EF}+\overrightarrow{FG}\right|}=\sqrt[3]{\left|\overrightarrow{AF}+\overrightarrow{FG}\right|}=\sqrt[3]{\left|\overrightarrow{AG}\right|}=\sqrt[3]{\sqrt{3}}=\sqrt[6]{3}$$

1.88: Vector Addition is Associative

$$(\vec{a}+\vec{b})+\vec{c}=\vec{a}+(\vec{b}+\vec{c})$$



$$\overrightarrow{PR}=\vec{a}+\vec{b}$$
$$\overrightarrow{QS}=\vec{b}+\vec{c}$$

$$\overrightarrow{PS}=\overrightarrow{PR}+\overrightarrow{RS}=(\vec{a}+\vec{b})+\vec{c}$$
$$\overrightarrow{PS}=\overrightarrow{PQ}+\overrightarrow{QS}=\vec{a}+(\vec{b}+\vec{c})$$

From the above two statements:

$$(\vec{a}+\vec{b})+\vec{c}=\vec{a}+(\vec{b}+\vec{c})$$

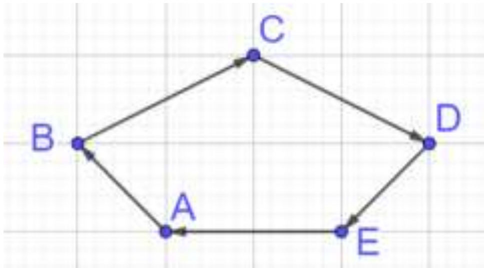
1.89: Geometrical Addition: Parallelogram Law

The above also establishes the parallelogram law of vector addition.

Example 1.90

Consider pentagon $ABCDE$. Find $\vec{x}$ in each case below (answer separately):

A. $\overrightarrow{AB}+\overrightarrow{BC}=\vec{x}$
B. $\overrightarrow{AB}+\overrightarrow{EA}+\overrightarrow{DE}=\vec{x}$
C. $\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CD}+\overrightarrow{DE}+\vec{x}=0$



$$\vec{x}=\overrightarrow{AB}+\overrightarrow{BC}=\overrightarrow{AC}$$
$$\vec{x}=\overrightarrow{AB}+\overrightarrow{EA}+\overrightarrow{DE}=\overrightarrow{DB}$$
$$\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CD}+\overrightarrow{DE}+\overrightarrow{EA}=0\Rightarrow\vec{x}=\overrightarrow{EA}$$

D. Geometric Method: Subtraction

1.91: Geometrical Subtraction: Cancellation

Vectors that are equal in magnitude value and opposite in direction add up to zero. In particular

$$\vec{a}+(-\vec{a})=\mathbf{0}$$

Example 1.92

Simplify:

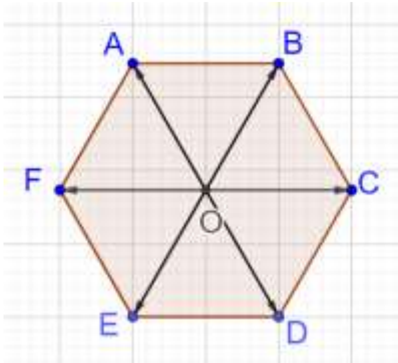
A. $\vec{b}+(-\vec{b})$

$$\vec{b}+(-\vec{b})=\mathbf{0}$$

Example 1.93

Consider Regular Hexagon $ABCDEF$. Find

A. the sum of the six vectors originating from the center of the hexagon and terminating at the vertices.
B. $\overrightarrow{EC}+\overrightarrow{CA}+\overrightarrow{CD}$



Part A

$$\overrightarrow{OA} + \overrightarrow{OD} = \vec{0}$$
$$\overrightarrow{OB} + \overrightarrow{OE} = \vec{0}$$
$$\overrightarrow{OF} + \overrightarrow{OC} = \vec{0}$$

Adding the above the sum of the six vectors is:

$$\overrightarrow{OA} + \overrightarrow{OD} + \overrightarrow{OB} + \overrightarrow{OE} + \overrightarrow{OF} + \overrightarrow{OC} = \vec{0}$$

Part B

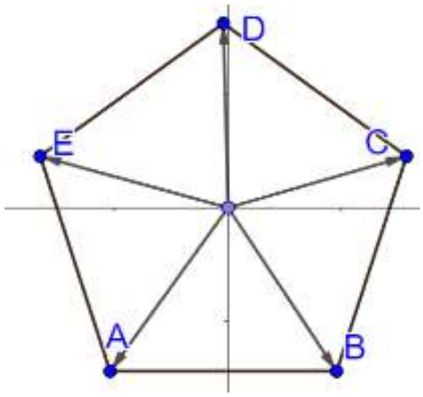
$$\overrightarrow{EC} + \overrightarrow{CA} + \overrightarrow{CD} = \overrightarrow{EA} + \overrightarrow{CD} = \overrightarrow{EA} + \overrightarrow{AF} = \overrightarrow{EF}$$

Example 1.94

Consider regular pentagon $ABCDE$. Find the sum of the five vectors originating from the center of the pentagon and terminating at the vertices.

Draw regular pentagon $ABCDE$. We want the sum:

$$\vec{a} = \overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC} + \overrightarrow{OD} + \overrightarrow{OE}$$



Rotate the vectors so that each vector moves over to its next vertex counterclockwise.
For example,

$$\textit{Rotate } \overrightarrow{OA} \textit{ so that it becomes } \overrightarrow{OB}$$
$$\textit{Rotate } \overrightarrow{OB} \textit{ so that it becomes } \overrightarrow{OC}$$
$$.$$
$$.$$
$$\textit{Rotate } \overrightarrow{OE} \textit{ so that it becomes } \overrightarrow{OA}$$

The sum is now:

$$\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC} + \overrightarrow{OD} + \overrightarrow{OE} = \vec{a}$$

The sum of the vectors did not change under a rotation which was not 360°. Hence, the vector must be the zero vector.

1.95: Sum of Vectors to Vertices of Regular Polygon

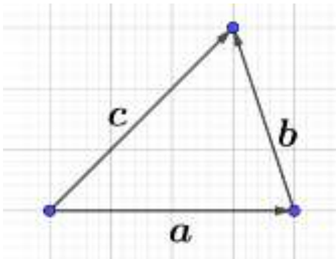
The sum of the vectors drawn from the center of a regular polygon to the vertices is zero.

Consider a regular polygon with n sides.
Let the sum of the vectors be $\vec{a}$.
Rotate the vectors so that each vector moves over to its corresponding vertex counterclockwise.
The sum remains $\vec{a}$. Since, the sum of the vectors did not change under a rotation which was not 360°. Hence, the vector must be the zero vector.

1.96: Geometrical Subtraction: Triangle Approach

Draw a vector triangle such that

$$\vec{a} + \vec{b} = \vec{c}$$



Subtract $\vec{b}$ from sides of the above:

$$\vec{a} = \vec{c} - \vec{b}$$
$$\vec{b} = \vec{c} - \vec{a}$$

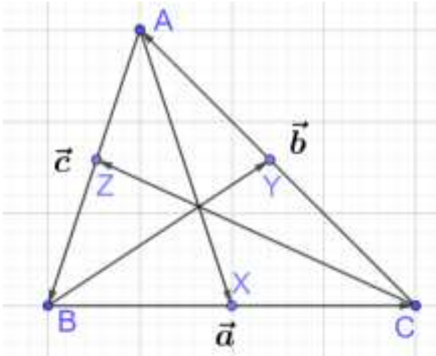
Example 1.97

In the diagram

$$\overrightarrow{BC} = \vec{a}, \overrightarrow{CA} = \vec{b}, \overrightarrow{AB} = \vec{c}, \overrightarrow{BX} = \vec{x}, \overrightarrow{CY} = \vec{y}, \overrightarrow{AZ} = \vec{z}$$

Express the vectors below in terms of the vectors above:

A. $\vec{a} + \vec{b}$
B. $\overrightarrow{AX}$
C. $\overrightarrow{BY}$
D. $\overrightarrow{CZ}$
E. $\overrightarrow{CX}$
F. $\overrightarrow{XC}$



Part A

$$\vec{a} + \vec{b} = -\vec{c}$$

Part B

$$\overrightarrow{AX} = \overrightarrow{AB} + \overrightarrow{BX} = \vec{c} + \vec{x}$$
$$\overrightarrow{AX} = \overrightarrow{AC} + \overrightarrow{CX} = -\vec{b} + \overrightarrow{CX} = \overrightarrow{CX} - \vec{b}$$

Part C

$$\overrightarrow{BY} = \vec{a} + \vec{y}$$
$$\overrightarrow{BY} = -\vec{c} + \overrightarrow{AY}$$

Part D

$$\overrightarrow{CZ} = \vec{b} + \vec{z}$$
$$\overrightarrow{CZ} = -\vec{a} + \vec{z}$$

Part E

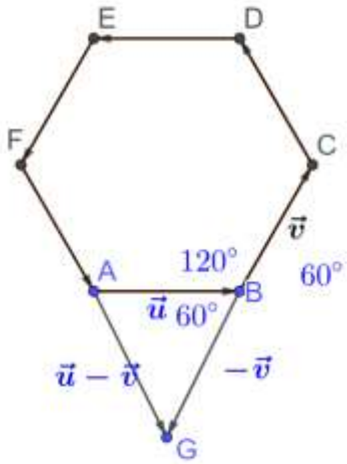
$$\overrightarrow{CX} = \overrightarrow{CB} + \overrightarrow{BX} = -\vec{a} + \vec{x}$$
$$\overrightarrow{CX} = \vec{b} + \vec{c} + \vec{x}$$

Part F

$$\vec{b} = \overrightarrow{CA}$$
$$-\vec{b} = \overrightarrow{AC}$$
$$\overrightarrow{XC} = -\overrightarrow{CX} = -(-\vec{a} + \vec{x}) = \vec{a} - \vec{x}$$
$$\overrightarrow{XC} = \overrightarrow{XA} + \overrightarrow{AC} = \overrightarrow{XA} - \vec{b}$$

Example 1.98

$ABCDEF$ is a regular hexagon with $\overrightarrow{AB} = \vec{u}$ and $\overrightarrow{BC} = \vec{v}$. Find $\overrightarrow{CD}, \overrightarrow{DE}, \overrightarrow{EF}$ and $\overrightarrow{FA}$ in terms of $\vec{u}$ and $\vec{v}$.



Each angle of a regular polygon with n sides

$$=\frac{180(n-2)}{n}=\frac{180(6-2)}{6}=\frac{180(4)}{6}=120°$$

$$\overrightarrow{ED}=-\vec{u}$$
$$\overrightarrow{EF}=-\vec{v}$$

Method I

Draw a triangle

$$\overrightarrow{FA}=\vec{u}-\vec{v}$$
$$\overrightarrow{CD}=-\overrightarrow{FA}=\vec{v}-\vec{u}$$

Method II

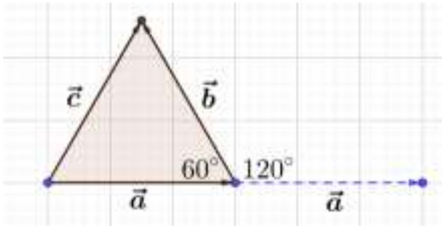
$$\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CD}=\overrightarrow{AD}$$
$$\vec{u}+\vec{v}+\overrightarrow{CD}=2\vec{v}$$
$$\overrightarrow{CD}=\vec{v}-\vec{u}$$

Example 1.99

The sum of two unit vectors is itself a unit vector.

A. Find the angle between the vectors.
B. Find the magnitude of the difference of the two vectors.

Part A



Let the unit vectors be $\hat{a}$ and $\hat{b}$.

$$\hat{a}+\hat{b}=\hat{c}$$
$$|\hat{a}|=|\hat{b}|=|\hat{c}|=1$$

Draw a diagram, and note that the triangle formed by the three vectors is equilateral.

Hence, the angle between the sides of the triangle

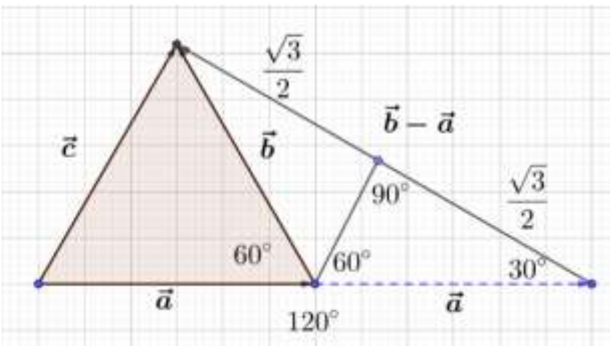
$$=60°$$

To find the angle between the vectors, we need to arrange them tail to tail. So, move $\hat{a}$ to the right so that its tail is at the tail of $\hat{b}$.

By angles in a straight line:

$$Angle\ between\ Vectors=180-60=120°$$

Part B



Note that the triangle formed by $\hat{a}$, $\hat{b}$ and $\hat{b}-\hat{a}$ is isosceles.

Hence, the angle shown is:

$$\frac{\mathbf{180-120}}{\mathbf{2}}=\frac{\mathbf{60}}{\mathbf{2}}=\mathbf{30°}$$

Drop a perpendicular from the vertex, to get a $30-60-90$ triangle.

In a $30-60-90$ triangle:

$$Side\ opp\ 60° = \frac{\sqrt{3}}{2} \times Hyp = \frac{\sqrt{3}}{2} \times 1 = \frac{\sqrt{3}}{2}$$

Similarly, the other half is also:

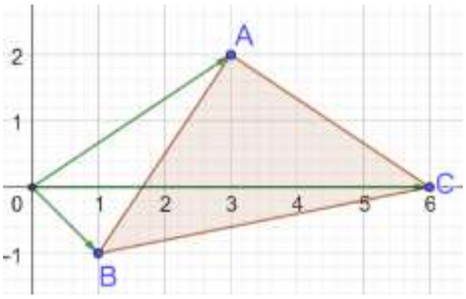
$$\frac{\sqrt{3}}{2}$$

Hence, the magnitude of the vector we want is:

$$\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = \sqrt{3}$$

Example 1.100

If $\hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}}, \hat{\boldsymbol{\jmath}} + \hat{\boldsymbol{k}}, \hat{\boldsymbol{\imath}} + \hat{\boldsymbol{k}}$ are the position vectors of the vertices of a triangle ABC taken in order, then $\angle A$ is equal to (BITSAT 2018)



Draw a diagram and let A, B, C be the vertices of the triangle.
The vectors are three-dimensional, but a 2D is still useful for reference purposes.

Let

$$\overrightarrow{\boldsymbol{OA}} = \hat{\imath} + \hat{\jmath}, \overrightarrow{\boldsymbol{OC}} = \hat{\imath} + \hat{\boldsymbol{k}}, \overrightarrow{\boldsymbol{OB}} = \hat{\jmath} + \hat{\boldsymbol{k}}$$

Then using vector addition and subtraction:

$$\overrightarrow{\boldsymbol{CA}} = \overrightarrow{\boldsymbol{OA}} + \overrightarrow{\boldsymbol{CO}} = \overrightarrow{\boldsymbol{OA}} - \overrightarrow{\boldsymbol{OC}} = \hat{\imath} + \hat{\jmath} - (\hat{\imath} + \hat{\boldsymbol{k}}) = \hat{\jmath} - \hat{\boldsymbol{k}}$$
$$\overrightarrow{\boldsymbol{AB}} = \overrightarrow{\boldsymbol{OB}} - \overrightarrow{\boldsymbol{OA}} = (\hat{\jmath} + \hat{\boldsymbol{k}}) - (\hat{\imath} + \hat{\jmath}) = \hat{\boldsymbol{k}} - \hat{\imath}$$
$$\overrightarrow{\boldsymbol{BC}} = \overrightarrow{\boldsymbol{OC}} - \overrightarrow{\boldsymbol{OB}} = \hat{\imath} - \hat{\jmath}$$

Calculate the length of the sides of the triangle by calculating the magnitude of the vectors that make up the sides of the triangle:

$$|\overrightarrow{\boldsymbol{AB}}| = |\overrightarrow{\boldsymbol{BC}}| = \overrightarrow{\boldsymbol{CA}} = \sqrt{1^2 + 1^2} = \sqrt{2}$$

Since the sides of the triangle are equal:

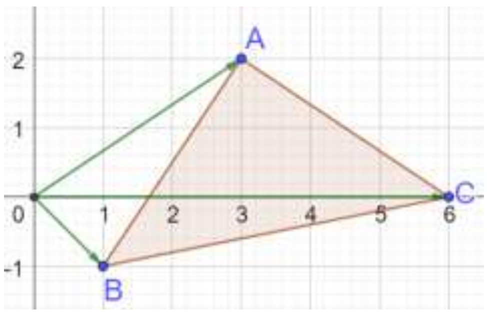
$$\Delta ABC\ is\ equilateral \Rightarrow \angle A = 60°$$

Example 1.101

Mark all correct options

If the position vectors of the vertices A, B, C of a triangle ABC are $7\hat{\boldsymbol{\jmath}} + 10\hat{\boldsymbol{k}}, -\hat{\boldsymbol{\imath}} + 6\hat{\boldsymbol{\jmath}} + 6\hat{\boldsymbol{k}}$ and $-4\hat{\boldsymbol{\imath}} + 9\hat{\boldsymbol{\jmath}} + 6\hat{\boldsymbol{k}}$ respectively, the triangle is:

A. Equilateral
B. Isosceles
C. Scalene
D. Right Angled
E. Acute Angled
F. Obtuse Angled (BITSAT 2008, Adapted)



$$\overrightarrow{\boldsymbol{AB}} = \overrightarrow{\boldsymbol{OB}} - \overrightarrow{\boldsymbol{OA}} = -\hat{\imath} - \hat{\jmath} - 4\hat{\boldsymbol{k}} \Rightarrow |\overrightarrow{\boldsymbol{AB}}| = \sqrt{1^2 + 1^2 + 4^2} = \sqrt{18}$$
$$\overrightarrow{\boldsymbol{BC}} = \overrightarrow{\boldsymbol{OC}} - \overrightarrow{\boldsymbol{OB}} = -3\hat{\imath} + 3\hat{\jmath} \Rightarrow |\overrightarrow{\boldsymbol{BC}}| = \sqrt{3^2 + 3^2} = \sqrt{18}$$
$$\overrightarrow{\boldsymbol{CA}} = \overrightarrow{\boldsymbol{OA}} - \overrightarrow{\boldsymbol{OC}} = 4\hat{\imath} - 2\hat{\jmath} + 4\hat{\boldsymbol{k}} \Rightarrow |\overrightarrow{\boldsymbol{CA}}| = \sqrt{4^2 + 2^2 + 4^2} = \sqrt{36} = 6$$

Since

$$\left|\overrightarrow{\boldsymbol{AB}}\right| = \left|\overrightarrow{\boldsymbol{BC}}\right| = \sqrt{18} \Rightarrow \Delta ABC \text{ is isosceles}$$

Check whether the triangle is right-angled by checking whether is satisfies Pythagoras Theorem

$$\left|\overrightarrow{\boldsymbol{AB}}\right|^2 + \left|\overrightarrow{\boldsymbol{BC}}\right|^2 = 18 + 18 = 36 = \left|\overrightarrow{\boldsymbol{OC}}\right|^2 \Rightarrow \Delta ABC \text{ is right} - \text{angled}$$

Options B and D

E. Physics Applications

1.102: Resultant Forces

1.103: Free Body Diagrams

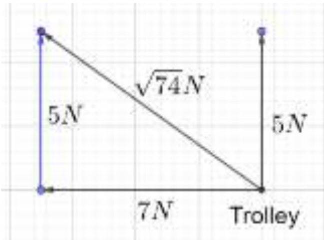
Example 1.104

A trolley being dragged along the ground is subjected to a force of $5N$ in the upward direction, and 7N in the negative x direction.

A. Draw a diagram showing the trolley, the applied forces, the resultant force
B. Calculate the magnitude of the resultant force.

Magnitude of Resultant

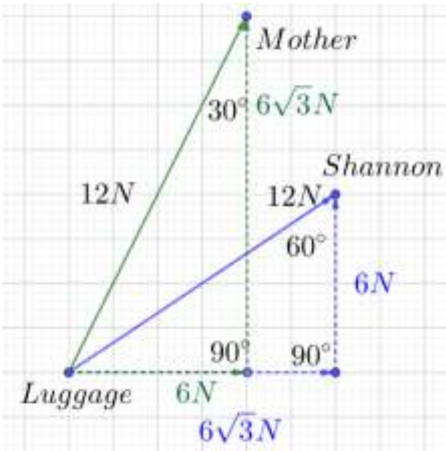
$$= \sqrt{7^2 + 5^2} = \sqrt{74}N$$



Example 1.105

Shannon is dragging his luggage when going to vacation in the Bahamas with a force of 12N. The handle of the luggage is at an angle of 30° to the ground. When, he gets tired, his mother pulls the luggage, with the same force as Shannon, but at an angle of 60° to the ground (since she is taller).

A. Draw a diagram for the force vectors for Shannon and his mother.
B. Determine the components of each force vector.
C. Who will pull the luggage faster? Shannon or his mother?



Part B

$$\overrightarrow{\boldsymbol{Shannon}} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 6\sqrt{3} \\ 6 \end{pmatrix}$$

$$\overrightarrow{\boldsymbol{Mother}} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 6 \\ 6\sqrt{3} \end{pmatrix}$$

Part C

Shannon will pull the luggage faster because the force in the horizontal direction is greater for him.

1.106: F

$$Work = Net\ change\ in\ energy = \Delta E$$

$$Potential\ Energy$$
$$Kinetic\ Energy$$

$$TME = KE + PME$$

Example 1.107

Ignore friction. Shannon drags his $\underbrace{trolley\ from\ rest}_{State\ A}$ until it has a $\underbrace{\text{speed of } 10\frac{miles}{hour}}_{State\ B}$.

A. Is there a change in energy for the trolley from State A to State B?
B. What caused it?

Part A

$$State\ A\colon Trolley\ has\ no\ energy$$
$$State\ B\colon Trolley\ has\ Kinetic\ Energy$$

Part B

$$Work\ done\ on\ the\ trolley\ increased\ the\ kinetic\ energy$$
$$Work = \Delta E$$

F. Triangle Inequality

1.108: Triangle Inequality: Geometric Interpretation

$$|\vec{\boldsymbol{x}} + \vec{\boldsymbol{y}}| \le |\vec{\boldsymbol{x}}| + |\vec{\boldsymbol{y}}|$$

In the 2-dimensional case, the above statement is equal to the statement from geometry:

The length of the third side of a triangle is less than the sum of the other two sides.

We can demonstrate by drawing a vector triangle.
Note that equality is achieved exactly when the two vectors both point in the same direction.

1.109: Triangle Inequality: Algebraic Proof

The magnitude of the sum of two vectors is less than or equal to the sum of the magnitudes of the two vectors.

$$|\vec{\boldsymbol{x}} + \vec{\boldsymbol{y}}| \le |\vec{\boldsymbol{x}}| + |\vec{\boldsymbol{y}}|$$

Assumptions

Let

$$\vec{\boldsymbol{x}} = (a_1, a_2, \dots, a_n), \qquad \vec{\boldsymbol{y}} = (b_1, b_2, \dots, b_n)$$

We can add the two vectors above to get:

$$\vec{\boldsymbol{x}} + \vec{\boldsymbol{y}} = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$$

Left Hand Side

Using the definition of magnitude, the LHS is:

$$|\vec{\boldsymbol{x}} + \vec{\boldsymbol{y}}| = \sqrt{(a_1 + b_1)^2 + (a_2 + b_2)^2 + \cdots + (a_n + b_n)^2}$$

Square both sides of the above:

$$(|\vec{\boldsymbol{x}} + \vec{\boldsymbol{y}}|)^2 = (a_1 + b_1)^2 + (a_2 + b_2)^2 + \cdots + (a_n + b_n)^2$$

Expand the RHS to get:

$$= (a_1^2 + 2a_1b_1 + b_1^2) + (a_2^2 + 2a_2b_2 + b_2^2) + \cdots + (a_n^2 + 2a_nb_n + b_n^2)$$

Rearrange to get:

$$= \underbrace{(a_1^2 + a_2^2 + \cdots + a_n^2) + (b_1^2 + b_2^2 + \cdots + b_n^2)}_{\textit{Call this X}} + (2a_1b_1 + 2a_2b_2 + \cdots + 2a_nb_n)$$

Right Hand Side

The square of the right-hand side is:

$$(|\vec{x}| + |\vec{y}|)^2 = |\vec{x}|^2 + |\vec{y}|^2 + 2|\vec{x}||\vec{y}|$$

Substitute using the definition of magnitude:

$$= \underbrace{(a_1^2 + a_2^2 + \cdots + a_n^2) + (b_1^2 + b_2^2 + \cdots + b_n^2)}_{\textit{Call this X}} + 2\sqrt{a_1^2 + a_2^2 + \cdots + a_n^2}\sqrt{b_1^2 + b_2^2 + \cdots + b_n^2}$$

Bringing it together

Comparing the two above, we see that **X** is common to both. Hence, we need to show inequality in the remaining terms. Start with the Cauchy-Schwarz inequality:

$$(a_1b_1 + a_2b_2 + \cdots + a_nb_n)^2 \le (a_1^2 + a_2^2 + \cdots + a_n^2)(b_1^2 + b_2^2 + \cdots + b_n^2)$$

Take the square root both sides and multiply by 2:

$$2(a_1b_1 + a_2b_2 + \cdots + a_nb_n) \le 2\sqrt{a_1^2 + a_2^2 + \cdots + a_n^2}\sqrt{b_1^2 + b_2^2 + \cdots + b_n^2}$$

And the above is exactly what we need.

1.4 Scalar Multiplication

A. Multiplication in Vectors

Consider the following operation with the real numbers 2 and 3:

$$(2)(3) = 2 \cdot 3 = 2 \times 3 = 6$$

The above four all refer to the same quantity.

However, when we come to vector multiplication, there are different kinds of multiplication.

$$\lambda\vec{a} \to \textit{Scalar Multiplication}$$

For scalar multiplication, the input is a scalar and a vector, and the output is a vector.

$$\vec{a} \cdot \vec{b} \to \textit{Scalar Product (Dot Product)}$$

For the scalar product, the input is two vectors, and the output is a real number(scalar).

$$\vec{a} \times \vec{b} \to \textit{Vector Product(Cross Product)}$$

For the vector product, the input is two vectors, and the output is a vector.

B. Basics

1.110: Algebraic Definition: Scalar Multiplication

Given a vector $\vec{a} = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$, and a scalar λ, a scalar multiple of the vector is given by:

$$\lambda\,\vec{a} = \lambda\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} \lambda x_1 \\ \lambda y_1 \\ \lambda z_1 \end{pmatrix}$$

Scalar multiplication:

- Is achieved by multiplying each component of the vector by λ.
- Changes the magnitude of a vector ($\lambda \ne 1$).

- Does not change the direction of the vector.

Example 1.111: Finding Scalar Multiples

A. Given that $\vec{\boldsymbol{a}} = (2,3,4)$, and $\lambda = \frac{1}{5}$, find $\lambda\boldsymbol{a}$

$$\lambda\boldsymbol{a} = \frac{1}{5}(2,3,4) = \left(\frac{2}{5}, \frac{3}{5}, \frac{4}{5}\right)$$

Example 1.112: Equations

Find the values of λ and μ given that $\vec{\boldsymbol{a}} = (2,3)$, $\vec{\boldsymbol{b}} = (3,4)$, $\vec{\boldsymbol{c}} = (4,5)$, $\lambda\vec{\boldsymbol{a}} + \mu\vec{\boldsymbol{b}} = \vec{\boldsymbol{c}}$

$$\lambda\vec{\boldsymbol{a}} + \mu\vec{\boldsymbol{b}} = \vec{\boldsymbol{c}}$$
$$\lambda\begin{pmatrix}2\\3\end{pmatrix} + \mu\begin{pmatrix}3\\4\end{pmatrix} = \begin{pmatrix}4\\5\end{pmatrix}$$
$$\begin{pmatrix}2\lambda\\3\lambda\end{pmatrix} + \begin{pmatrix}3\mu\\4\mu\end{pmatrix} = \begin{pmatrix}4\\5\end{pmatrix}$$
$$\begin{pmatrix}2\lambda + 3\mu\\3\lambda + 4\mu\end{pmatrix} = \begin{pmatrix}4\\5\end{pmatrix}$$

Since the vectors are equal, their components must be equal:

$$2\lambda + 3\mu = 4 \Rightarrow \underbrace{6\lambda + 9\mu = 12}_{\text{Equation I}}$$
$$3\lambda + 4\mu = 5 \Rightarrow \underbrace{6\lambda + 8\mu = 10}_{\text{Equation II}}$$

Subtract Equation II from Equation I:

$$\mu = 2$$

Substitute $\mu = 2$ in $2\lambda + 3\mu = 4$:

$$2\lambda + 3(2) = 4 \Rightarrow \lambda = -1$$

Example 1.113

The position vectors of point A, B and C are:

$$\overrightarrow{\boldsymbol{OA}} = (4,7), \qquad \overrightarrow{\boldsymbol{OB}} = (-2,3), \qquad \overrightarrow{\boldsymbol{OC}} = (6,-4)$$

A. Vector $\boldsymbol{a}$ is parallel to $\boldsymbol{OA}$ and has magnitude twice of OA. Find the position vector of $\boldsymbol{a}$.
B. Find the vector $\boldsymbol{b}$ which is parallel to BC and has thrice the magnitude of BC.

Part A

$$\vec{\boldsymbol{a}} = 2(\boldsymbol{OA}) = 2(4,7) = (8,14)$$

Part B

$$\boldsymbol{BC} = \boldsymbol{OC} - \boldsymbol{OB} = \begin{pmatrix}6\\-4\end{pmatrix} - \begin{pmatrix}-2\\3\end{pmatrix} = \begin{pmatrix}8\\-7\end{pmatrix}$$
$$\boldsymbol{b} = 3\boldsymbol{BC} = 3\begin{pmatrix}8\\-7\end{pmatrix} = \begin{pmatrix}24\\-21\end{pmatrix}$$

1.114: Unit Vector in a Direction[1]

The unit vector in the direction of $\vec{\boldsymbol{a}}$ is given by

$$\frac{\vec{\boldsymbol{a}}}{|\vec{\boldsymbol{a}}|}$$

[1] Unit Vector is a very important concept, and gets used in many places going forward.

- Multiplying a vector with the reciprocal of its *length* scales the vector, and gives us a unit vector.

Example 1.115

Find the unit vector that is in the same direction as the vector from $A(6,8)$ to $B(14,23)$.

$$\overrightarrow{\boldsymbol{AB}} = (14-6, 23-8) = (8,15)$$
$$\left|\overrightarrow{\boldsymbol{AB}}\right| = \sqrt{8^2 + 15^2} = 17$$

The unit vector in the direction of $\overrightarrow{\boldsymbol{AB}}$ is:

$$\frac{\overrightarrow{\boldsymbol{AB}}}{\left|\overrightarrow{\boldsymbol{AB}}\right|} = \frac{(8,15)}{17} = \left(\frac{8}{17}, \frac{15}{17}\right)$$

Example 1.116: Back Calculations

A. If $\vec{\boldsymbol{a}} = (1,2,3)$, find λ such that $\lambda\vec{\boldsymbol{a}}$ is a unit vector.

$$|\vec{\boldsymbol{a}}| = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{1+4+9} = \sqrt{14}$$
$$\lambda|\vec{\boldsymbol{a}}| = 1$$
$$\lambda\sqrt{14} = 1$$
$$\lambda = \frac{1}{\sqrt{14}} = \frac{\sqrt{14}}{14}$$

Example 1.117

The position vectors of points X and Y are $\begin{pmatrix} 2 \\ -4 \end{pmatrix}$ and $\begin{pmatrix} 7 \\ 3 \end{pmatrix}$ respectively.

A. Find $\left|\overrightarrow{\boldsymbol{XY}}\right|$.
B. Find the smallest value of $\mu > 1$ such that $\left|\mu\left(\overrightarrow{\boldsymbol{XY}}\right)\right| \in \mathbb{N}$
C. Find the unit vectors parallel and antiparallel to $\overrightarrow{\boldsymbol{XY}}$.

Part A

$$\overrightarrow{\boldsymbol{XY}} = \begin{pmatrix} 7 \\ 3 \end{pmatrix} - \begin{pmatrix} 2 \\ -4 \end{pmatrix} = \begin{pmatrix} 5 \\ 7 \end{pmatrix}$$
$$|\boldsymbol{XY}| = \sqrt{5^2 + 7^2} = \sqrt{25+49} = \sqrt{74}$$

Part B

$$\mu = \sqrt{74}$$

Part C

Unit vector parallel to $\overrightarrow{\boldsymbol{XY}}$ is:

$$\frac{1}{\sqrt{74}}\begin{pmatrix} 5 \\ 7 \end{pmatrix}$$

Unit vector antiparallel to $\overrightarrow{\boldsymbol{XY}}$ is:

$$-\frac{1}{\sqrt{74}}\begin{pmatrix} 5 \\ 7 \end{pmatrix}$$

1.118: Decomposition into Length and Unit Vector

$$\vec{\boldsymbol{a}} = \underbrace{|\vec{\boldsymbol{a}}|}_{Magnitude} \underbrace{\frac{\vec{\boldsymbol{a}}}{|\vec{\boldsymbol{a}}|}}_{Direction}$$

Given a vector $\vec{\boldsymbol{a}}$**,** we know that we can find a unit vector in the direction of $\vec{\boldsymbol{a}}$ by using the formula**:**

$$\hat{\boldsymbol{a}} = \frac{\vec{\boldsymbol{a}}}{|\vec{\boldsymbol{a}}|}$$

Using the idea above, we can decompose any vector into a product of its length, and its direction:

$$\vec{\boldsymbol{a}} = \vec{\boldsymbol{a}}\frac{|\vec{\boldsymbol{a}}|}{|\vec{\boldsymbol{a}}|} = \underbrace{|\vec{\boldsymbol{a}}|}_{Magnitude} \underbrace{\frac{\vec{\boldsymbol{a}}}{|\vec{\boldsymbol{a}}|}}_{Direction}$$

Example 1.119

Write the following vectors as the product of a scalar indicating its direction, and a unit vector.

A. $\vec{\boldsymbol{a}} = (5,12)$
B. $\vec{\boldsymbol{b}} = (3,4)$
C. $\vec{\boldsymbol{c}} = (8,15)$
D. $\vec{\boldsymbol{d}} = (4,5)$

Part A

The length of the vector is:

$$\sqrt{5^2 + 12^2} = \sqrt{169} = 13\ (Pythagorean\ Triplet: 5,12,13)$$

Then:

$$\vec{\boldsymbol{a}} = (13)\left(\frac{1}{13}\right)(5,12) = \underbrace{(13)}_{Length} \underbrace{\left(\frac{5}{13}, \frac{12}{13}\right)}_{Unit\ Vector}$$

Part B

$$\vec{\boldsymbol{b}} = (3,4) = (5)\left(\frac{1}{5}\right)(3,4) = (5)\left(\frac{3}{5}, \frac{4}{5}\right)$$

Part C

$$\vec{\boldsymbol{c}} = (17)\left(\frac{8}{17}, \frac{15}{17}\right)$$

Part D

$$\vec{\boldsymbol{d}} = \left(\frac{\sqrt{4^2 + 5^2}}{\sqrt{4^2 + 5^2}}\right)(4,5) = \sqrt{41}\left(\frac{4}{\sqrt{41}}, \frac{5}{\sqrt{41}}\right)$$

1.120: Multiplication by a Negative Scalar

Multiplication by a negative scalar reverses the direction of a vector.

By reverses, we mean that the direction changes by 180°

Example 1.121

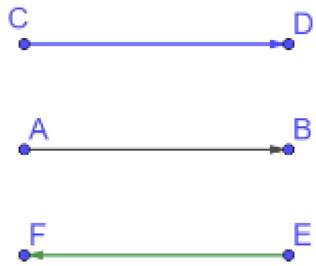
Position vector $\vec{\boldsymbol{a}}$ makes an angle of 34° with the positive direction of the x −axis. Find the angle made by $-\vec{\boldsymbol{a}}$ with the positive direction of $x - axis$.

C. Parallel Vectors

1.122: Parallel Vectors

Two vectors are parallel or anti-parallel when one vector is a non-zero scalar multiple of the other.

Example 1.123: Collinearity



Angle between $\boldsymbol{AB}$ and $\boldsymbol{CD}$

$$= 0°$$

Angle between $\boldsymbol{AB}$ and $\boldsymbol{FE}$

$$= 180°$$

Example 1.124

The non-zero vectors $\vec{\boldsymbol{a}}, \vec{\boldsymbol{b}}$ and $\vec{\boldsymbol{c}}$ are related by $\vec{\boldsymbol{a}} = 8\vec{\boldsymbol{b}}$, and $\vec{\boldsymbol{c}} = -7\vec{\boldsymbol{b}}$. Then, the angle between $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{c}}$ is: (JEE Main 2008)

Since $\vec{\boldsymbol{a}}$ is a scalar multiple of $\vec{\boldsymbol{b}}, \vec{\boldsymbol{a}}$ is parallel to $\vec{\boldsymbol{b}}$.

$$\vec{\boldsymbol{a}} \textit{ goes in the same direction as } \vec{\boldsymbol{b}}$$

Since $\vec{\boldsymbol{c}}$ is a negative scalar multiple of $\vec{\boldsymbol{b}}$, it is anti-parallel to $\vec{\boldsymbol{b}}$.

$$\vec{\boldsymbol{c}} \textit{ goes in the opposite direction as } \vec{\boldsymbol{b}}$$
$$\vec{\boldsymbol{c}} \textit{ goes in the opposite direction as } \vec{\boldsymbol{a}}$$

So, angle between $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{c}}$ is π.

Example 1.125

If vectors $\overrightarrow{\boldsymbol{a_1}} = x\hat{\boldsymbol{\imath}} - \hat{\boldsymbol{\jmath}} + \widehat{\boldsymbol{k}}$ and $\overrightarrow{\boldsymbol{a_2}} = \hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}} + z\widehat{\boldsymbol{k}}$ are collinear, then a possible unit vector parallel to the vector $x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}} + z\widehat{\boldsymbol{k}}$ is:

A. $\frac{1}{\sqrt{2}}(-\hat{\boldsymbol{\jmath}} + \widehat{\boldsymbol{k}})$
B. $\frac{1}{\sqrt{2}}(\hat{\boldsymbol{\imath}} - \hat{\boldsymbol{\jmath}})$
C. $\frac{1}{\sqrt{3}}(\hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}} - \widehat{\boldsymbol{k}})$
D. $\frac{1}{\sqrt{3}}(\hat{\boldsymbol{\imath}} - \hat{\boldsymbol{\jmath}} + \widehat{\boldsymbol{k}})$ (JEE Main 2021, 26 Feb, Shift-II)

Since the two vectors are collinear, one must a non-zero scalar multiple of the other. Let

$$\overrightarrow{\boldsymbol{a_1}} = \lambda\overrightarrow{\boldsymbol{a_2}}$$
$$(x, -1, 1) = \lambda(1, y, z)$$
$$(x, -1, 1) = (\lambda, \lambda y, \lambda z)$$

Equate components to get $x = \lambda, y = -\frac{1}{\lambda}, z = \frac{1}{\lambda}$ and substitute in $x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}} + z\widehat{\boldsymbol{k}}$ to get:

$$= \lambda\hat{\boldsymbol{\imath}} + \left(-\frac{1}{\lambda}\right)\hat{\boldsymbol{\jmath}} + \left(\frac{1}{\lambda}\right)\widehat{\boldsymbol{k}} = \frac{1}{\lambda}(\lambda^2\hat{\boldsymbol{\imath}} - \hat{\boldsymbol{\jmath}} + \widehat{\boldsymbol{k}})$$

The magnitude of $x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}} + z\widehat{\boldsymbol{k}}$ is:

$$\sqrt{\lambda^2 + \left(-\frac{1}{\lambda}\right)^2 + \left(\frac{1}{\lambda}\right)^2} = \sqrt{\lambda^2 + \frac{1}{\lambda^2} + \frac{1}{\lambda^2}} = \sqrt{\frac{\lambda^4 + 2}{\lambda^2}} = \frac{1}{\lambda}\sqrt{\lambda^4 + 2}$$

A unit vector in the direction of $x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}} + z\widehat{\boldsymbol{k}}$ is:

$$\frac{\frac{1}{\lambda}(\lambda^2\hat{\imath}-\hat{\jmath}+\hat{k})}{\frac{1}{\lambda}\sqrt{\lambda^4+2}}=\frac{(\lambda^2\hat{\imath}-\hat{\jmath}+\hat{k})}{\sqrt{\lambda^4+2}}$$

We do not know the value of the parameter λ. Any non-zero value will work. Try:

$$\lambda=1\Rightarrow\frac{\hat{\imath}-\hat{\jmath}+\hat{k}}{\sqrt{1+2}}=\frac{\hat{\imath}-\hat{\jmath}+\hat{k}}{\sqrt{3}}\Rightarrow \mathit{Option\ D}$$

Example 1.126

A. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-zero vectors which are pairwise non-collinear. If $\vec{a}+3\vec{b}$ is collinear with $\vec{c}$ and $\vec{b}+2\vec{c}$ is collinear with $\vec{a}$, then $\vec{a}+3\vec{b}+6\vec{c}$ is: (JEE Main 2011)

B. $\vec{a}, \vec{b}$ and $\vec{c}$ are three non-zero vectors such that no two of these are collinear. If the vector $\vec{a}+2\vec{b}$ is collinear with $\vec{c}$ and $\vec{b}+3\vec{c}$ is collinear with $\vec{a}$ (λ being some non-zero scalar), then $\vec{a}+2\vec{b}+6\vec{c}$ is: (JEE Main 2004)

Part A

From the collinearity conditions:

$$\vec{a}+3\vec{b}=\lambda\vec{c}\Rightarrow\underbrace{\vec{a}+3\vec{b}+6\vec{c}=(\lambda+6)\vec{c}}_{\mathit{Equation\ I}}$$

$$\vec{b}+2\vec{c}=\mu\vec{a}\Rightarrow 3\vec{b}+6\vec{c}=3\mu\vec{a}\Rightarrow\underbrace{\vec{a}+3\vec{b}+6\vec{c}=(1+3\mu)\vec{a}}_{\mathit{Equation\ II}}$$

From Equation I and II:

$$(\lambda+6)\vec{c}=(1+3\mu)\vec{a}$$

But the vectors are non-collinear and non-zero. Hence:

$$\lambda+6=1+3\mu=0$$

$$\vec{a}+3\vec{b}+6\vec{c}=(\lambda+6)\vec{c}=0\vec{c}=0$$

Part B

From the collinearity conditions:

$$\vec{a}+2\vec{b}=\lambda\vec{c}\Rightarrow\underbrace{\vec{a}+2\vec{b}+6\vec{c}=(\lambda+6)\vec{c}}_{\mathit{Equation\ I}}$$

$$\vec{b}+3\vec{c}=\mu\vec{a}\Rightarrow 2\vec{b}+6\vec{c}=2\mu\vec{a}\Rightarrow\underbrace{\vec{a}+2\vec{b}+6\vec{c}=(1+2\mu)\vec{a}}_{\mathit{Equation\ II}}$$

From Equation I and II:

$$(\lambda+6)\vec{c}=(1+2\mu)\vec{a}$$

But the vectors are non-collinear and non-zero. Hence:

$$\lambda+6=1+2\mu=0$$

$$\vec{a}+3\vec{b}+6\vec{c}=(\lambda+6)\vec{c}=0\vec{c}=0$$

D. Physics with Scalar Multiplication

1.127: Force

$$\vec{F}=m\vec{a}$$

Force is a vector quantity given by the product of mass and acceleration where:

- Mass is a scalar quantity (with unit, for example, kg)

- Acceleration is a vector quantity (with unit, for example, $\frac{m}{s^2}$)

Note the dimensional analysis for the units:

$$\underbrace{\vec{\boldsymbol{F}}}_{kg\frac{m}{s^2}} = \underbrace{m}_{kg}\underbrace{\vec{\boldsymbol{a}}}_{\frac{m}{s^2}}$$

Note the unit of force in the SI system:

$$1\ N = 1\ kg\frac{m}{s^2}$$

1 Newton is equivalent to a force acting on an object with a mass of 1 kg and accelerating it by 1 $\frac{m}{s^2}$.

Note certain assumptions:

- Unless otherwise specified, assume friction to be zero

Example 1.128

A. On a frictionless surface, an object of mass 3 kg is accelerated by 2 $\frac{m}{s^2}$. Calculate the force in Newtons.

B. A force of 72 Newtons accelerates an object with an integer number of kg in its mass on a frictionless surface an integer number of meters per second squared. Calculate the product of the possible values of the mass of the object.

C. A spaceship with a mass of 10^7 kg fires a thruster that pushes the spaceship with a force of 72 Newtons. Calculate the acceleration of the spaceship in $\frac{m}{s^2}$. Write your answer in scientific notation.

Part A

$$\vec{\boldsymbol{F}} = m\vec{\boldsymbol{a}} = (3)(2) = 6N$$

Part B

$$\{1 \times 72, 2 \times 36, 3 \times 24, 4 \times 18, 6 \times 12, 8 \times 9\}$$
$$72^6 = (2^3 \times 3^2)^6 = 2^{18} \times 3^{12}$$

You can also do this using the formula for product of factors of a number. The product of factors of 72

$$= 72^{\frac{\tau(72)}{2}} = 72^{\frac{\tau(2^3 \times 3^2)}{2}} = 72^{\frac{(3+1)(2+1)}{2}} = 72^6 = 2^{18} \times 3^{12}$$

Part C

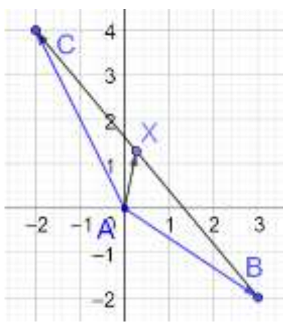
$$72 = 10^7 \times \vec{\boldsymbol{a}}$$
$$\vec{\boldsymbol{a}} = \frac{72}{10^7} = 72 \times 10^{-7} = 7.2 \times 10^{-6}\frac{m}{s^2}$$

1.5 Midpoint, Section Formulas; Linear Combination

A. Midpoint Formula

Example 1.129

Given that $\overrightarrow{\boldsymbol{AB}} = (3, -2), \overrightarrow{\boldsymbol{AC}} = (-2, 4)$, determine $\overrightarrow{\boldsymbol{CX}}$, where X is the midpoint of BC.



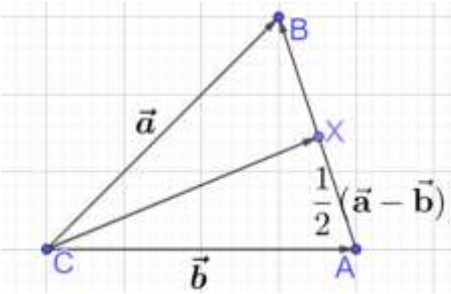
Introduce an origin at $A(0,0)$ and draw a diagram.

$$\overrightarrow{\boldsymbol{CX}} = -\overrightarrow{\boldsymbol{BX}} = -\frac{\mathbf{1}}{\mathbf{2}}(\overrightarrow{\boldsymbol{BA}} + \overrightarrow{\boldsymbol{AC}}) = -\frac{\mathbf{1}}{\mathbf{2}}(-\overrightarrow{\boldsymbol{AB}} + \overrightarrow{\boldsymbol{AC}}) = \frac{\mathbf{1}}{\mathbf{2}}(\overrightarrow{\boldsymbol{AB}} - \overrightarrow{\boldsymbol{AC}})$$
$$= \frac{1}{2}\left[\begin{pmatrix}3\\-2\end{pmatrix} - \begin{pmatrix}-2\\4\end{pmatrix}\right] = \frac{1}{2}\left[\begin{pmatrix}5\\-6\end{pmatrix}\right] = \begin{pmatrix}\frac{5}{2}\\-3\end{pmatrix}$$

1.130: Midpoint Formula

See the diagram alongside, with the usual notation.
If X is the midpoint of BA, then:

$$\overrightarrow{CX} = \frac{\vec{a} + \vec{b}}{2}$$



Using Vectors

From the triangle:

$$\overrightarrow{AB} = \overrightarrow{AC} + \overrightarrow{CB} = -\vec{b} + \vec{a} = \vec{a} - \vec{b}$$

And since X is the midpoint of AB:

$$\overrightarrow{AX} = \frac{1}{2}\overrightarrow{AB} = \frac{1}{2}(\vec{a} - \vec{b})$$

Finally, the expression that we want is:

$$\overrightarrow{CX} = \overrightarrow{CA} + \overrightarrow{AX} = \vec{b} + \frac{1}{2}(\vec{a} - \vec{b}) = \vec{b} + \frac{1}{2}\vec{a} - \frac{1}{2}\vec{b} = \frac{1}{2}\vec{b} + \frac{1}{2}\vec{a} = \frac{\vec{a} + \vec{b}}{2}$$

Coordinate Geometry

Introduce an origin at $C = (0,0)$. Let the position vectors of A and B be:

$$A(x_1, y_1)\ and\ B(x_2, y_2)$$

By the midpoint formula:

$$X = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) = \frac{1}{2}(x_1 + x_2, y_1 + y_2) = \frac{1}{2}[(x_1, y_1) + (x_2, y_2)] = \frac{1}{2}[\overrightarrow{CA} + \overrightarrow{CB}] = \frac{1}{2}[\vec{b} + \vec{a}]$$

Example 1.131: Midpoint

Mark the correct option

If C is the midpoint of AB and P is any point outside AB, then:

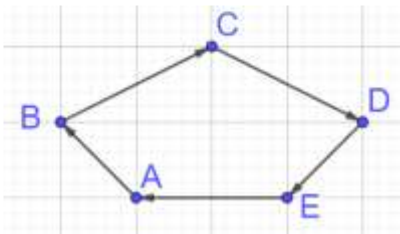
A. $\overrightarrow{PA} + \overrightarrow{PB} + \overrightarrow{PC} = 0$
B. $\overrightarrow{PA} + \overrightarrow{PB} + 2\overrightarrow{PC} = 0$
C. $\overrightarrow{PA} + \overrightarrow{PB} = \overrightarrow{PC}$
D. $\overrightarrow{PA} + \overrightarrow{PB} = 2\overrightarrow{PC}$ (JEE Main 2005)

Use the midpoint formula in ΔAPB:

$$\frac{\overrightarrow{PA} + \overrightarrow{PB}}{2} = \overrightarrow{PC} \Rightarrow \overrightarrow{PA} + \overrightarrow{PB} = 2\overrightarrow{PC} \Rightarrow \textit{Option D}$$

Example 1.132

X is the midpoint of CD in pentagon $ABCDE$. Using the midpoint formula, write the vectors below in terms of vectors that have start and end points on consecutive vertices of the pentagon:



A. $\overrightarrow{AX}$ (without using $\overrightarrow{CD}$)
B. $\overrightarrow{EX}$ (without using $\overrightarrow{CD}$)
C. $\overrightarrow{BX}$(without using $\overrightarrow{CD}$)
D. Simplify $\frac{1}{2}(\overrightarrow{CA} + \overrightarrow{AD})$

Part A

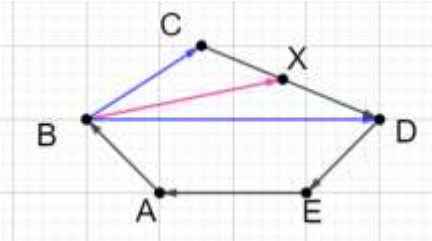
$$\overrightarrow{AX} = \frac{1}{2}(\overrightarrow{AC} + \overrightarrow{AD}) = \frac{1}{2}(\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{AE} + \overrightarrow{ED})$$

Part B

$$\overrightarrow{EX}=\frac{1}{2}\left(\overrightarrow{EC}+\overrightarrow{ED}\right)=\frac{1}{2}\left(\overrightarrow{EA}+\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{ED}\right)$$

Part C

$$\overrightarrow{BX}=\frac{1}{2}\left(\overrightarrow{BC}+\overrightarrow{BD}\right)=\frac{1}{2}\left(\overrightarrow{BC}+\overrightarrow{BA}+\overrightarrow{AE}+\overrightarrow{ED}\right)$$



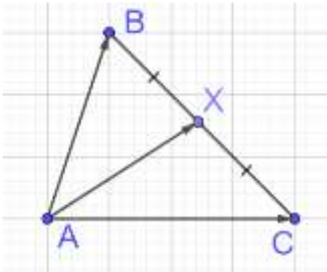
Part D

$$\frac{1}{2}\left(\overrightarrow{CA}+\overrightarrow{AD}\right)=\frac{1}{2}\left(\overrightarrow{CD}\right)=\overrightarrow{CX}$$

Example 1.133

If the vectors $\overrightarrow{AB}=3\hat{\imath}+4\hat{k}$ and $\overrightarrow{AC}=5\hat{\imath}-2\hat{\jmath}+4\hat{k}$ are the sides of a ΔABC, then the length of the median through A is: (JEE Main 2003, 2013)

$$\boldsymbol{AM}=\frac{\overrightarrow{AB}+\overrightarrow{AC}}{2}=\frac{(3,0,4)+(5,-2,4)}{2}=,(4,-1,4)$$
$$|\boldsymbol{AM}|=\sqrt{4^2+(-1)^2+4^2}=\sqrt{33}$$

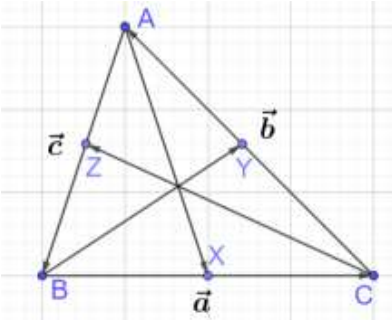


Example 1.134

Show, using the midpoint formula, that the sum of the medians of a triangle, with the vertices as the start point for each median, is zero.

Draw ΔABC, and let

$$\overrightarrow{AB}=\vec{c},\overrightarrow{BC}=\vec{a},\overrightarrow{CA}=\vec{b}$$



Then, by the midpoint formula, the medians are:

$$\underbrace{\overrightarrow{AX}=\frac{1}{2}\left(\vec{c}-\vec{b}\right)}_{Equation\ I},\quad\underbrace{\overrightarrow{BY}=\frac{1}{2}\left(\vec{a}-\vec{c}\right)}_{Equation\ II},\quad\underbrace{\overrightarrow{CZ}=\frac{1}{2}\left(\vec{b}-\vec{a}\right)}_{Equation\ III}$$

Add Equations I, II and III[2]:

$$\overrightarrow{AX}+\overrightarrow{BY}+\overrightarrow{CZ}=\frac{1}{2}\left(\vec{c}-\vec{b}+\vec{a}-\vec{c}+\vec{b}-\vec{a}\right)=\frac{1}{2}(0)=0$$

B. Linear Combination: Algebraic

1.135: Linear Combination of Vectors

Given vectors $\vec{a}$ and $\vec{b}$, we can find a linear combination of the vectors as:

$$\lambda\vec{a}+\mu\vec{b}=\vec{c}$$

- It is called a linear combination since there are no powers (squares, cubes, etc) involved.

[2] This works because the equations are cyclical. Adding them gives a zero.

Example 1.136

Find the linear combination of the first two vectors that results in the third vector:

$$\vec{\boldsymbol{a}} = \binom{1}{3}, \vec{\boldsymbol{b}} = \binom{2}{5}, \vec{\boldsymbol{c}} = \binom{8}{21}$$

Write as a vector equation:

$$\lambda\vec{\boldsymbol{a}} + \mu\vec{\boldsymbol{b}} = \vec{\boldsymbol{c}}$$

Substitute the known values:

$$\lambda\binom{1}{3} + \mu\binom{2}{5} = \binom{8}{21}$$

Simplify:

$$\binom{\lambda + 2\mu}{3\lambda + 5\mu} = \binom{8}{21}$$

Solve the above system of equations:

$$\mu = 3, \lambda = 2$$
$$2\vec{\boldsymbol{a}} + 3\vec{\boldsymbol{b}} = \vec{\boldsymbol{c}}$$

Concept Check 1.137

Mark all correct options

Patrick attempted to solve an equation of the form $\lambda\vec{\boldsymbol{a}} + \mu\vec{\boldsymbol{b}} = \vec{\boldsymbol{c}}$ to find the values of the constants λ and μ. He found that there were no such values. Then, which of the following is a possible explanation for this:

A. $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$ have the same magnitude
B. $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$ have different magnitudes
C. $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$ are parallel
D. $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$ are anti-parallel
E. $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$ are unit vectors
F. $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$ are not unit vectors
G. $\vec{\boldsymbol{c}}$ is a zero vector
H. One of $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$ is a zero vector
I. Patrick made a mistake.

Consider Option G:

$$\lambda = \mu = 0 \Rightarrow Valid\ Solution \Rightarrow G\ is\ not\ correct$$

$$Correct\ Options\ C, D, H, I$$

1.138: Uniqueness of Linear Combination of Vectors

Linear combination of vectors is unique.

Example 1.139

What can we conclude about the coefficients if $m\vec{\boldsymbol{a}} + n\vec{\boldsymbol{b}} = p\vec{\boldsymbol{a}} + q\vec{\boldsymbol{b}}$?

Since the vectors on both sides are equal, the coefficients must also be the same.

$$m = p$$
$$n = q$$

Example 1.140

Let $\vec{\boldsymbol{\alpha}} = (\lambda - 2)\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}}$ and $\vec{\boldsymbol{\beta}} = (4\lambda - 2)\vec{\boldsymbol{a}} + 3\vec{\boldsymbol{b}}$ be two given vectors where $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$ are non-collinear. The value of λ for which vectors $\vec{\boldsymbol{\alpha}}$ and $\vec{\boldsymbol{\beta}}$ are collinear is: (JEE Main 2019, 10 Jan, Shift-II)

If the vectors $\vec{\boldsymbol{\alpha}}$ and $\vec{\boldsymbol{\beta}}$ are collinear, then one vector must be a non-zero scalar multiple of the other:

$$\vec{\boldsymbol{\alpha}} = \mu\vec{\boldsymbol{\beta}}$$

Substitute the values from the question:

$$(\lambda - 2)\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}} = \mu\left[(4\lambda - 2)\vec{\boldsymbol{a}} + 3\vec{b}\right]$$

Use the distributive property on the RHS:

$$(\lambda - 2)\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}} = \mu(4\lambda - 2)\vec{\boldsymbol{a}} + 3\mu\vec{\boldsymbol{b}}$$

Since the vectors on both sides are equal, the coefficients must also be the same.

$$1 = 3\mu \Rightarrow \mu = \frac{1}{3}$$
$$\lambda - 2 = \mu(4\lambda - 2)$$
$$\lambda - 2 = \frac{1}{3}(4\lambda - 2)$$
$$3\lambda - 6 = 4\lambda - 2$$
$$\lambda = -4$$

Pending

C. Section Formula

Example 1.141

Find, using first principles.

1.142: Section Formula

Example 1.143

1.6 Geometry with Scalar Multiplication

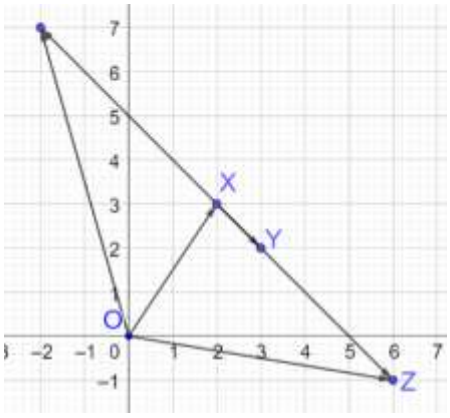
A. Geometry with Scalar Multiplication

Example 1.144

Relative to an origin O, the position vectors of points X and Y are given by $\binom{2}{3}$ and $\binom{3}{2}$. X, Y and Z lie on a straight line such that the length of $\overrightarrow{\boldsymbol{XZ}}$ is four times the length of $\overrightarrow{\boldsymbol{XY}}$. The sum of the possible values of the length of $\overrightarrow{\boldsymbol{OZ}}$ can be written as $\sqrt{a} + \sqrt{b}$. Find the sum of the digits of $a + b$.

From the diagram, we see that:

$$\underbrace{\overrightarrow{OZ} = \overrightarrow{OX} + \overrightarrow{XZ}}_{Equation\ I}$$



There are two possible locations for Z, based on which direction we go in, giving us two cases:

$$\underbrace{\overrightarrow{XZ} = \pm 4\overrightarrow{XY} = \pm 4\left[\begin{pmatrix}3\\2\end{pmatrix} - \begin{pmatrix}2\\3\end{pmatrix}\right] = \pm \mathbf{4}\begin{pmatrix}1\\-1\end{pmatrix} = \pm\begin{pmatrix}4\\-4\end{pmatrix}}_{Equation\ II}$$

Substitute Equation II into Equation I:

$$\overrightarrow{OZ} = \overrightarrow{OX} \pm 4\overrightarrow{XY} = \begin{pmatrix}2\\3\end{pmatrix} \pm \begin{pmatrix}4\\-4\end{pmatrix}$$

Find the value of each case separately:

$$\overrightarrow{OZ} = \begin{pmatrix}6\\-1\end{pmatrix} \Rightarrow |\boldsymbol{OZ}| = \sqrt{6^2 + (-1)^2} = \sqrt{36+1} = \sqrt{37}$$

$$OR\ \overrightarrow{OZ} = \begin{pmatrix}-2\\7\end{pmatrix} \Rightarrow |\boldsymbol{OZ}| = \sqrt{(-2)^2 + 7^2} = \sqrt{4+49} = \sqrt{53}$$

The possible values for the length of $\overrightarrow{OZ}$ are

$$\{\sqrt{37}, \sqrt{53}\}$$

$$a = 37, b = 53 \Rightarrow a + b = 37 + 53 = 90 \Rightarrow Sum\ of\ Digits = 9$$

1.145: Distributive Property

$$\lambda(\vec{a} + \vec{b}) = \lambda\vec{a} + \lambda\vec{b}$$

$$\lambda(\vec{a} - \vec{b}) = \lambda\vec{a} - \lambda\vec{b}$$

The distributive property works for scalar multiplication.

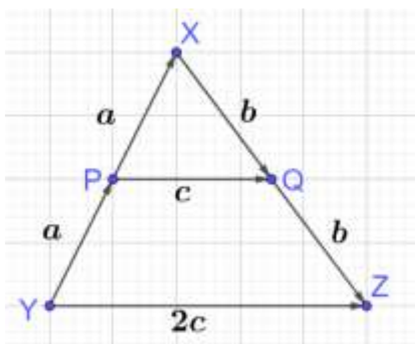
Example 1.146: Midpoint Theorem

Prove that the line connecting the midpoints of two sides of a triangle is parallel to the third side, and half of the third side.

Draw ΔXYZ.

Let P be the midpoint of XY, and Q be the midpoint of XZ. Then:

$$\overrightarrow{YP} = \overrightarrow{PX} = \vec{a}\ (say), \qquad \overrightarrow{XQ} = \overrightarrow{QZ} = \vec{b}\ (say), \qquad \overrightarrow{PQ} = \vec{c}$$



Because

$$YP = PX\ since\ P\ is\ the\ midpoint \Rightarrow\ Same\ Magnitude$$

$$Y, P, X\ are\ collinear \Rightarrow Same\ Direction$$

Note that:

$$\overrightarrow{PQ} = \vec{c} = \vec{a} + \vec{b}$$

$$\overrightarrow{YZ} = \overrightarrow{YX} + \overrightarrow{XZ} = 2\vec{a} + 2\vec{b} = 2(\vec{a} + \vec{b}) = 2\vec{c}$$

Hence,

$$YZ\ is\ double\ of\ PQ$$
$$YZ\ is\ parallel\ to\ PQ$$

Example 1.147

Show, without using the midpoint theorem, that the quadrilateral formed by the midpoints of a convex quadrilateral, taken in order, is a parallelogram.

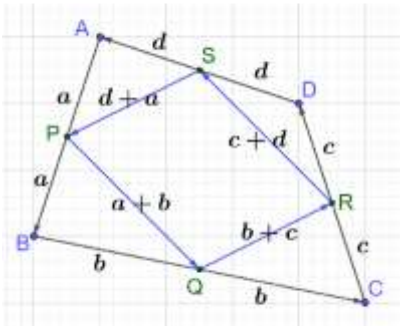
Hint: A quadrilateral is a parallelogram if a pair of its opposite sides is parallel and congruent.

Draw convex quadrilateral $ABCD$.

- Note that it is a general convex quadrilateral, with no special properties.
- Let the midpoints of the sides be P, Q, R, S, as shown.

Let:

$$\overrightarrow{AP} = \overrightarrow{PB} = \vec{a}$$
$$\overrightarrow{BQ} = \overrightarrow{QC} = \vec{b}$$
$$\overrightarrow{CR} = \overrightarrow{RD} = \vec{c}$$
$$\overrightarrow{DS} = \overrightarrow{DA} = \vec{d}$$



Hence,

$$\overrightarrow{PQ} = \overrightarrow{PB} + \overrightarrow{BQ} = \vec{a} + \vec{b}$$
$$\overrightarrow{QR} = \overrightarrow{QC} + \overrightarrow{CR} = \vec{b} + \vec{c}$$
$$\overrightarrow{RS} = \overrightarrow{RD} + \overrightarrow{DS} = \vec{c} + \vec{d}$$
$$\overrightarrow{SP} = \overrightarrow{SA} + \overrightarrow{AP} = \vec{d} + \vec{a}$$

Since the sum of the vectors in a vector polygon is zero:

$$\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{DA} = \vec{0}$$
$$2\vec{a} + 2\vec{b} + 2\vec{c} + 2\vec{d} = \vec{0}$$

Divide both sides by 2:

$$\vec{a} + \vec{b} + \vec{c} + \vec{d} = \vec{0}$$
$$\vec{a} + \vec{b} = -(\vec{c} + \vec{d})$$
$$\overrightarrow{PQ} = -(\overrightarrow{RS})$$
$$\overrightarrow{PQ} = \overrightarrow{SR}$$

Hence,

$$\overrightarrow{SR}\ is\ equal\ in\ magnitude\ to\ \overrightarrow{PQ} \Rightarrow \underbrace{SR \cong PQ}_{Conclusion\ I}$$
$$\underbrace{SR \parallel PQ}_{Conclusion\ II}$$

From Conclusions I and II:

$$PQRS\ is\ a\ parallelogram$$

B. Linear Combination: Geometric

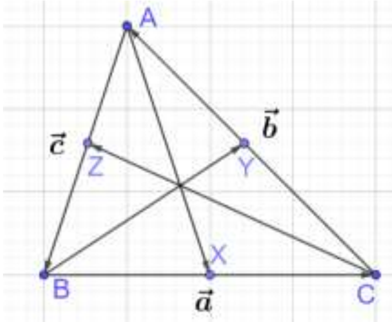
Example 1.148

Show that the sum of the medians of a triangle, taken with the vertices as the start point, is zero.

Note: This was shown using the midpoint formula earlier. You can now show this using a linear combination of vectors.

Draw ΔABC, and let

$$\overrightarrow{\boldsymbol{AB}} = \vec{c}, \overrightarrow{\boldsymbol{BC}} = \vec{a}, \overrightarrow{\boldsymbol{CA}} = \vec{b}$$



Then:

$$\underbrace{\overrightarrow{\boldsymbol{AX}} = \vec{c} + \frac{1}{2}\vec{a}}_{Equation\ I}, \quad \underbrace{\overrightarrow{\boldsymbol{BY}} = \vec{a} + \frac{1}{2}\vec{b}}_{Equation\ II}, \quad \underbrace{\overrightarrow{\boldsymbol{CZ}} = \vec{b} + \frac{1}{2}\vec{c}}_{Equation\ III}$$

Add Equations I, II and III:

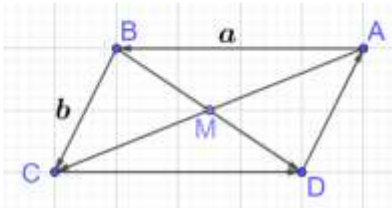
$$\overrightarrow{\boldsymbol{AX}} + \overrightarrow{\boldsymbol{BY}} + \overrightarrow{\boldsymbol{CZ}} = \frac{3}{2}\left(\vec{a} + \vec{b} + \vec{c}\right) = \frac{3}{2}(0) = 0$$

Where $\vec{a} + \vec{b} + \vec{c}$ is the sum of a vector polygon and is hence zero.

Example 1.149

Show, using vector methods, that the diagonals of a quadrilateral bisect each other $if\ and\ only\ if$ the quadrilateral is a parallelogram.

Since this is an $if\ and\ only\ if$ statement, we need to prove it both ways.



Part I: If the diagonals of a quadrilateral bisect each other, then the quadrilateral is a parallelogram

Draw quadrilateral $ABCD$. If the diagonals bisect each other at point M, then

$$\underbrace{\overrightarrow{\boldsymbol{BM}} = \overrightarrow{\boldsymbol{MD}}}_{Equation\ 1}, \quad \underbrace{\overrightarrow{\boldsymbol{MC}} = \overrightarrow{\boldsymbol{AM}}}_{Equation\ 2}$$

Add Equations 1 and 2:

$$\overrightarrow{\boldsymbol{BM}} + \overrightarrow{\boldsymbol{MC}} = \overrightarrow{\boldsymbol{AM}} + \overrightarrow{\boldsymbol{MD}}$$

By vector addition:

$$\overrightarrow{\boldsymbol{BC}} = \overrightarrow{\boldsymbol{AD}}$$

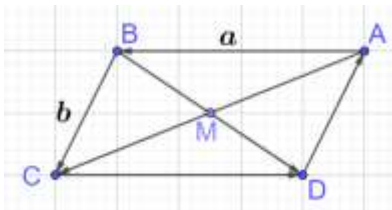
This means that $\overrightarrow{\boldsymbol{BC}}$ and $\overrightarrow{\boldsymbol{AD}}$ are equal in both magnitude and direction.
If a pair of opposite sides of a quadrilateral is parallel and congruent, then the quadrilateral is a parallelogram.
Hence,

$$ABCD\ is\ a\ parallelogram$$

Part II: If a quadrilateral is a parallelogram, then its diagonals bisect each other

Draw parallelogram $ABCD$ and let:

$$\overrightarrow{\boldsymbol{AB}} = \overrightarrow{\boldsymbol{DC}} = \vec{a}, \qquad \overrightarrow{\boldsymbol{BC}} = \overrightarrow{\boldsymbol{AD}} = \vec{b}$$



Using the triangle law of vector addition:

$$\overrightarrow{\boldsymbol{BD}} = \vec{b} - \vec{a} \Rightarrow \underbrace{\overrightarrow{\boldsymbol{BM}} = \mu\left(\vec{b} - \vec{a}\right)}_{Equation\ I}$$

$$\overrightarrow{\boldsymbol{AC}} = \vec{a} + \vec{b} \Rightarrow \underbrace{\overrightarrow{\boldsymbol{AM}} = \lambda\left(\overrightarrow{\vec{a} + \boldsymbol{b}}\right)}_{Equation\ II}$$

The sum of vectors in a vector polygon is zero:

$$\overrightarrow{\boldsymbol{AB}} + \overrightarrow{\boldsymbol{BM}} + \overrightarrow{\boldsymbol{MA}} = 0$$
$$\overrightarrow{\boldsymbol{AB}} + \overrightarrow{\boldsymbol{BM}} - \overrightarrow{\boldsymbol{AM}} = 0$$

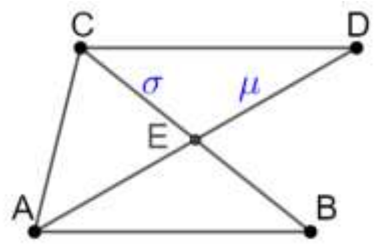
Substitute the values from Equations I and II:
$$\vec{\boldsymbol{a}} + \mu(\vec{\boldsymbol{b}} - \vec{\boldsymbol{a}}) - \lambda(\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}}) = 0$$

Rearrange:
$$\vec{\boldsymbol{a}}(1 - \mu - \lambda) + \vec{\boldsymbol{b}}(\mu - \lambda) = 0\vec{\boldsymbol{a}} + 0\vec{\boldsymbol{b}}$$

Use the method of undetermined coefficients. Equate coefficients on both sides:
$$1 - \mu - \lambda = 0 \Rightarrow \underbrace{\mu + \lambda = 1}_{\text{Equation III}}$$
$$\underbrace{\mu - \lambda = 0}_{\text{Equation IV}}$$
Add Equations III and IV:
$$2\mu = 1 \Rightarrow \mu = \lambda = \frac{1}{2}$$
Hence,
$$\textit{Diagonals bisect each other}$$

Example 1.150
In quadrilateral $CDBA$, the diagonals intersect at E, and CD is parallel to AB. Given that
$$\overrightarrow{\boldsymbol{CE}} = \sigma\overrightarrow{\boldsymbol{CB}}, \qquad \overrightarrow{\boldsymbol{AD}} = \mu\overrightarrow{\boldsymbol{AE}}, \qquad \overrightarrow{\boldsymbol{CD}} = \lambda\overrightarrow{\boldsymbol{AB}}$$
A. Determine the value of λ in term of σ.
B. Calculate the value of λ when $\sigma = \frac{4}{3}$. Explain what is right or wrong with your answer.

Part A
Express AD in two different ways
$$\overrightarrow{\boldsymbol{AD}} = \mu\overrightarrow{\boldsymbol{AE}} = \mu(\overrightarrow{\boldsymbol{AC}} + \overrightarrow{\boldsymbol{CE}})$$



Substitute $\overrightarrow{\boldsymbol{CE}} = \sigma\overrightarrow{\boldsymbol{CB}} = \sigma(\overrightarrow{\boldsymbol{CA}} + \overrightarrow{\boldsymbol{AB}}) = \sigma(\overrightarrow{\boldsymbol{AB}} - \overrightarrow{\boldsymbol{AC}})$
$$\overrightarrow{\boldsymbol{AD}} = \mu[\overrightarrow{\boldsymbol{AC}} + \sigma(\overrightarrow{\boldsymbol{AB}} - \overrightarrow{\boldsymbol{AC}})]$$

Simplify:
$$\underbrace{\overrightarrow{\boldsymbol{AD}} = \mu(1 - \sigma)\overrightarrow{\boldsymbol{AC}} + \mu\sigma\overrightarrow{\boldsymbol{AB}}}_{\text{Equation I}}$$

Since $CD \parallel AB$
$$\underbrace{\overrightarrow{\boldsymbol{AD}} = \overrightarrow{\boldsymbol{AC}} + \overrightarrow{\boldsymbol{CD}} = \overrightarrow{\boldsymbol{AC}} + \lambda\overrightarrow{\boldsymbol{AB}}}_{\text{Equation II}}$$

Use the relationship
From Equation I and II:
$$\mu(1 - \sigma)\overrightarrow{\boldsymbol{AC}} + \mu\sigma\overrightarrow{\boldsymbol{AB}} = \overrightarrow{\boldsymbol{AC}} + \lambda\overrightarrow{\boldsymbol{AB}}$$

Using the method of undetermined coefficients:

$$\mu\sigma = \lambda \Rightarrow \underbrace{\mu = \frac{\lambda}{\sigma}}_{\textit{Equation III}}, \quad \underbrace{\mu(1-\sigma) = 1}_{\textit{Equation IV}}$$

Substitute Equation III into Equation IV:

$$\frac{\lambda}{\sigma}(1-\sigma) = 1$$
$$\lambda = \frac{\sigma}{1-\sigma}$$

Part B

$$\lambda = \frac{\sigma}{1-\sigma} = \frac{\frac{4}{3}}{1-\frac{4}{3}} = \frac{\frac{4}{3}}{-\frac{1}{3}} = -4$$

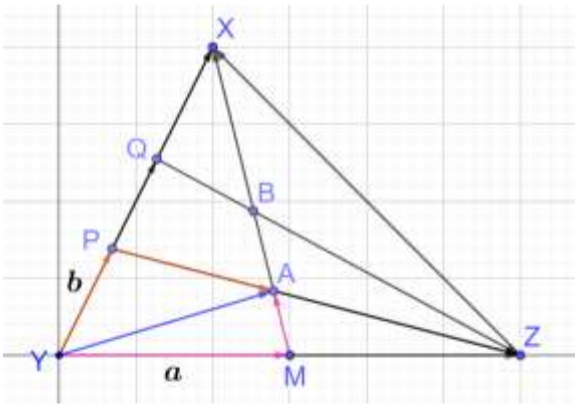
CE is a part of CB. Hence, to be meaningful, we must have:

$$0 < \sigma < 1$$

And since the above condition is violated, we get a meaningless result for λ.

Example 1.151

In ΔXYZ, lines ZP and ZQ trisect XY. The points Y, P, Q, X are collinear, in that order. The point of intersection of ZP with the median XM is point A. Find the ratio in which point A divides XM, and also the ratio in which point A divides ZP.



Draw a diagram. Let

$$\overrightarrow{\boldsymbol{YM}} = \vec{\boldsymbol{a}}, \qquad \overrightarrow{\boldsymbol{YP}} = \vec{\boldsymbol{b}}$$

Note that $\overrightarrow{\boldsymbol{MA}}$ must be some fraction of $\overrightarrow{\boldsymbol{MX}}$:

$$\overrightarrow{\boldsymbol{YA}} = \overrightarrow{\boldsymbol{YM}} + \overrightarrow{\boldsymbol{MA}} = \boldsymbol{a} + \lambda\overrightarrow{\boldsymbol{MX}}$$

Substitute:

$$= \boldsymbol{a} + \lambda(3\boldsymbol{b} - \boldsymbol{a}) = \boldsymbol{a} + 3\lambda\boldsymbol{b} - \lambda\boldsymbol{a} = \underbrace{\boldsymbol{a}(1-\lambda) + 3\lambda\boldsymbol{b}}_{\textit{Expression I}}$$

Similarly, note that $\overrightarrow{\boldsymbol{PA}}$ must be some fraction of $\overrightarrow{\boldsymbol{PZ}}$:

$$\overrightarrow{\boldsymbol{YA}} = \overrightarrow{\boldsymbol{YP}} + \overrightarrow{\boldsymbol{PA}} = \vec{\boldsymbol{b}} + \mu\overrightarrow{\boldsymbol{PZ}}$$

Substitute:

$$\vec{\boldsymbol{b}} + \mu\left(2\vec{\boldsymbol{a}} - \vec{\boldsymbol{b}}\right) = \vec{\boldsymbol{b}} + 2\mu\vec{\boldsymbol{a}} - \mu\vec{\boldsymbol{b}} = \underbrace{2\mu\vec{\boldsymbol{a}} + (1-\mu)\boldsymbol{b}}_{\textit{Expression II}}$$

Use the method of undetermined coefficients.

$$\underbrace{\boldsymbol{a}(1-\lambda) + 3\lambda\boldsymbol{b}}_{\textit{Expression I}} = \underbrace{2\mu\boldsymbol{a} + \boldsymbol{b}(1-\mu)}_{\textit{Expression II}}$$

Equating coefficients for a:

$$1 - \lambda = 2\mu \Rightarrow \lambda = 1 - 2\mu \Rightarrow 3\lambda = 3 - 6\mu$$

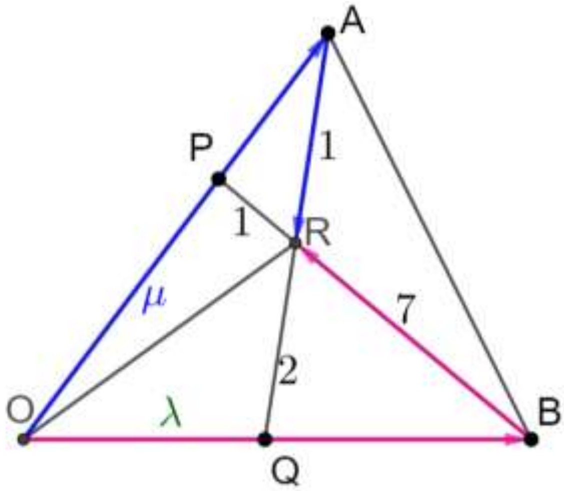
Equating coefficients for b:

$$3\lambda = 1 - \mu \Rightarrow 3 - 6\mu = 1 - \mu \Rightarrow \mu = \frac{2}{5}$$

$$\lambda = 1 - 2\mu = 1 - \left(\frac{2}{5}\right) = \frac{1}{5}$$

Example 1.152

In ΔOAB, P lies on OA and Q lies on OB. The point R divides AQ in the ratio $1:2$, with the smaller part closer to A. Also, the point R divides BP in the ratio $7:1$ with the larger part closer to B. Determine $\overrightarrow{OR}$ in terms of $\overrightarrow{OA}$ and $\overrightarrow{OB}$.



$$\overrightarrow{\boldsymbol{OR}} = \overrightarrow{\boldsymbol{OA}} + \overrightarrow{\boldsymbol{AR}} = \overrightarrow{\boldsymbol{OB}} + \overrightarrow{\boldsymbol{BR}}$$

$$\overrightarrow{\boldsymbol{AR}} = \frac{1}{3}\overrightarrow{\boldsymbol{AQ}} = \frac{1}{3}\left(\overrightarrow{\boldsymbol{AO}} + \overrightarrow{\boldsymbol{OQ}}\right) = \frac{1}{3}\left(\lambda\overrightarrow{\boldsymbol{OB}} - \overrightarrow{\boldsymbol{OA}}\right)$$
$$\overrightarrow{\boldsymbol{BR}} = \frac{7}{8}\overrightarrow{\boldsymbol{BP}} = \frac{7}{8}\left(\overrightarrow{\boldsymbol{BO}} + \overrightarrow{\boldsymbol{OP}}\right) = \frac{7}{8}\left(\mu\overrightarrow{\boldsymbol{OA}} - \overrightarrow{\boldsymbol{OB}}\right)$$

Substitute the values that we calculated:

$$\overrightarrow{\boldsymbol{OA}} + \frac{1}{3}\left(\lambda\overrightarrow{\boldsymbol{OB}} - \overrightarrow{\boldsymbol{OA}}\right) = \overrightarrow{\boldsymbol{OB}} + \frac{7}{8}\left(\mu\overrightarrow{\boldsymbol{OA}} - \overrightarrow{\boldsymbol{OB}}\right)$$
$$\frac{2}{3}\overrightarrow{\boldsymbol{OA}} + \frac{1}{3}\lambda\overrightarrow{\boldsymbol{OB}} = \frac{1}{8}\overrightarrow{\boldsymbol{OB}} + \frac{7}{8}\mu\overrightarrow{\boldsymbol{OA}}$$

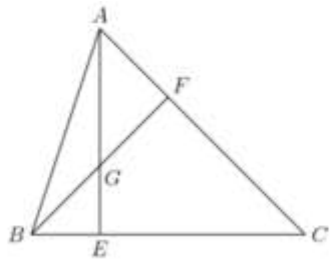
Equating coefficients:

$$\frac{2}{3} = \frac{7}{8}\mu \Rightarrow \mu = \frac{16}{21}$$
$$\frac{1}{3}\lambda = \frac{1}{8} \Rightarrow \lambda = \frac{3}{8}$$

$$\overrightarrow{\boldsymbol{OR}} = \frac{2}{3}\overrightarrow{\boldsymbol{OA}} + \frac{1}{3}\lambda\overrightarrow{\boldsymbol{OB}} = \frac{2}{3}\overrightarrow{\boldsymbol{OA}} + \frac{1}{3}\cdot\frac{3}{8}\overrightarrow{\boldsymbol{OB}} = \frac{2}{3}\overrightarrow{\boldsymbol{OA}} + \frac{1}{8}\overrightarrow{\boldsymbol{OB}}$$

Example 1.153

In ΔABC, point F divides side AC in the ratio 1:2. Let E be the point of intersection of side BC and AG where G is the midpoint of BF. The point E divides side BC in the ratio ____ **(AHSME 1971/26)**



Let

$$\overrightarrow{\boldsymbol{FA}} = \vec{\boldsymbol{a}} \Rightarrow \overrightarrow{\boldsymbol{CF}} = \overrightarrow{\boldsymbol{2a}}, \qquad \overrightarrow{\boldsymbol{BG}} = \overrightarrow{\boldsymbol{GF}} = \vec{\boldsymbol{b}}$$

Also:

$$\overrightarrow{\boldsymbol{BG}} = \overrightarrow{\boldsymbol{BE}} + \overrightarrow{\boldsymbol{EG}} = \vec{\boldsymbol{b}}$$

Substitute $\overrightarrow{\boldsymbol{EG}} = \mu\overrightarrow{\boldsymbol{GA}} = \mu\left(\overrightarrow{\boldsymbol{GF}} + \overrightarrow{\boldsymbol{FA}}\right) = \mu\left(\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}}\right)$, $\overrightarrow{\boldsymbol{BE}} = \lambda\overrightarrow{\boldsymbol{BC}} = \lambda\left(\overrightarrow{\boldsymbol{BF}} + \overrightarrow{\boldsymbol{FC}}\right) = \lambda\left(2\vec{\boldsymbol{b}} - 2\vec{\boldsymbol{a}}\right)$

$$\lambda\left(2\vec{\boldsymbol{b}} - 2\vec{\boldsymbol{a}}\right) + \mu\left(\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}}\right) = \vec{\boldsymbol{b}}$$

Rearrange:

$$a(\mu - 2\lambda) + b(\mu + 2\lambda) = 0\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}}$$

Equating coefficients:

$$\underbrace{\mu-2\lambda=0}_{Equation\ I},\qquad \underbrace{\mu+2\lambda=1}_{Equation\ II}$$

Add Equations I and II:

$$2\mu=1\Rightarrow\mu=\frac{1}{2}$$

Substitute $\mu=\frac{1}{2}$ in Equation II:

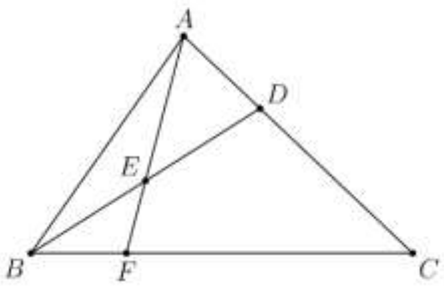
$$\frac{1}{2}+2\lambda=1\Rightarrow\lambda=\frac{1}{4}$$

E divides BC in the ratio

$$1:3$$

Also, G divides EA in the ratio

$$1:2$$



(Continuation) Example 1.154

In triangle ABC, point D divides side $\overline{AC}$ so that $AD:DC=1:2$. Let E be the midpoint of $\overline{BD}$ and let F be the point of intersection of line BC and line AE. Given that the area of $\triangle ABC$ is 360, what is the area of $\triangle EBF$? (AMC 8 2019/24)

Note that we have the same triangle as in the previous example (except for names of vertices). Use the ratios from before:

$$BF=\frac{1}{4}BC,\qquad EF=\frac{1}{3}AF$$

Since $\angle AFC=\angle EFC$:

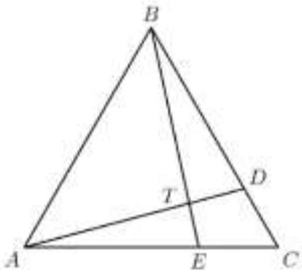
$$Height(\Delta BEF)=\frac{1}{3}Height(\Delta ABC\ )$$

Using theorems on area of triangles:

$$A(\Delta EBF)=A(\Delta ABC)\times\underbrace{\frac{1}{4}}_{\substack{Ratio\ of\\ Bases}}\times\underbrace{\frac{1}{3}}_{\substack{Ratio\ of\\ Heights}}$$

Simplify:

$$A(\Delta EBF)=360\times\frac{1}{4}\times\frac{1}{3}=30$$



Example 1.155

In ΔABC, points D and E lie on BC and AC, respectively. If AD and BE intersect at T so that $\frac{AT}{DT}=3$ and $\frac{BT}{ET}=4$, what is $\frac{CD}{BD}$? (AMC 10B 2004/20)

Let:

$$\overrightarrow{ET}=\vec{\boldsymbol{a}}\Rightarrow\overrightarrow{TB}\ =4\vec{\boldsymbol{a}}$$
$$\overrightarrow{TD}=\vec{\boldsymbol{b}}\Rightarrow\overrightarrow{AT}=3\vec{\boldsymbol{b}}$$

Let $\frac{\overrightarrow{CD}}{\overrightarrow{DB}}=\mu$:

$$\overrightarrow{CD}=\mu\overrightarrow{DB}=\mu\left(\overrightarrow{TB}-\overrightarrow{TD}\right)=\mu\left(4\vec{\boldsymbol{a}}-\vec{\boldsymbol{b}}\right)$$

Let $\frac{\overrightarrow{CE}}{\overrightarrow{AE}}=\lambda$:

$$\overrightarrow{\boldsymbol{EC}} = \lambda\overrightarrow{\boldsymbol{AE}} = \lambda\left(\overrightarrow{\boldsymbol{AT}} - \overrightarrow{\boldsymbol{ET}}\right) = \lambda\left(3\vec{\boldsymbol{b}} - \vec{\boldsymbol{a}}\right)$$

Calculate $\overrightarrow{\boldsymbol{ED}}$ in two different ways in Quadrilateral $ETDC$:

$$\overrightarrow{\boldsymbol{ED}} = \vec{\boldsymbol{a}} + \vec{\boldsymbol{b}}$$

$$\overrightarrow{\boldsymbol{ED}} = \overrightarrow{\boldsymbol{EC}} + \overrightarrow{\boldsymbol{CD}} = \lambda\left(3\vec{\boldsymbol{b}} - \vec{\boldsymbol{a}}\right) + \mu\left(4\vec{\boldsymbol{a}} - \vec{\boldsymbol{b}}\right) = \vec{\boldsymbol{a}}(4\mu - \lambda) + \vec{\boldsymbol{b}}(-\mu + 3\lambda)$$

Equate coefficients:

$$4\mu - \lambda = 1 \Rightarrow \underbrace{12\mu - 3\lambda = 3}_{\textbf{Equation I}}, \qquad \underbrace{-\mu + 3\lambda = 1}_{\textbf{Equation II}}$$

Add Equations I and II:

$$11\mu = 4 \Rightarrow \mu = \frac{4}{11}$$

C. Centroid

Example 1.156
Medians of a triangle are concurrent.
Centroid of a triangle is 2/3 of the way along the median (from the vertices)

Example 1.157
Centroid of a triangle is the average of the position vectors of the vertices of the triangle.

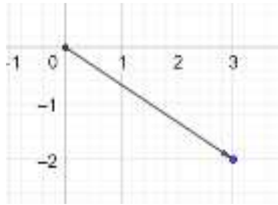
D. Angle Bisector

Example 1.158
Angle Bisector

Example 1.159
Incenter of a triangle using angle bisector

2. Dot Product

2.1 Polar Form; Resolving Components



A. Polar Form

We have seen the algebraic form of a vector where the components are specified:

$$\vec{a} = (x, y) = (3, -2) = 3\hat{\imath} - 2\hat{\jmath}$$

We have also seen the geometric form of a vector where the vector is represented as a directed line segment. There is one more form of a vector, where it is specified using the magnitude and angle.

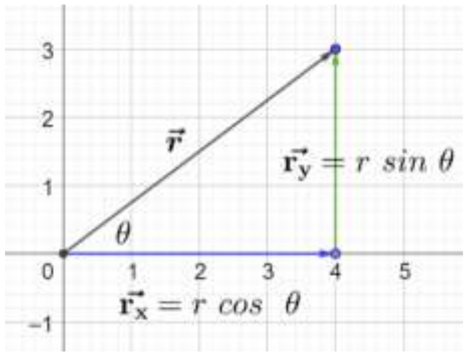
2.1: Polar Form of a Vector

A vector $\vec{r}$ with its tail at the origin can be defined in terms of (r, θ) where:

$$r = |\vec{r}| = Magnitude\ of\ \vec{r}$$

$$\theta = Angle\ made\ by\ \vec{r}\ with\ positive\ direction\ of\ x - axis$$

- If the tail of the vector is at the origin, then the polar form of a vector gives the position of the tip in polar coordinates. Hence, the two are related.
- Angle is considered:
  - ✓ positive in the counterclockwise direction
  - ✓ negative in the clockwise direction



2.2: Converting into component form

$$\vec{r} = \underbrace{(r, \theta)}_{Polar\ Form} = \underbrace{(r\cos\theta, r\sin\theta)}_{Component\ Form}$$

We can determine the magnitude of $\vec{r_y}$:

$$\sin\theta = \frac{r_y}{r} \Rightarrow r_y = r\sin\theta$$

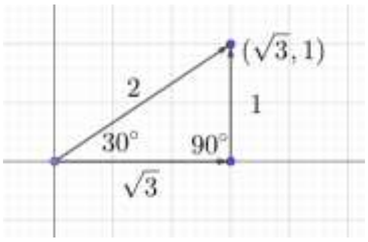
$$\cos\theta = \frac{r_x}{r} \Rightarrow r_x = r\cos\theta$$

Example 2.3

Plot the vectors given in polar form on the coordinate plane. Write them in component form. State the quadrant in which the vectors lie.

A. $\vec{a} = (2, 30°)$
B. $\vec{b} = (5, 45°)$
C. $\vec{c} = (\sqrt{3}, -60°)$
D. $\vec{d} = \left(3, \frac{\pi}{3}\right)$
E. $\vec{a} = \left(4, \frac{\pi}{6}\right)$
F. $\vec{b} = \left(7, -\frac{\pi}{3}\right)$
G. $\vec{c} = \left(2, \frac{9\pi}{4}\right)$

$$\vec{a} = (\sqrt{3}, 1), Quadrant\ I$$
$$\vec{b} = \left(\frac{5\sqrt{2}}{2}, \frac{5\sqrt{2}}{2}\right), Quadrant\ I$$
$$\vec{c} = \left(\frac{\sqrt{3}}{2}, -\frac{3}{2}\right), Quadrant\ IV$$



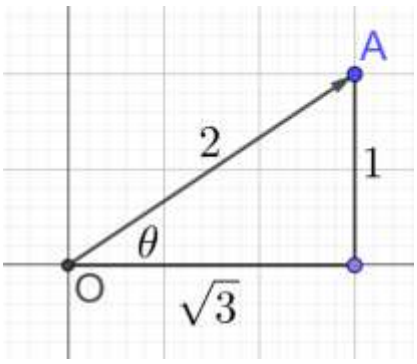
$$\vec{d} = \left(\frac{3}{2}, \frac{3\sqrt{3}}{2}\right), Quadrant\ I$$
$$\vec{a} = \left(4, \frac{\pi}{6}\right) = (4, 30°) = (2\sqrt{3}, 2)$$

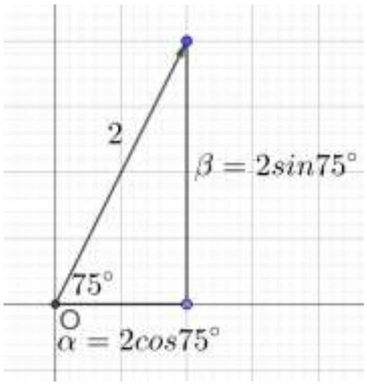
Example 2.4

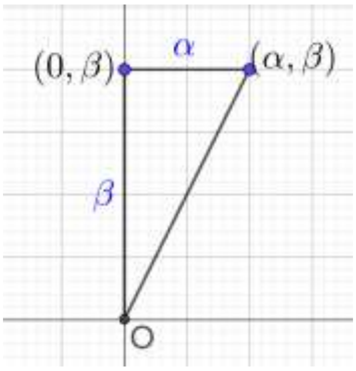
Let a vector $\alpha\boldsymbol{i} + \beta\boldsymbol{j}$ be obtained by rotating the vector $\sqrt{3}\boldsymbol{i} + \boldsymbol{j}$ by an angle 45° about the origin in counter clockwise direction in the first quadrant. Then, the area of the triangle having vertices (α, β), $(0, \beta)$ and $(0,0)$ is equal to: **(JEE Main 2021, 16 March, Shift-I)**

Let the vector be $\overrightarrow{OA}$. (First diagram on the left). Note that:

$$\tan\theta = \frac{1}{\sqrt{3}} \Rightarrow \theta = 30°, \qquad |\overrightarrow{OA}| = \sqrt{(\sqrt{3})^2 + 1^2} = \sqrt{3+1} = \sqrt{4} = 2$$







Rotate the vector by 45° counterclockwise (middle diagram) and note that the magnitude remains 2.
The components are

$$\alpha = 2\cos 75°, \beta = 2\sin 75°$$

Draw the required triangle (rightmost diagram), and hence, the area of that triangle=

$$= \frac{1}{2}hb = \frac{\alpha\beta}{2} = \frac{(2\cos 75°)(2\sin 75°)}{2}$$

Simplify:

$$= 2\cos 75° \sin 75°$$

Use the double angle identity for $\sin\theta$:

$$= \sin 150° = \sin 30° = \frac{1}{2}$$

2.5: Standard Polar Form of a Vector

A vector $\vec{r}$ in polar form (r, θ) is said to be in standard form if

$$\underbrace{0° \le \theta < 360°}_{Degrees} \Leftrightarrow \underbrace{0 \le \theta < 2\pi}_{Radians}$$

Example 2.6

Write the following position vectors in standard polar form. If the angles are given in radians, convert into degrees. Also, state the quadrant in which they lie.

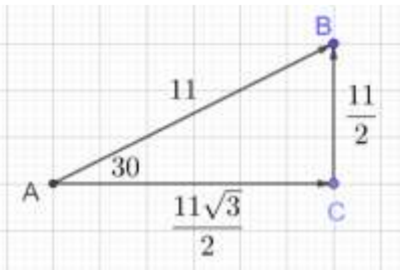
A. $\vec{a} = (2, -30°)$
B. $\vec{b} = (5, -210°)$
C. $\vec{c} = (9{,}410°)$
D. $\vec{d} = \left(1, \frac{\pi}{2}\right)$
E. $\vec{e} = \left(10, -\frac{\pi}{6}\right)$

$$\vec{a} = (2, -30°) = (2{,}330°)$$
$$\vec{b} = (5, -210°) = (5{,}150°)$$
$$\vec{c} = (9{,}410°) = (9{,}50°)$$
$$\vec{d} = \left(1, \frac{\pi}{2}\right) = (1{,}90°)$$
$$\vec{e} = \left(10, -\frac{\pi}{6}\right) = (10, -30°) = (10{,}330°)$$

Example 2.7

A vector has a magnitude of 11, and makes an angle of 30° with the horizontal in the right direction. Determine the exact value of its horizontal and vertical components.

$$\sin 30° = \frac{BC}{AB} \Rightarrow BC = AB \sin 30° = BC \times \frac{1}{2} = \frac{11}{2}$$
$$\cos 30° = \frac{AC}{AB} \Rightarrow AC = AB \times \cos 30° = 11 \times \frac{\sqrt{3}}{2} = \frac{11\sqrt{3}}{2}$$



$$\left(\frac{11\sqrt{3}}{2}, \frac{11}{2}\right)$$

Example 2.8

A. Rory the dog is running towards his favorite ball on the beach, at a velocity of $34 \frac{miles}{hour}$ at an angle of α to the positive direction of x axis. Find his speed in the horizontal and vertical direction.
B. The x component of $\vec{a}$ is -3. $\vec{a}$ makes an angle γ (in the counterclockwise direction) with the positive direction of the x axis. Given that $\vec{a}$ lies in the second quadrant, determine the magnitude of $\vec{a}$, and the vector which is the y component of $\vec{a}$.
C. Find x given that $\vec{a} = x\hat{\imath} + 4\hat{\jmath}$ (lying in the first quadrant) and $\vec{a}$ makes an angle γ (in the clockwise direction) with the positive direction of the y axis.

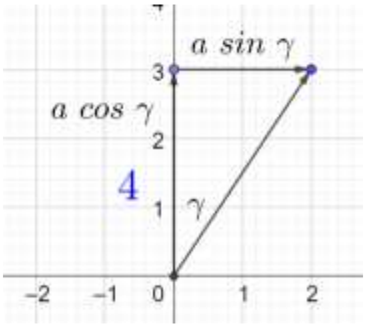
Part A

$$Vertical\ Speed = 34 \sin \alpha$$
$$Horizontal\ Speed = 34 \cos \alpha$$

Part B

$$\cos(180 - \gamma) = -\frac{3}{|\vec{a}|} \Rightarrow |\vec{a}| = -\frac{3}{\cos(180 - \gamma)} = \frac{3}{\cos \gamma}$$
$$\tan(180 - \gamma) = \frac{|\boldsymbol{y}|}{-3} \Rightarrow |\boldsymbol{y}| = 3 \tan \gamma \Rightarrow \boldsymbol{y} = 3 \tan \gamma\, \hat{\jmath}$$

Part C



$$a\cos\gamma = 4 \Rightarrow a = \frac{4}{\cos\gamma} = 4\sec\gamma$$
$$\tan\gamma = \frac{x}{4} \Rightarrow x = 4\tan\gamma$$
$$a + x = 4(\sec\gamma + \tan\gamma)$$

Example 2.9: Interpreting Components

A. A military tank in an exercise is going towards its objective. It has a velocity of $30\,\frac{miles}{hr}$, and a bearing of 100° from true north. Identify whether its x component and y component will be positive or negative. (Using the default coordinate plane). Overhead view of ground
B. Side view of object in air/on ground
C. Interpretation of positive and negative x component
D. Interpretation of positive and negative y component

Example 2.10: Concept Check

$\vec{b}$ is in the second quadrant. Then:

A. Its horizontal component is positive, and its vertical component is negative.
B. Its horizontal component is negative, and its vertical component is positive.
C. Its horizontal component and vertical component are both negative.
D. Its horizontal component and vertical component are both positive.

$$Option\ B$$

Example 2.11: Concept Check

$\boldsymbol{x}$ makes an angle θ with the positive direction of the x axis, in the counterclockwise direction. $\sin\theta$ is negative. What quadrant(s) can the vertor be in?

$\sin\theta$ is negative in the 3rd and 4th quadrant.

$$3rd\ \ or\ 4th\ Quadrant$$

B. Converting to Polar Form

2.12: Converting into polar form

$$\vec{r} = \underbrace{(x, y)}_{\substack{Component\\Form}} = \underbrace{\left(\sqrt{x^2+y^2}, \tan^{-1}\left(\frac{y}{x}\right)\right)}_{Polar\ Form}$$

Quadrant III Vector

$$\frac{y}{x} > 0 \Rightarrow \theta = \tan^{-1}\left(\frac{y}{x}\right) > 0 \Rightarrow \theta \in QI$$

But if $\vec{r}$ **is in** Quadrant III, then you need to add 180° to get the angle in QIII.

$$180° + \tan^{-1}\left(\frac{y}{x}\right)$$

Quadrant II Vector

$$\frac{y}{x} < 0 \Rightarrow \theta = \tan^{-1}\left(\frac{y}{x}\right) < 0 \Rightarrow \theta \in QIV$$

But if $\vec{r}$ **is in** Quadrant II, then you need to add 180° to get the angle in QIII.

$$180° + \tan^{-1}\left(\frac{y}{x}\right)$$

To generalize:

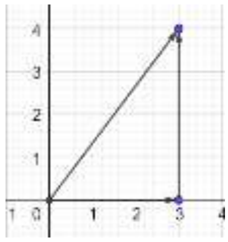
- For vectors in QI and QIV use $\tan^{-1}\left(\frac{y}{x}\right)$
- For vectors in QII and QIII use $180° + \tan^{-1}\left(\frac{y}{x}\right)$

Example 2.13

The vectors below are given in component form. Convert into polar form using an exact expression.

A. $\vec{a} = (3,4)$
B. $\vec{b} = (-3,-4)$

Part A



$$|\vec{a}| = \sqrt{x^2 + y^2} = \sqrt{3^2 + 4^2} = 5$$

$$\tan\theta = \frac{4}{3} \Rightarrow \theta = \tan^{-1}\left(\frac{4}{3}\right)$$

$$\vec{a} = \left(5, \tan^{-1}\left(\frac{4}{3}\right)\right)$$

Part B

$|\vec{b}|$ has the same magnitude as $|\vec{a}|$, but it goes in the opposite direction.

$$|\vec{b}| = \sqrt{x^2 + y^2} = \sqrt{3^2 + 4^2} = 5$$

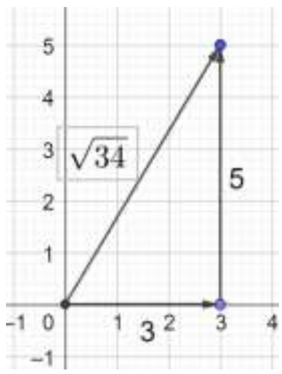
$$\tan\theta = \frac{4}{3} \Rightarrow \theta = 180° + \tan^{-1}\left(\frac{4}{3}\right)$$

$$\vec{b} = \left(5, 180° + \tan^{-1}\left(\frac{4}{3}\right)\right)$$

Example 2.14

A vector starts at the origin, and ends at (3,5). Find its

A. x component
B. y component
C. Magnitude
D. Direction



$$Horizontal\ component = 3\ Right$$

$$Vertical\ component = 5\ Right$$

$$Magnitude = \sqrt{5^2 + 3^2} = \sqrt{25 + 9} = \sqrt{34}$$

$$Direction = \left(\tan^{-1}\frac{5}{3}\right)°\ with\ the\ horizontal$$

C. Addition and Subtraction

We have already seen addition and subtraction in component form. However, vectors in polar form cannot be added directly. Hence, one method of adding them is convert them to component form, add them, and then convert back to polar form, if needed.

2.15: Component Resolution

If $\vec{\boldsymbol{c}} = \vec{\boldsymbol{a}} + \vec{\boldsymbol{b}}$

$$\boldsymbol{c} = (c_x, c_y) = (a_x + b_x, a_y + b_y)$$

Example 2.16

$$\vec{\boldsymbol{a}} = \left(2, \frac{\pi}{6}\right), \qquad \vec{\boldsymbol{b}} = \left(3, \frac{\pi}{4}\right)$$

The vectors above are given in polar form. Find each part below. State your answer in polar form. Also state the quadrant they lie in.

A. $\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}}$
B. $\vec{\boldsymbol{a}} - \vec{\boldsymbol{b}}$

Part A

$$\vec{\boldsymbol{a}} = \left(2, \frac{\pi}{6}\right) = (\underbrace{\sqrt{3}}, 1)$$

$$\vec{\boldsymbol{b}} = \left(3, \frac{\pi}{4}\right) = \left(\frac{3\sqrt{2}}{2}, \frac{3\sqrt{2}}{2}\right)$$

Find the vectors in component form:

$$\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}} = \left(\underbrace{\sqrt{3}}_{a_x} + \underbrace{\frac{3\sqrt{2}}{2}}_{b_x}, \underbrace{1}_{a_y} + \underbrace{\frac{3\sqrt{2}}{2}}_{b_y}\right) = \left(\frac{2\sqrt{3} + 3\sqrt{2}}{2}, \frac{2 + 3\sqrt{2}}{2}\right) = (+ve, +ve) \Rightarrow Quadrant\ I$$

Convert to polar form:

$$|\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}}| = \sqrt{\left(\frac{2\sqrt{3} + 3\sqrt{2}}{2}\right)^2 + \left(\frac{2 + 3\sqrt{2}}{2}\right)^2}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{\frac{2 + 3\sqrt{2}}{2}}{\frac{2\sqrt{3} + 3\sqrt{2}}{2}}\right) = \tan^{-1}\left(\frac{2 + 3\sqrt{2}}{2\sqrt{3} + 3\sqrt{2}}\right)$$

Part B

Find the vectors in component form:

$$\vec{\boldsymbol{a}} - \vec{\boldsymbol{b}} = \left(\frac{2\sqrt{3} - 3\sqrt{2}}{2}, \frac{2 - 3\sqrt{2}}{2}\right) \approx \left(\frac{2(1.7) - 3(1.4)}{2}, \frac{2 - 3(1.4)}{2}\right) = (-ve, -ve) \Rightarrow Q\ III$$

Convert to polar form:

$$|\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}}| = \sqrt{\left(\frac{2\sqrt{3} - 3\sqrt{2}}{2}\right)^2 + \left(\frac{2 - 3\sqrt{2}}{2}\right)^2}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{\frac{2 - 3\sqrt{2}}{2}}{\frac{2\sqrt{3} - 3\sqrt{2}}{2}}\right) = \tan^{-1}\left(\frac{2 - 3\sqrt{2}}{2\sqrt{3} - 3\sqrt{2}}\right)$$

(Calculator) Example 2.17

A ship leaves port at a bearing of 30° from true north and travels 30 km. It then travels 50 km at a bearing of 130° from true north. Find the bearing and the distance of the final position from the port.

$$D^2 = 30^2 + 50^2 - 2(30)(50)(\cos 80°) =$$
$$D =$$

Use the sine rule to find the angle:

Type equation here.

(Calculator) Example 2.18

An airplane is flying at a speed of 200 miles/hour at a bearing of 30 degrees from true north (before taking the wind into account). The wind is 40 miles/hour, at a bearing of 120 degrees from true north. Find the net velocity vector of the plane, taking into consideration the wind.

$$D^2 = 200^2 + 40^2 - 2(200)(40)(\cos 90°) =$$
$$D =$$

Use the sine rule to find the angle:

Type equation here.

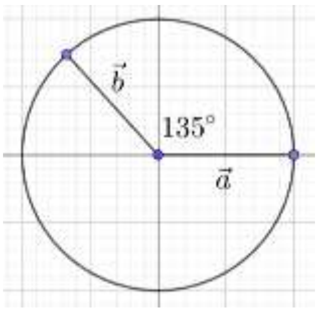
Example 2.19

An airplane takes off from Point A, travels on a circular path with radius r around an island, and lands at point B. The length of the path travelled by the plane is $\frac{3}{4}\pi r$. Find, using vector methods only, the distance between point A and point B in terms of r.

$$\vec{\boldsymbol{a}} = \overrightarrow{\boldsymbol{a_x}} + \overrightarrow{\boldsymbol{a_y}} = -r\hat{\boldsymbol{\imath}} + \boldsymbol{0} = -r\hat{\boldsymbol{\jmath}}$$

$$\vec{\boldsymbol{b}} = \overrightarrow{\boldsymbol{b_x}} + \overrightarrow{\boldsymbol{b_y}} = r\cos 135°\,\hat{\boldsymbol{\imath}} + r\sin 135°\,\hat{\boldsymbol{\jmath}} = r\left(-\frac{\sqrt{2}}{2}\right)\hat{\boldsymbol{\imath}} + r\left(\frac{\sqrt{2}}{2}\right)\hat{\boldsymbol{\jmath}}$$

$$\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}} = -r\hat{\boldsymbol{\imath}} + r\left(-\frac{\sqrt{2}}{2}\right)\hat{\boldsymbol{\imath}} + r\left(\frac{\sqrt{2}}{2}\right)\hat{\boldsymbol{\jmath}} = -r\left(1 + \frac{\sqrt{2}}{2}\right)\hat{\boldsymbol{\imath}} + r\left(\frac{\sqrt{2}}{2}\right)\hat{\boldsymbol{\jmath}}$$



Example 2.20

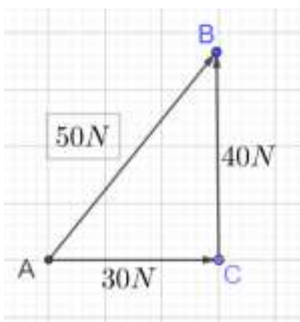
Two forces of $30N$ and $40N$ act on an object in the x-direction and the y-direction, respectively.

A. Find the angle between the two forces
B. Find the magnitude and direction of the resultant force
C. Find the angle that the resultant makes with the larger force.

Part A

Part B

$$Magnitude = \sqrt{30^2 + 40^2} = 50$$
$$\sin\theta = \frac{BC}{AB} \Rightarrow \theta = \sin^{-1}\frac{40}{50} = \sin^{-1}\frac{4}{5}$$



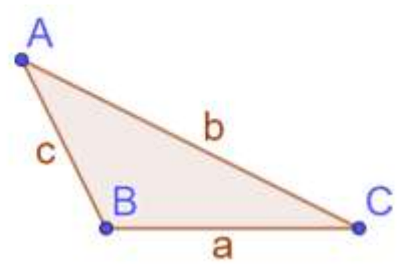
Part B

$$\sin\theta = \frac{AC}{AB} \Rightarrow \theta = \sin^{-1}\frac{30}{50} = \sin^{-1}\frac{3}{5}$$

D. Finding Angles

2.21: Law of Sines

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{Sin\ C}$$

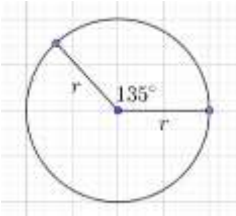


2.22: Law of Cosines

$$a^2 = \underbrace{b^2 + c^2 - 2bc\cos A}_{\substack{Simplifies\ to\ Pythagoras \\ for\ right-angled\ triangles \\ \cos 90 = 0}} \Leftrightarrow \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

Example 2.23

An airplane takes off from Point A, travels on a circular path with radius r around an island, and lands at point B. The length of the path travelled by the plane is $\frac{3}{4}\pi r$. Find, using vector methods only, the distance between point A and point B in terms of r.



The angle traversed is:

$$135°$$

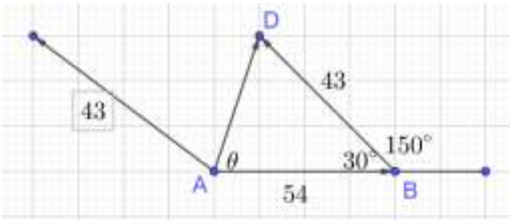
Using the law of cosines, the distance is:

$$D^2 = r^2 + r^2 - 2r^2\cos 135 = 2r^2(1 - \cos 135°) = 2r^2\left(1 + \frac{\sqrt{2}}{2}\right)$$

$$D = r\sqrt{2 + \sqrt{2}}$$

(Calculator) Example 2.24

Consider two vectors with length 54 and 43 respectively, which have an angle of 150° between them. Find the resultant vector. Give the magnitude of the vector, and the angle that it makes with the first vector.



By Law of Cosines:

$$|u + v|^2 = u^2 + v^2 - 2\,uv\cos\theta$$

Substituting and taking square roots:

$$|u + v| = \sqrt{54^2 + 43^2 - 2(54)(43)(\cos 30)} = 27.26$$

By the Law of Sines:

$$\frac{\sin\theta}{43} = \frac{\sin 30}{27.26} \Rightarrow \sin\theta = \frac{\sin 30}{27.26} \times 43 \Rightarrow \theta = \sin^{-1}\left(\frac{\sin 30}{27.26} \times 43\right) = 52.06$$

E. Physics

2.25: Components of Force and Acceleration

$$F_x = ma_x, \quad F_y = ma_y, \quad F_z = ma_z$$

We know that:

$$\vec{\boldsymbol{F}} = m\vec{\boldsymbol{a}}$$

Write each vector in component form:

$$(F_x, F_y, F_z) = m(a_x, a_y, a_z)$$

Carry out the scalar multiplication:

$$(F_x, F_y, F_z) = (ma_x, ma_y, ma_z)$$

If two vectors are equal, then their individual components must be equal.
Equating components gives us the result that we want:

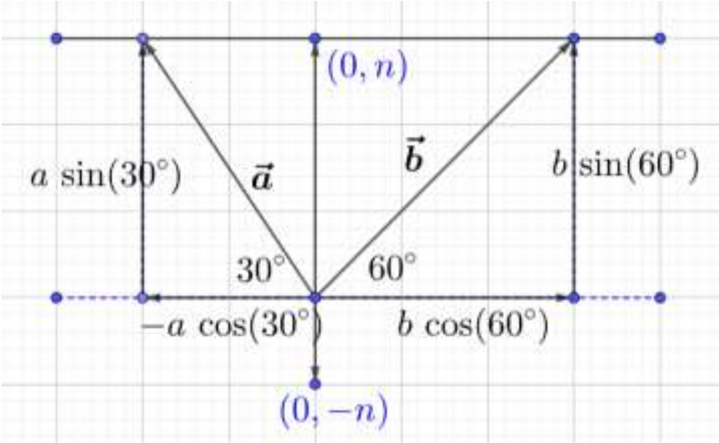
$$F_x = ma_x, \qquad F_y = ma_y, \qquad F_z = ma_z$$

Example 2.26

A. Find the force in each direction given mass and acceleration.

Example 2.27: Systems of Equations

An object of weight 100 Newtons is suspended from two taut wires attached to hooks in the ceiling to its left and right. At the point of suspension, the left and right wires make angles $\frac{\pi}{6}$ and $\frac{\pi}{3}$ with the horizontal. Determine the forces acting in the two wires in both polar and component form.



Consider the y component of the force.
Since the object is in equilibrium, the net force is zero:

$$Upward\ Force + Downward\ Force = 0$$

$$Upward\ Force - 100 = 0$$

$$Upward\ Force = 100$$

Convert from radians to degrees:

$$\frac{\pi}{6} = 30°, \frac{\pi}{3} = 60°$$

Calculate the components of each vector:

$$\vec{a} = (-a\cos 30°, a\sin 30°) = \left(-\frac{\sqrt{3}a}{2}, \frac{1}{2}a\right)$$

$$\vec{b} = (b\cos 60°, b\sin 60°) = \left(\frac{b}{2}, \frac{\sqrt{3}b}{2}\right)$$

Add the vectors:

$$\vec{a} + \vec{b} = \left(\frac{b}{2} - \frac{\sqrt{3}a}{2}, \frac{\sqrt{3}b}{2} + \frac{1}{2}a\right) = (0,100)$$

Since the two vectors are equal, their individual components are equal.
Equate the components to get:

$$\frac{b}{2} - \frac{\sqrt{3}a}{2} = 0 \Rightarrow \underbrace{b = \sqrt{3}a}_{Equation\ I}, \qquad \underbrace{\frac{\sqrt{3}}{2}b + \frac{1}{2}a = 100}_{Equation\ II}$$

Substitute the value of b from Equation I:

$$LHS = \frac{\sqrt{3}}{2}(\sqrt{3}a) + \frac{1}{2}a = \frac{3a + a}{2} = \frac{4a}{2} = 2a$$

Then, the equation becomes:

$$2a = 100 \Rightarrow a = 50$$

Substitute the above value of a into Equation I:

$$b = \sqrt{3}a = 50\sqrt{3}$$

We can write the vectors in polar form as:

$$\vec{\boldsymbol{a}} = (50,150), \qquad \vec{\boldsymbol{b}} = \left(50\sqrt{3}, 60\right)$$

And we can write the same vectors in component form as:

$$\vec{\boldsymbol{a}} = (-a\ cos\ 30°, a\ sin\ 30°) = \left(50 \times \frac{\sqrt{3}}{2}, 50 \times \frac{1}{2}\right) = \left(-25\sqrt{3}, 25\right)$$

$$\vec{\boldsymbol{b}} = (b\ cos\ 60°, b\ sin\ 60°) = \left(50\sqrt{3} \times \frac{1}{2}, 50\sqrt{3} \times \frac{\sqrt{3}}{2}\right) = \left(25\sqrt{3}, 75\right)$$

Example 2.28: Systems of Equations

Answer the above question if the left wire makes an angle $\frac{\pi}{4}$ with the horizontal, and all else remains the same.

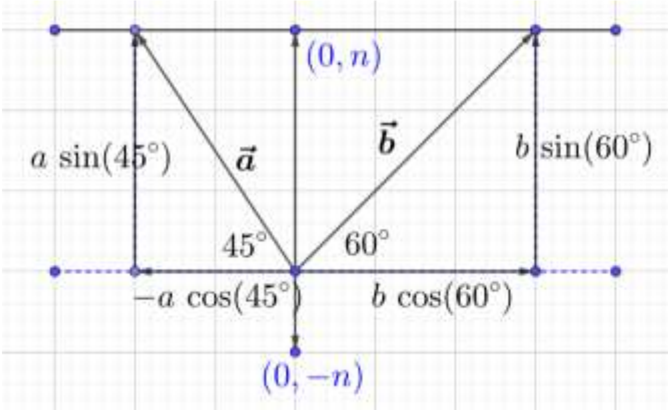
Convert from radians to degrees:

$$\frac{\pi}{4} = 45°, \frac{\pi}{3} = 60°$$

Calculate the components of each vector:

$$\vec{\boldsymbol{a}} = (-a\cos 45°, a\ sin\ 45°) = \left(-\frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}\right)$$

$$\vec{\boldsymbol{b}} = (b\ cos\ 60°, b\ sin\ 60°) = \left(\frac{b}{2}, \frac{\sqrt{3}b}{2}\right)$$



Add the vectors:

$$\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}} = \left(\frac{b}{2} - \frac{a}{\sqrt{2}}, \frac{\sqrt{3}b}{2} + \frac{a}{\sqrt{2}}\right) = (0,100)$$

Since the two vectors are equal, their individual components are equal.

Equate the components to get:

$$\frac{b}{2} - \frac{a}{\sqrt{2}} = 0 \Rightarrow \underbrace{b = \frac{2}{\sqrt{2}}a}_{Equation\ I}, \qquad \underbrace{\frac{\sqrt{3}}{2}b + \frac{a}{\sqrt{2}} = 100}_{Equation\ II}$$

Substitute the value of b from Equation I:

$$LHS = \frac{\sqrt{3}}{2} \cdot \frac{2a}{\sqrt{2}} + \frac{a}{\sqrt{2}} = \frac{\sqrt{3}a}{\sqrt{2}} + \frac{a}{\sqrt{2}} = \frac{\sqrt{3}a + a}{\sqrt{2}} = \frac{a\left(1 + \sqrt{3}\right)}{\sqrt{2}}$$

Then, the equation becomes:

$$\frac{a\left(1 + \sqrt{3}\right)}{\sqrt{2}} = 100$$

And we can solve for a:

$$a = 100 \times \frac{\sqrt{2}}{1 + \sqrt{3}} = \frac{100\sqrt{2}}{1 + \sqrt{3}} = \frac{100\sqrt{2}\left(1 - \sqrt{3}\right)}{-2} = \frac{100\sqrt{2} - 100\sqrt{6}}{-2} = 50\sqrt{6} - 50\sqrt{2}$$

Substitute the above value of a into Equation I:

$$b = \frac{2}{\sqrt{2}} 50\sqrt{6} - 50\sqrt{2} = 100\sqrt{3} - 100$$

We can write the vectors in polar form as:

$$\vec{\boldsymbol{a}} = \left(50\sqrt{6} - 50\sqrt{2}, 135\right), \qquad \vec{\boldsymbol{b}} = \left(100\sqrt{3} - 100, 60\right)$$

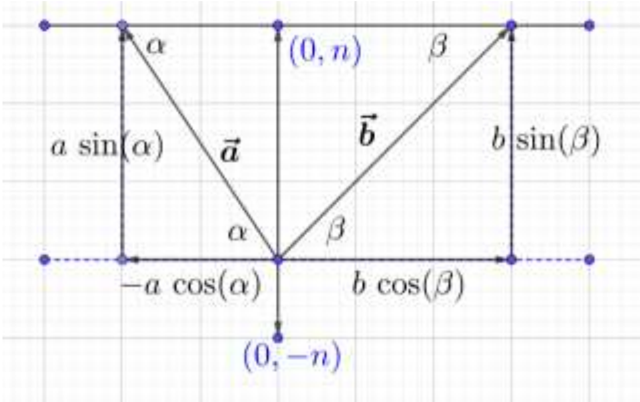
And we can write the same vectors in component form as:

$$\vec{\boldsymbol{a}} = (-a\ cos\ 45°, a\ sin\ 45°) = \left(\left(-50\sqrt{6} - 50\sqrt{2}\right)\frac{1}{\sqrt{2}}, \left(50\sqrt{6} - 50\sqrt{2}\right)\frac{1}{\sqrt{2}}\right) = \left(-50\sqrt{3} - 50, 50\sqrt{3} - 50\sqrt{2}\right)$$

$$\vec{\boldsymbol{b}} = (b\ cos\ 60°, b\ sin\ 60°) = \left((100\sqrt{3}-100)\frac{1}{2}, (100\sqrt{3}-100)\frac{\sqrt{3}}{2}\right) = (50\sqrt{3}-50, 150-50\sqrt{3})$$

Example 2.29: Systems of Equations

A weight of mass n Newtons is suspended from two taut wires attached to hooks in the ceiling to its left and right. At the point of suspension, the left and right wires make angles α and β with the horizontal. Determine the forces acting in the two wires.



$$\vec{\boldsymbol{a}} = (-a\ cos\ \alpha, a\ sin\ \alpha)$$
$$\vec{\boldsymbol{b}} = (b\ cos\ \beta, b\ sin\ \beta)$$
$$\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}} = (0, n)$$

x components:

$$b\cos\beta - a\cos\alpha = 0 \Rightarrow \underbrace{b = \frac{a\cos\alpha}{\cos\beta}}_{\textit{Equation I}}$$

y components:

$$a\sin\alpha + b\sin\beta = n$$

Substitute the value of b from Equation I:

$$a\sin\alpha + \left(\frac{a\cos\alpha}{\cos\beta}\right)\sin\beta = n$$

Factor a and add the fractions:

$$a\left[\frac{\sin\alpha\cos\beta + \cos\alpha\sin\beta}{\cos\beta}\right] = n$$

Use the identity $\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$:

$$a\left[\frac{\sin(\alpha+\beta)}{\cos\beta}\right] = n \Rightarrow a = \frac{n\cos\beta}{\sin(\alpha+\beta)}$$

Substitute the above value of a into Equation I:

$$b = a\frac{\cos\alpha}{\cos\beta} = \frac{n\ \cos\beta}{\sin(\alpha+\beta)}\left(\frac{\cos\alpha}{\cos\beta}\right) = \frac{n\cos\alpha}{\sin(\alpha+\beta)}$$

We can write the vectors in polar form as:

$$\vec{\boldsymbol{a}} = \left(\frac{n\ \cos\beta}{\sin(\alpha+\beta)}, 180-\alpha\right), \qquad \vec{\boldsymbol{b}} = \left(\frac{n\cos\alpha}{\sin(\alpha+\beta)}, \beta\right)$$

And we can write the same vectors in component form as:

$$\vec{\boldsymbol{a}} = (-a\ cos\ \alpha, a\ sin\ \alpha) = \left(-\frac{n\cos\beta\cos\alpha}{\sin(\alpha+\beta)}, \frac{n\cos\beta\sin\alpha}{\sin(\alpha+\beta)}\right)$$
$$\vec{\boldsymbol{b}} = (b\ cos\ \beta, b\ sin\ \beta) = \left(\frac{n\cos\alpha\ cos\ \beta}{\sin(\alpha+\beta)}, \frac{n\cos\alpha\ sin\ \beta}{\sin(\alpha+\beta)}\right)$$

Example 2.30: Kinematics

At time t, an insect is attempting to outrun a book which has been thrown at it. It runs with a velocity of $v\frac{inches}{minute}$, at an angle of θ (in clockwise direction) to true North such that $0 < \theta < 90$. The book is rectangular, with a width ($x\ component$) of 9 cm and a length ($y\ component$) of 20 cm, and currently the insect is exactly below the middle of the book length-wise and width-wise. The book is oriented so that the length is parallel to the y axis. The book falls straight down, and hits the ground at time $t+2$ seconds. Determine the value of v such that the insect can escape the book. Ignore the length and width of the cockroach.

Since the angle is given clockwise with respect to true North, the usual interpretations will reverse.

$$\cos\theta \to vertical$$
$$\sin\theta \to horizontal$$

$$Vertical\ Speed\ Required = \frac{Distance}{Time} = \frac{10\ cm}{(t+2)-t} = 5\frac{cm}{second}$$
$$Actual\ Vertical\ Speed = v\cos\theta\frac{inches}{minute}$$

$$v\cos\theta\frac{inches}{minute} = 5\frac{cm}{second}$$
$$v\cos\theta\frac{2.54\ cm}{60\ seconds} = 5\frac{cm}{second}$$
$$v = \frac{300}{2.54\cos\theta} = \frac{30000}{254\cos\theta} = \frac{15000}{127\cos\theta}$$

$$Horizontal\ Speed\ Required = \frac{Distance}{Time} = \frac{4.5\ cm}{(t+2)-t} = 2.25\frac{cm}{second}$$
$$Actual\ Horizontal\ Speed = v\sin\theta\frac{inches}{minute}$$

$$v\sin\theta\frac{inches}{minute} = 2.25\frac{cm}{second}$$
$$v\sin\theta\frac{2.54\ cm}{60\ seconds} = 2.25\frac{cm}{second}$$
$$v = \frac{135}{2.54\sin\theta}$$

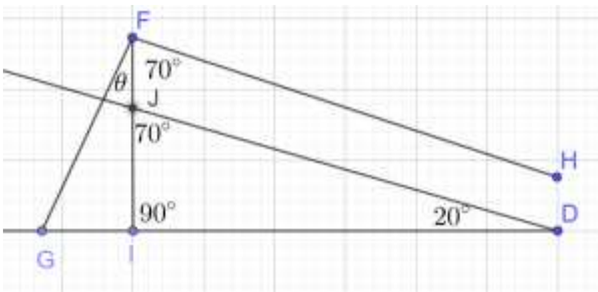
To escape the book, the cockroach can escape either horizontally, or vertically. Hence, the final answer is:

$$Minimum\left(\frac{15000}{127\cos\theta}, \frac{135}{2.54\sin\theta}\right)$$

Example 2.31

A man on a wheelchair is being pushed up a hospital ramp by an attendant. The man weighs 70 kg, his wheelchair 15 kg, and the attendant weighs 60 kg. $\frac{2}{3}rd$ of the way up the ramp, the attendant stops briefly. At this point, what are the components of the system's mass parallel and perpendicular to the ramp if the ramp makes an angle of 20° with the horizontal.
Consider the man, his wheelchair, and the attendant to be a point mass.

Step I: Focus on the Geometry



The ramp makes an angle of 20° with the horizontal. Hence

$$\angle JDI = 20°$$

Let the man/wheelchair/attendant be a point mass at F. We want the components of the system's mass parallel and perpendicular to the ramp.
Hence, draw

$$FH \parallel JD, \quad FH \perp FG, \quad FI \perp GD$$
$$\angle FID = 90° \Rightarrow \angle IJD = 70°$$

By corresponding angles in parallel lines FH and JD:

$$\angle JFH = \angle IHD = 70°$$

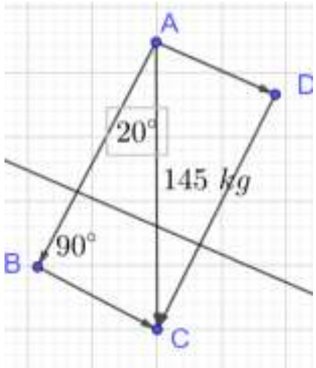
Since $FH \perp FG$

$$\angle GFH = 90° \Rightarrow \angle GFI = 90 - \angle IFH = 90 - 70 = 20°$$

Step II: Focus on the Vector

$$\sin A \ = \frac{opp}{hyp} \Rightarrow \sin 20° = \frac{BC}{145} \Rightarrow BC = \sin 20\,° \times 145 = 45.59\ kg$$

$$\cos A \ = \frac{adj}{hyp} \Rightarrow \cos 20° \ = \frac{AB}{145} \Rightarrow AB = \cos 20\,° \times 145 = 136.26\ kg$$



2.2 Projections

A. Introduction

We now look at multiplication in the context of vectors, beginning with the concept of projection of vectors to introduce the idea.

In terms of multiplication with real numbers, recall that:

$$\underbrace{3 \cdot 4}_{Dot\ Product} \quad = \quad \underbrace{3 \times 4}_{Cross\ Product} \quad = 12$$

In the context of real numbers, both products are the same.

However, in the context of vectors, the two products are not the same.

B. Scalar Projection

In the previous section, we introduce the idea of components of a vector. We now generalize this idea to projections.

2.32: Projection along axes

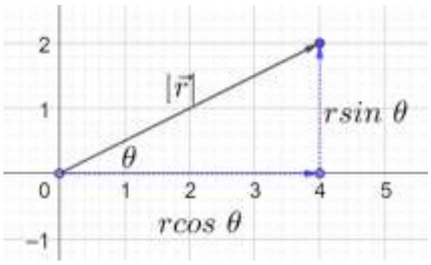
The projection of a vector along the x-axis gives the component of the vector in the $x - direction$

The projection of a vector along the y-axis gives the component of the vector in the $y - direction$

- $|\vec{r}| = r$
- θ is the angle made by the vector with the positive direction of the $x - axis$.

$$\sin \theta = \frac{opp}{hyp} \Rightarrow opp = (\sin \theta)(hyp) = r \sin \theta$$

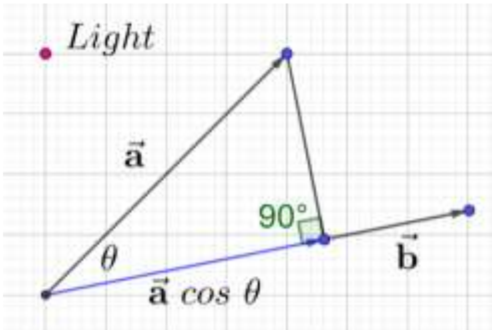
$$\cos \theta = \frac{adj}{hyp} \Rightarrow adj = (\cos \theta)(hyp) = r \cos \theta$$



2.33: Length of Projection along a unit vector

If $\vec{a}$ is a vector, $\hat{b}$ is a unit vector, and the angle between the two vectors is θ, then the length of the projection of $\vec{a}$ along $\hat{b}$ is given by:

$$\vec{a} \cdot \hat{b} = |\vec{a}| \cos \theta$$



$$\underbrace{|\vec{a}|}_{Scalar} \underbrace{\cos \theta}_{Scalar} = \underbrace{x}_{Scalar}$$

As each term on the left is a scalar, the answer on the right is also a scalar.

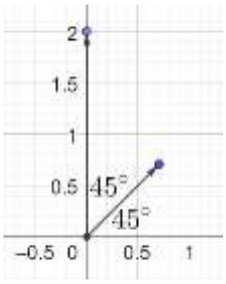
- The length of the projection of $\vec{a}$ **along** $\vec{b}$ is the length of the blue line in the diagram.
- Imagine a light shining somewhere to the left of vector $\vec{a}$. The projection of vector $\vec{a}$ along $\hat{b}$ is the shadow cast by $\vec{a}$ upon $\hat{b}$.

➢ The projection also gives the component of $\vec{\boldsymbol{a}}$ that is in the direction of $\hat{\boldsymbol{b}}$.

Example 2.34

A. Find the length of the projection of $\vec{\boldsymbol{x}} = (\mathbf{0}, \mathbf{2})$ along $\vec{\boldsymbol{y}} = \left(\frac{\mathbf{1}}{\sqrt{\mathbf{2}}}, \frac{\mathbf{1}}{\sqrt{\mathbf{2}}}\right)$

B. Find the length of the projection of $\vec{\boldsymbol{x}} = (\mathbf{3}, \mathbf{0})$ along $\vec{\boldsymbol{y}} = \left(\frac{\mathbf{1}}{\sqrt{\mathbf{5}}}, -\frac{\mathbf{2}}{\sqrt{\mathbf{5}}}\right)$



Part A

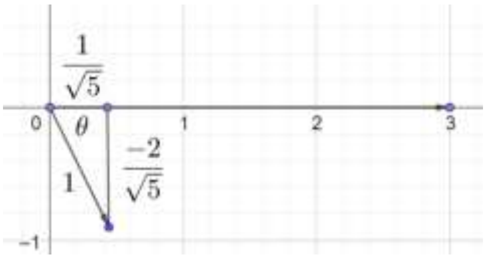
Note that:

$$\theta = 45°$$

Hence, the projection is:

$$\vec{\boldsymbol{x}} \cdot \hat{\boldsymbol{y}} = |\vec{\boldsymbol{x}}| \cos\theta = \sqrt{2^2 + 0^2} \cos 45° = 2 \times \frac{1}{\sqrt{2}} = \sqrt{2}$$

Part B



$$\cos\theta = \frac{adj}{hyp} = \frac{\frac{1}{\sqrt{5}}}{1} = \frac{1}{\sqrt{5}}$$

Hence, the projection is:

$$\vec{\boldsymbol{x}} \cdot \hat{\boldsymbol{y}} = |\vec{\boldsymbol{x}}| \cos\theta = \sqrt{3^2 + 0^2} \times \frac{1}{\sqrt{5}} = \frac{3}{\sqrt{5}}$$

2.35: Scaling a vector to get a unit vector

If we wish to find $\vec{\boldsymbol{a}} \cdot \hat{\boldsymbol{b}}$ and $\vec{\boldsymbol{b}}$ is not a unit vector, we can scale it to be a unit vector using the formula:

$$\hat{\boldsymbol{b}} = \frac{\vec{\boldsymbol{b}}}{|\vec{\boldsymbol{b}}|}$$

Example 2.36

Find the length of the projection of $\vec{\boldsymbol{x}} = \left(0, -\frac{5}{2}\right)$ along $\vec{\boldsymbol{y}} = (1.5, -4)$

Find the magnitude of:

$$|\vec{\boldsymbol{y}}| = \sqrt{\left(\frac{3}{2}\right)^2 + (-4)^2} = \sqrt{\frac{9}{4} + 16} = \sqrt{\frac{73}{4}} = \frac{\sqrt{73}}{2}$$

Scale $\vec{\boldsymbol{y}}$ to be a unit vector:

$$\hat{\boldsymbol{z}} = \frac{1}{\frac{\sqrt{73}}{2}} \vec{\boldsymbol{y}} = \frac{2}{\sqrt{73}}\left(\frac{3}{2}, -4\right) = \left(\frac{3}{\sqrt{73}}, -\frac{8}{\sqrt{73}}\right)$$

We can check that the vector we have found is indeed a unit vector:

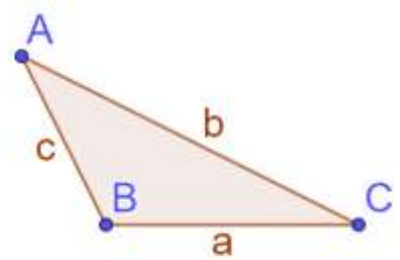
$$|\hat{\boldsymbol{z}}| = \sqrt{\left(\frac{3}{\sqrt{73}}\right)^2 + \left(-\frac{8}{\sqrt{73}}\right)} = \sqrt{\frac{9}{73} + \frac{64}{73}} = \sqrt{\frac{73}{73}} = \sqrt{1} = 1$$

Hence, the projection is:

$$\vec{\boldsymbol{x}} \cdot \hat{\boldsymbol{z}} = |\vec{\boldsymbol{x}}| \cos\theta = \sqrt{0^2 + \left(-\frac{5}{2}\right)^2} \left(\frac{8}{\sqrt{73}}\right) = \frac{5}{2} \times \frac{8}{\sqrt{73}} = \frac{20}{\sqrt{73}}$$

2.37: Law of Cosines

$$a^2 = b^2 + c^2 - 2bc\cos A \Leftrightarrow \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$



Example 2.38

A. In ΔABC, if $|\overrightarrow{BC}| = 3, |\overrightarrow{CA}| = 5$ and $|\overrightarrow{BA}| = 7$, then the projection of the vector $\overrightarrow{BA}$ on $\overrightarrow{BC}$ is equal to: (JEE Main 2021, 20 July, Shift-II)

B. In a triangle ABC, if $|\overrightarrow{BC}| = 8, |\overrightarrow{CA}| = 7$ and $|\overrightarrow{AB}| = 10$, then the projection of the vector $\overrightarrow{AB}$ on $\overrightarrow{AC}$ is equal to: (JEE Main 2021, 18 March, Shift-II)

Part A

$$\cos\angle ABC = \frac{7^2 + 3^2 - 5^2}{2(7)(3)} = \frac{49 + 9 - 25}{42} = \frac{33}{42} = \frac{11}{14}$$

Projection of the vector $\overrightarrow{BA}$ on $\overrightarrow{BC}$

$$= |\overrightarrow{BA}|\cos\angle ABC = 7 \cdot \frac{11}{14} = \frac{11}{2}$$

Part B

Projection of the vector $\overrightarrow{AB}$ on $\overrightarrow{AC}$ is equal to:

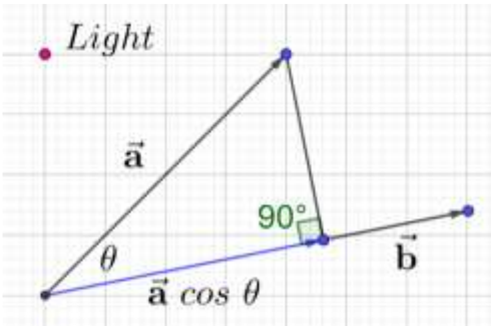
$$\overrightarrow{AB}\cos\theta = 10\left(\frac{10^2 + 7^2 - 8^2}{2(7)(10)}\right) = \frac{85}{14}$$

2.39: Restating the projection definition

$$\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|} = |\vec{a}|\cos\theta$$

$$\vec{a}\cdot\hat{b} = \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|} = \underbrace{|\vec{a}|\cos\theta}$$

C. Vector Projection



2.40: Projection along a unit vector

If $\vec{a}$ is a vector, $\hat{b}$ is a unit vector, and the angle between the two vectors is θ, then the projection of $\vec{a}$ along $\hat{b}$ is given by:

$$\underbrace{(|\vec{a}|\cos\theta)}_{Magnitude}\ \underbrace{\hat{b}}_{Direction} = \underbrace{(|\vec{a}|\cos\theta)}_{Magnitude}\ \underbrace{\frac{\vec{b}}{|\vec{b}|}}_{Direction}$$

Example 2.41

Compare the formula for a vector projection with the formula for the length of the projection, and determine the relation between the two:

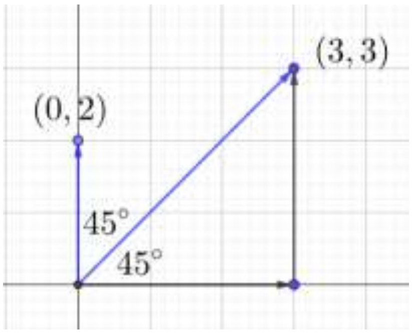
$$\underbrace{(|\vec{a}|\cos\theta)\left(\frac{\vec{b}}{|\vec{b}|}\right)}_{Vector\ Projection}, \qquad \underbrace{|\vec{a}|\cos\theta}_{Length\ of\ Projection}$$

$$Vector\ Projection = (Length\ of\ Projection)(Unit\ Vector\ along\ length\ of\ projection)$$

Example 2.42
Find the projection vector for the following projections. Also, find the length of the projection.

A. $\vec{a} = (0,2)$ along $\vec{b} = (3,3)$
B. $\vec{a} = (0,-3)$ along $\vec{b} = \left(-\frac{1}{\sqrt{7}}, \frac{\sqrt{3}}{\sqrt{7}}\right)$

Part A



Length of the projection

$$= |\vec{a}| \cos\theta = 2\cos 45° = 2\left(\frac{\sqrt{2}}{2}\right) = \sqrt{2}$$

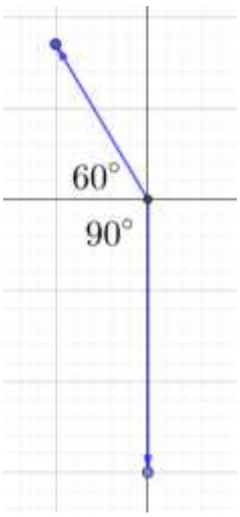
$$|\vec{b}| = \sqrt{3^2 + 3^2} = \sqrt{18} = 3\sqrt{2}$$

The vector is:

$$|\vec{a}| \cos\theta \left(\frac{\vec{b}}{|\vec{b}|}\right) = \sqrt{2}\frac{(3,3)}{3\sqrt{2}} = (1,1)$$

2.43: Negative Projection

Example 2.44
Find the projection vector for the following projections. Also, find the length of the projection.



The ratio of the components of $\vec{b}$ is the same as the ratio of the legs of a $30-60-90$ triangle:

$$-\frac{1}{\sqrt{7}} : \frac{\sqrt{3}}{\sqrt{7}} = -1 : \sqrt{3}$$

We can

$$\theta = 60° + 90° = 150° \Rightarrow \cos 150° = \cos(180-30) = -\cos 30° = -\frac{\sqrt{3}}{2}$$

$$|\vec{a}| \cos\theta = (3)\left(-\frac{\sqrt{3}}{2}\right) = -\frac{3\sqrt{3}}{2}$$

The length of the projection is

$$\frac{3\sqrt{3}}{2}\ in\ the\ direction\ opposite\ to\ \vec{b} = \left(-\frac{1}{\sqrt{7}}, \frac{\sqrt{3}}{\sqrt{7}}\right)$$

We can now find the unit vector in the direction of the vector that we are taking the projection along:

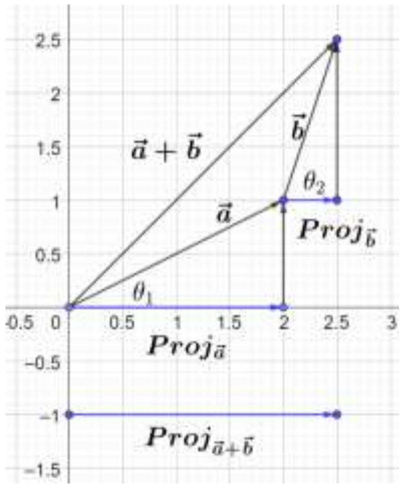
$$|(-1, \sqrt{3})| = 2 \Rightarrow \left|\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)\right| = 1$$

The projection vector is:

$$|\vec{a}| \cos\theta \left(\frac{\vec{b}}{|\vec{b}|}\right) = -\frac{3\sqrt{3}}{2}\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right) = \left(\frac{3\sqrt{3}}{4}, -\frac{9}{4}\right)$$

2.45: Distributive Property

$$Proj(\vec{a} + \vec{b}) = Proj(\vec{a}) + Proj(\vec{b})$$



2.3 Angles and Perpendicularity

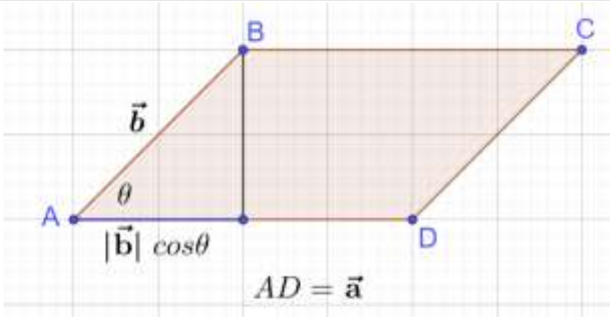
A. Geometrical Interpretation of Dot Product

2.46: Dot Product

$$\vec{a}\cdot\vec{b}=|\vec{a}||\vec{b}|\cos\theta$$

Where

$$\theta = angle\ between\ two\ vectors$$



Let

$$\overrightarrow{AB}=b, \overrightarrow{AD}=a$$

Projection of b upon a is:

$$|\vec{b}|\cos\theta$$

Then, we define the dot product of the vectors $\vec{a}$ and $\vec{b}$ as:

$$\vec{a}\cdot\vec{b}=|\vec{a}||\vec{b}|\cos\theta$$

Notes:

- Scalar multiplication has a vector multiplied with a scalar. The magnitude changes, but the direction is unchanged. The output of scalar multiplication is a vector.
- The dot product (also called scalar product) of two vectors has a scalar output.
- Since the output is a scalar, it is also called a scalar product.
- The scalar product is in contrast to the cross product which has a vector for an output. We will learn the cross product later.

Example 2.47

Consider the same diagram as above.

A. What is the domain of $\cos\theta$ in this context? What quadrants does it correspond to on the unit circle?
B. What is the range of $\cos\theta$ in this context?
C. In which quadrants is $\cos\theta$ positive? In which quadrants is it negative?

The domain is given by the possible values for θ. Note that θ can be obtuse$\left(\frac{\pi}{2}<\theta<\pi\right)$, but not reflex.

$$0\le\theta\le\pi$$

$$Quadrant\ I\ and\ II$$

$$-1 \le \cos\theta \le 1$$

Example 2.48

A. Determine the sign of $|\vec{a}|$, the sign of $|\vec{b}|$
B. Hence, for non-zero vectors $\vec{a}$ and $\vec{b}$, what does the sign of $\vec{a}\cdot\vec{b}$ depend on?
C. Determine the sign of $\cos\theta$

Part A

The magnitude of a vector is the length of the vector, which is always a nonnegative quantity:

$$|\vec{a}| \ge 0$$
$$|\vec{b}| \ge 0$$

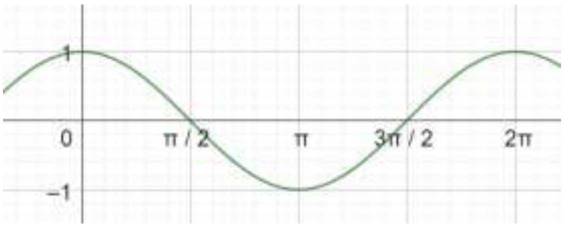
Part B

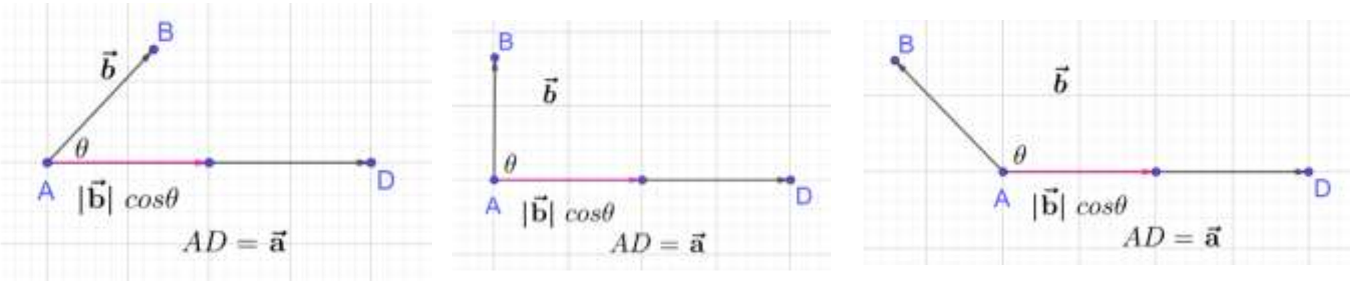
$$\vec{a}\cdot\vec{b} = \underbrace{|\vec{a}|}_{+ve}\underbrace{|\vec{b}|}_{+ve}\cos\theta$$

Since the first two terms are positive, the value of $\vec{a}\cdot\vec{b}$ depends on the sign of $\cos\theta$

Part C

$$\cos\theta > 0 \Rightarrow \theta \in Quadrant\ I$$
$$\cos\theta = 0 \Rightarrow \theta = \frac{\pi}{2}$$
$$\cos\theta < 0 \Rightarrow \theta \in Quadrant\ II$$





Example 2.49

Find the dot product in each case below between $\vec{a}$ and $\vec{b}$, in terms of the magnitude of the vectors, if the angle between the two vectors is:

A. zero
B. $\frac{\pi}{2}$
C. π

$$|\vec{a}||\vec{b}| \times 1 = |\vec{a}||\vec{b}|$$
$$|\vec{a}||\vec{b}| \times 0 = 0$$

$$|\vec{a}||\vec{b}| \times (-1) = -|\vec{a}||\vec{b}|$$

2.50: Dot Product

The dot product is:

- Positive when $0 \le \theta < \frac{\pi}{2}$
- Zero when $\theta = \frac{\pi}{2}$
- Negative when $\frac{\pi}{2} < \theta \le \pi$

B. Work

2.51: Work-Energy Theorem

The net work done by the forces on an object equals the change in its kinetic energy (KE).

- If KE increases, work done is positive
  - ✓ For example, when a force applied in the direction of movement of the object increases the velocity of the object, the work done is positive.
- If KE remains the same, work done is zero
  - ✓ If the direction of force is perpendicular to the direction of movement of the object, the work done is zero.
- If KE decreases, work done is negative
  - ✓ For example, when a force is applied opposite to the direction of movement of the object, the force slows the object down, reducing KE, and doing negative work.

Example 2.52

The moon orbits the Earth. For this question, assume that the orbit is circular. Explain why the work done by the Earth on the moon is zero:

A. Mathematically
B. Using the concept of KE

Part A: Mathematical Explanation

The tangent to a circle is perpendicular to the radius at the point of tangency.
Hence, the angle between the force of gravity, and the direction of movement of the moon is always a right angle.
Hence

$$\cos\theta = \cos 90° = 0$$

Part B: Kinetic Energy

Gravity changes the direction of the moon, but does not change the magnitude of the velocity.
Hence, there is no change in the kinetic energy.
Hence,

$$Work = 0$$

Example 2.53

Decide whether work done is positive, negative, or zero:

A. The net force on an object increases its velocity
B. The net force on an object decreases its velocity
C. The net force on an object does not change its velocity.

$$Work\ done\ is\ positive$$
$$Work\ done\ is\ negative$$
$$Work\ done\ is\ zero$$

2.54: Work done by Individual Force

The sum total of work done is the work done by each individual force acting on an object

$$W_S = W_1 + W_2 + \cdots + W_n$$

Example 2.55

Decide whether work done is positive, negative, or zero:

A. The force on an object increases its velocity
B. The force on an object decreases its velocity
C. The force on an object does not change its velocity.

$$Work\ done\ is\ positive$$
$$Work\ done\ is\ negative$$
$$Work\ done\ is\ zero$$

2.56: Work

Work done also equals change in energy, not just kinetic energy.

$$Total\ Energy = Kinetic\ Energy + Potential\ Energy$$

2.57: Work done when lifting an object against gravity

$$W = mgh$$

Suppose you move an object h feet higher then where it was. The increase in potential energy of the object comes from:

- The height h the object was moved up. The greater the height, the more the energy.
- The mass of the object. The greater the mass, the more the energy
- The resistance provided by gravity. Earth has higher gravity, compared to the moon, for instance.

Example 2.58

Work done by gravity in left/right movement

Example 2.59

Resolution into components for dragging a trolley, and then ignoring the vertical component

2.60: Work in 3D

$$W = \vec{\boldsymbol{F}} \cdot \vec{\boldsymbol{s}} = Fs\cos\theta$$

- We have already seen that a force perpendicular to an object does zero work.
- Hence, for a two-dimensional force, we only want the component of the force along the direction of movement.

Example 2.61

A particle is acted upon by constant forces $4\hat{\boldsymbol{\imath}}+\hat{\boldsymbol{\jmath}}-3\hat{\boldsymbol{k}}$ and $3\hat{\boldsymbol{\imath}}+\hat{\boldsymbol{\jmath}}-\hat{\boldsymbol{k}}$ which displace it from a point $\hat{\boldsymbol{\imath}}+2\hat{\boldsymbol{\jmath}}+3\hat{\boldsymbol{k}}$ to the point $5\hat{\boldsymbol{\imath}}+4\hat{\boldsymbol{\jmath}}+\hat{\boldsymbol{k}}$. The work done in standard units by the forces is given by: (JEE Main 2004)

Calculate:

$$Force=\vec{\boldsymbol{F}}=(4,1,-3)+(3,1,-1)=(7,2,-4)$$
$$Displacement=\vec{\boldsymbol{s}}=\underbrace{(5,4,1)}_{\substack{End\\Point}}-\underbrace{(1,2,3)}_{\substack{Initial\\Point}}=(4,2,-2)$$

Then, using the definition of work:

$$Work=\vec{\boldsymbol{F}}\cdot\vec{\boldsymbol{s}}=(7,2,-4)\cdot(4,2,-2)=28+4+8=40\ units$$

C. Dot Product: Angle Definition

2.62: Dot Product

Given two vectors, $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$ their dot product is:

$$\vec{\boldsymbol{a}}\cdot\vec{\boldsymbol{b}}=|\vec{\boldsymbol{a}}||\vec{\boldsymbol{b}}|\cos\theta$$

Where

$$\theta\ is\ the\ smaller\ angle\ between\ the\ vectors$$

- The input for the dot product is two vectors.
- The output for the dot product is a scalar.

Example 2.63

Find the dot product for each part below:

A. $\vec{\boldsymbol{a}}$ has a length of 3 units, $\vec{\boldsymbol{b}}$ has a length of 5 units. The angle between them is 120°.
B. $\vec{\boldsymbol{a}}$ has a length of 7 units, $\vec{\boldsymbol{b}}$ has a length of 2 units. The angle between them is 45°.
C. $|\vec{\boldsymbol{a}}|=\sqrt{3},|\vec{\boldsymbol{b}}|=2$ and angle between $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$ is 60. (CBSE 2011)

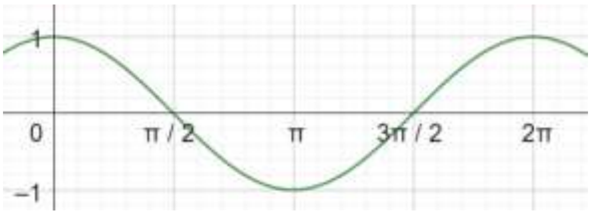
$$\vec{\boldsymbol{a}}\cdot\vec{\boldsymbol{b}}=|\vec{\boldsymbol{a}}||\vec{\boldsymbol{b}}|\cos\theta=(3)(5)\left(-\frac{1}{2}\right)=-\frac{15}{2}$$
$$\vec{\boldsymbol{a}}\cdot\vec{\boldsymbol{b}}=|\vec{\boldsymbol{a}}||\vec{\boldsymbol{b}}|\cos\theta=(7)(2)\left(\frac{\sqrt{2}}{2}\right)=7\sqrt{2}$$
$$\vec{\boldsymbol{a}}\cdot\vec{\boldsymbol{b}}=|\vec{\boldsymbol{a}}||\vec{\boldsymbol{b}}|\cos\theta=(\sqrt{3})(2)\left(\frac{1}{2}\right)=\sqrt{3}$$

Example 2.64

Mark all correct options

The dot product of two vectors is 12. The magnitude of one of the vectors is 6. The magnitude of the other vector is 3. Then, the angle between the two vectors, in degrees, is between:

A. 0 *to* 30
B. 30 *to* 60
C. 60 *to* 90
D. 90 *to* 180
E. 270 *to* 360
F. Cannot be determined



$$|\vec{a}||\vec{b}|\cos\theta = 12$$
$$(6)(3)\cos\theta = 12$$
$$\cos\theta = \frac{2}{3} = 0.66 \Rightarrow In\ Quadrant\ I$$
$$\cos 60 = \frac{1}{2} = 0.5, \qquad \cos 30 = \frac{\sqrt{3}}{2} \approx \frac{1.71}{2} = 0.855 \Rightarrow Option\ B$$

Example 2.65

Find the dot product for each part below:

A. The magnitude of $\vec{a}$ is 10, the magnitude of $\vec{b}$ is 7, and the larger angle between them is 225°.
B. $\vec{a} = (3,0), \vec{b} = (2,2)$
C. $\vec{a} = (4,4), \vec{b} = (\quad)$

Part A

$$\theta = 360 - 225 = 135°$$
$$\cos 135° = \cos(180 - 45°) = -\cos 45° = -\frac{\sqrt{2}}{2}$$
$$\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta = (10)(7)\left(\cos -\frac{\sqrt{2}}{2}\right) = -35\sqrt{2}$$

Part B

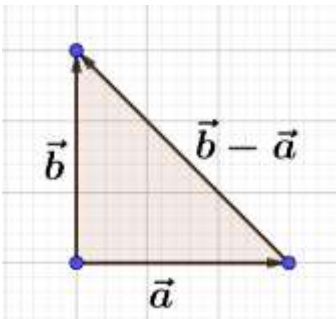
$$\theta = 45° \Rightarrow \cos\theta = \frac{\sqrt{2}}{2}$$
$$\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta = 3(2\sqrt{2})\frac{\sqrt{2}}{2} = 6$$

Example 2.66

A. $\vec{a}$ and $\vec{b}$ are vectors each with a magnitude of $2\sqrt{2}$, and $\vec{b} - \vec{a}$ has a magnitude of 4. Find the dot product of the two vectors.
B. Find the dot product of vectors $\vec{a}$ and $\vec{b}$ given that $\vec{a}$ is a unit vector, $\vec{b}$ has a magnitude of 2 units, and $\vec{b} - \vec{a}$ has a magnitude of $\sqrt{3}$.

Part A

Draw a diagram. Note that the difference of the vectors gives the third side of a triangle (if they form one).



$$(2\sqrt{2})^2 = 4 \times 2 = 8$$
$$2\sqrt{2} \times \sqrt{2} = 4$$

Since the magnitude of the vectors $\vec{a}$ and $\vec{b}$ when multiplied by $\sqrt{2}$ gives the magnitude of the third side of the triangle formed by them, the triangle is a $45 - 45 - 90$ triangle.

Hence, the angle between $\vec{a}$ and $\vec{b}$ is 90°:

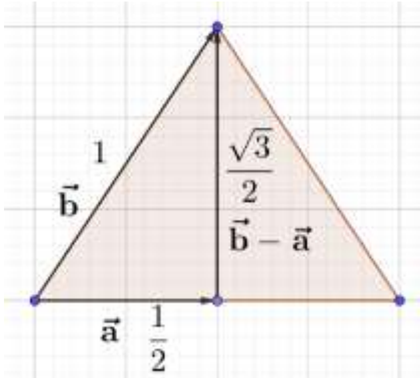
$$Dot\ product = 0$$

Part B

We know that the vectors $\vec{a}, \vec{b}, \vec{b}-\vec{a}$ form a triangle**.** The ratios of the magnitudes are:

$$|\boldsymbol{a}|:\left|\vec{b}-\vec{a}\right|:\left|\vec{b}\right| = 1:\sqrt{3}:2 = \frac{1}{2}:\frac{\sqrt{3}}{2}:1$$

But, these are the ratios of a $30-60-90$ *triangle* (see diagram). Hence:
Therefore, the angle between the two vectors is 60°.



Dot product:

$$= |\vec{a}|\left|\vec{b}\right|\cos\theta = (1)(2)\cos 60° = 2\times\frac{1}{2} = 1$$

Example 2.67

Find the magnitude of each of the two vectors $\vec{a}$ and $\vec{b}$, having the same magnitude such that the angle between them is 60, and their scalar product is $\frac{9}{2}$. **(CBSE 2018)**

$$\vec{a}\cdot\vec{b} = |\vec{a}|\left|\vec{b}\right|\cos\theta$$

$$|\vec{a}||\vec{a}|\cos 60° = \frac{9}{2}$$
$$\frac{1}{2}|\vec{a}|^2 = \frac{9}{2}$$
$$|\vec{a}|^2 = 9$$
$$|\vec{a}| = 3$$

Example 2.68

Given two vectors $\vec{a}$ and $\vec{b}$, determine

A. the maximum and the minimum value of their dot product as θ varies. Answer in terms of $\vec{a}, \vec{b}$.
B. the angle θ, if the dot product of the two vectors has the maximum possible value? the minimum possible value.

Part A

The dot product is given by:

$$\vec{a}\cdot\vec{b} = |\vec{a}|\left|\vec{b}\right|\cos\theta$$

We know that:

$$-1 \le \cos\theta \le 1$$

The dot product will be maximum when $\cos\theta = 1$:

$$\vec{a}\cdot\vec{b} = |\vec{a}|\left|\vec{b}\right|\cos\theta = |\vec{a}|\left|\vec{b}\right|$$

The dot product will be minimum when $\cos\theta = -1$:

$$\vec{a}\cdot\vec{b} = |\vec{a}|\left|\vec{b}\right|\cos\theta = -|\vec{a}|\left|\vec{b}\right|$$

Part B

Maximum:

$$\cos\theta = 1 \Rightarrow \theta = 0$$

Minimum:

$$\cos\theta = -1 \Rightarrow \theta = \pi$$

Example 2.69

Determine whether the following is possible. If not possible, explain why.

A. $\vec{a}$ and $\vec{b}$ are position vectors such that $\vec{a} \cdot \vec{b}$ is nonnegative, the tip of $\vec{a}$ lies in Quadrant II, and the tip of $\vec{b}$ lies in Quadrant IV.

Since $\vec{a}$ lies in Quadrant II, and $\vec{b}$ lies in Quadrant IV

$$90 < \theta \le 180 \Rightarrow \cos\theta < 0 \Rightarrow Dot\ Product < 0 \Rightarrow Contradiction \Rightarrow Not\ possible$$

Example 2.70

The dot product of two vectors is 12. The magnitude of one of the vectors is $\sqrt{2}$. Find the magnitude of the other vector if the angle between the vectors is 135°.

Method I

The angle between the two vectors is obtuse

$$\theta = 135° \Rightarrow \cos\theta < 0 \Rightarrow Dot\ Product < 0 \Rightarrow Contradiction \Rightarrow No\ such\ vector$$

Method II

$$|\vec{a}||\vec{b}|\cos\theta = 12$$
$$\sqrt{2}|\vec{b}|\left(-\frac{\sqrt{2}}{2}\right) = 12$$
$$|\vec{b}| = -12$$

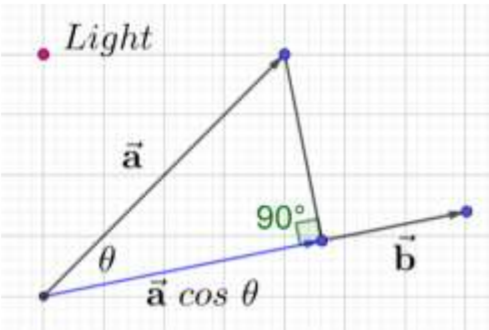
Since magnitude is never negative, there is no such vector.

D. Projection

2.71: Restating the projection definition

The projection of $\vec{a}$ along $\vec{b}$ is

$$\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|} = |\vec{a}|\cos\theta$$



$$LHS = \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|} = \frac{|\vec{a}||\vec{b}|\cos\theta}{|\vec{b}|} = |\vec{a}|\cos\theta = RHS$$

E. Perpendicularity

2.72: Dot Product of Zero Vector

Suppose $\vec{a}$ is a zero vector.

$$\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta = (0)(|\vec{b}|\cos\theta) = 0$$

Example 2.73

If $\vec{a}\cdot\vec{a} = 0$ and $\vec{a}\cdot\vec{b} = 0$, then what can be concluded about the vector $\vec{b}$? (CBSE 2011)

$$\vec{a}\cdot\vec{a} = 0$$
$$|\vec{a}||\vec{a}|\cos 0 = 0$$
$$|\vec{a}||\vec{a}| = 0$$

$$|\vec{\boldsymbol{a}}| = 0$$

$$(0)\left(\left|\vec{\boldsymbol{b}}\right|\cos\theta\right) = 0$$

Nothing can be concluded about $\vec{\boldsymbol{b}}$.

2.74: Dot product of standard basis vectors

Given that $\hat{\boldsymbol{\imath}} = (1,0,0), \hat{\boldsymbol{\jmath}} = (0,1,0), \hat{\boldsymbol{k}} = (0,0,1)$

$$\hat{\boldsymbol{\imath}}\cdot\hat{\boldsymbol{\jmath}} = \hat{\boldsymbol{\jmath}}\cdot\hat{\boldsymbol{k}} = \hat{\boldsymbol{k}}\cdot\hat{\boldsymbol{\imath}} = \boldsymbol{0}$$
$$\hat{\boldsymbol{\jmath}}\cdot\hat{\boldsymbol{\imath}} = \hat{\boldsymbol{k}}\cdot\hat{\boldsymbol{\jmath}} = \hat{\boldsymbol{\imath}}\cdot\hat{\boldsymbol{k}} = \boldsymbol{0}$$

Recall that $\hat{\boldsymbol{\imath}}, \hat{\boldsymbol{\jmath}}$ and $\hat{\boldsymbol{k}}$ are the unit vectors in the x, y and z direction respectively.

$$\hat{\boldsymbol{\imath}}\cdot\hat{\boldsymbol{\jmath}} = |\hat{\boldsymbol{\imath}}||\hat{\boldsymbol{\jmath}}|\cos\theta = |\hat{\boldsymbol{\imath}}||\hat{\boldsymbol{\jmath}}|\cos\left(\frac{\pi}{2}\right) = |\hat{\boldsymbol{\imath}}||\hat{\boldsymbol{\jmath}}|(0) = 0$$

2.75: Associative Property (with two vectors and a scalar)

$$(k\vec{\boldsymbol{a}})\cdot\vec{\boldsymbol{b}} = k\left(\vec{\boldsymbol{a}}\cdot\vec{\boldsymbol{b}}\right) = \vec{\boldsymbol{a}}\cdot\left(k\vec{\boldsymbol{b}}\right)$$

Example 2.76

Given that $\vec{\boldsymbol{a}} = 5\hat{\boldsymbol{\imath}}$ and $\vec{\boldsymbol{b}} = 7\hat{\boldsymbol{\jmath}}$ and $\vec{\boldsymbol{c}} = \boldsymbol{4}\hat{\boldsymbol{k}}$, find $\vec{\boldsymbol{a}}\cdot\vec{\boldsymbol{b}} + \vec{\boldsymbol{b}}\cdot\vec{\boldsymbol{c}} + \vec{\boldsymbol{c}}\cdot\vec{\boldsymbol{a}}$

Note that there are three terms in the expression $\vec{\boldsymbol{a}}\cdot\vec{\boldsymbol{b}} + \vec{\boldsymbol{b}}\cdot\vec{\boldsymbol{c}} + \vec{\boldsymbol{c}}\cdot\vec{\boldsymbol{a}}$ and each term requires find the dot product of two standard basis vectors which are not in the same direction.

For example:

$$\vec{\boldsymbol{a}}\cdot\vec{\boldsymbol{b}} = 5\hat{\boldsymbol{\imath}}\cdot 7\hat{\boldsymbol{\jmath}} = 35\hat{\boldsymbol{\imath}}\cdot\hat{\boldsymbol{\jmath}} = (35)(0) = 0$$

Similarly, the other terms are also zero, and the final answer is:

$$0 + 0 + 0 = 0$$

2.77: Perpendicular Vectors

Given two vectors, $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$ their dot product $\vec{\boldsymbol{a}}\cdot\vec{\boldsymbol{b}}$ is zero *if and only if* one or more of the three conditions below holds:

- $\vec{\boldsymbol{a}} = \vec{\boldsymbol{0}}$
- $\vec{\boldsymbol{b}} = \vec{\boldsymbol{0}}$
- $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$ are perpendicular to each other

Dot product is given by:

$$\vec{\boldsymbol{a}}\cdot\vec{\boldsymbol{b}} = |\vec{\boldsymbol{a}}|\left|\vec{\boldsymbol{b}}\right|\cos\theta$$

By the zero-product property, the RHS will be zero when at least one of the terms is zero.

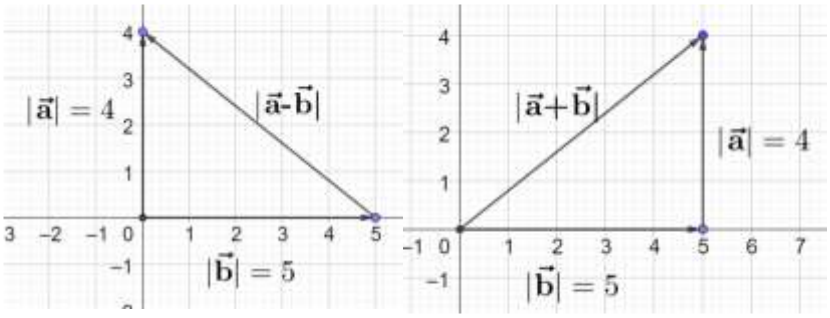
$$\vec{\boldsymbol{a}} = \vec{\boldsymbol{0}}$$
$$\vec{\boldsymbol{b}} = \vec{\boldsymbol{0}}$$
$$\cos\theta = 0 \Rightarrow \theta = \frac{\pi}{2} = 90°$$

Example 2.78

Two vectors have magnitudes 4 and 5 respectively. Their dot product is zero. Find the magnitude of the

A. Difference of the two vectors.
B. Sum of the two vectors

$$|\boldsymbol{a}-\boldsymbol{b}| = \sqrt{4^2+5^2} = \sqrt{41}$$
$$|\boldsymbol{a}+\boldsymbol{b}| = \sqrt{4^2+5^2} = \sqrt{41}$$



2.4 Properties

A. Angle between Vectors

2.79: Angle between Vectors

$$\vec{\boldsymbol{a}}\cdot\vec{\boldsymbol{b}} = |\vec{\boldsymbol{a}}||\vec{\boldsymbol{b}}|\cos\theta \Leftrightarrow \theta = \cos^{-1}\left(\frac{\vec{\boldsymbol{a}}\cdot\vec{\boldsymbol{b}}}{|\vec{\boldsymbol{a}}||\vec{\boldsymbol{b}}|}\right)$$

$$\vec{\boldsymbol{a}}\cdot\vec{\boldsymbol{b}} = |\vec{\boldsymbol{a}}||\vec{\boldsymbol{b}}|\cos\theta$$

Divide both sides by $|\vec{\boldsymbol{a}}||\vec{\boldsymbol{b}}|$:

$$\cos\theta = \frac{\vec{\boldsymbol{a}}\cdot\vec{\boldsymbol{b}}}{|\vec{\boldsymbol{a}}||\vec{\boldsymbol{b}}|}$$

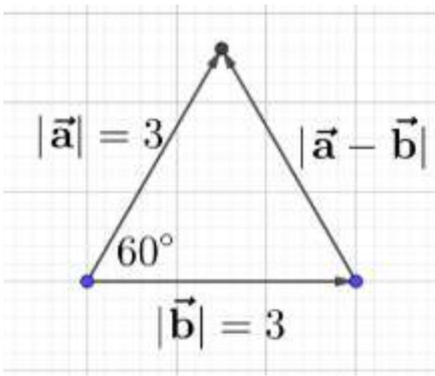
Take the cos inverse of both sides:

$$\theta = \cos^{-1}\left(\frac{\vec{\boldsymbol{a}}\cdot\vec{\boldsymbol{b}}}{|\vec{\boldsymbol{a}}||\vec{\boldsymbol{b}}|}\right)$$

Example 2.80

The dot product of two vectors, both with magnitude 3, is $\frac{9}{2}$. Find the length of the difference of the two vectors.

$$|\vec{\boldsymbol{a}}||\vec{\boldsymbol{b}}|\cos\theta = \frac{9}{2} \Rightarrow (3)(3)\cos\theta = \frac{9}{2} \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = 60°$$



$|\vec{\boldsymbol{a}}-\vec{\boldsymbol{b}}|$ is the third side of the triangle, and since the triangle is equilateral, the

$$Length\ of\ third\ side = 3$$

2.81: Dot product of standard basis vectors with themselves

$$\hat{\boldsymbol{\imath}}\cdot\hat{\boldsymbol{\imath}} = \hat{\boldsymbol{\jmath}}\cdot\hat{\boldsymbol{\jmath}} = \hat{\boldsymbol{k}}\cdot\hat{\boldsymbol{k}} = \boldsymbol{1}$$

$$\hat{\boldsymbol{\imath}}\cdot\hat{\boldsymbol{\imath}} = |\vec{\imath}||\vec{\imath}|\cos 0 = (1)(1)(1) = 1$$

Example 2.82

Let V be the set of standard basis vectors.

A. Write V
B. W is the Cartesian product of the set V with itself. Write W. Find the cardinality of W.
C. Find the sum of the dot product of all elements of W.
D. X is the set of elements formed by evaluating the dot product of each pair in W. Write X.

Hint: The Cartesian product of two sets is the set of all ordered pairs such that the first element belongs to the first set, and the second element belongs to the second set.

Part A

$$V = \{\hat{\boldsymbol{\imath}}, \hat{\boldsymbol{\jmath}}, \hat{\boldsymbol{k}}\}$$

Part B

$$W = \{(\hat{\imath},\hat{\imath})(\hat{\imath},\hat{\jmath})(\hat{\imath},\hat{\boldsymbol{k}})(\hat{\jmath},\hat{\imath})(\hat{\jmath},\hat{\jmath})(\hat{\jmath},\hat{\boldsymbol{k}})(\hat{\boldsymbol{k}},\hat{\imath})(\hat{\boldsymbol{k}},\hat{\jmath})(\hat{\boldsymbol{k}},\hat{\boldsymbol{k}})\} \Rightarrow n(W) = 9$$

$$\hat{\boldsymbol{\imath}} \cdot \hat{\boldsymbol{\imath}} = \hat{\boldsymbol{\jmath}} \cdot \hat{\boldsymbol{\jmath}} = \hat{\boldsymbol{k}} \cdot \hat{\boldsymbol{k}} = 1$$

All other elements involve vectors which are perpendicular to each other, and hence their dot product is zero.

Part C

Sum of dot product

$$= 1 + 1 + 1 + 0 + 0 + 0 + 0 + 0 + 0 = 3$$

Part D

$$X = \{0,1\}$$

2.83: Dot Product of Vector with Itself

$$\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{a}} = |\vec{\boldsymbol{a}}|^2$$

$$\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{a}} = |\vec{\boldsymbol{a}}||\vec{\boldsymbol{a}}| \cos\theta$$

But the angle that a vector makes with itself is $\theta = 0$:

$$= |\vec{\boldsymbol{a}}||\vec{\boldsymbol{a}}| \cos 0 = |\vec{\boldsymbol{a}}||\vec{\boldsymbol{a}}| = |\vec{\boldsymbol{a}}|^2$$

Example 2.84

A. Given that $\vec{\boldsymbol{x}} = (3, -4)$, find $\vec{\boldsymbol{x}} \cdot \vec{\boldsymbol{x}}$

$$\vec{\boldsymbol{x}} \cdot \vec{\boldsymbol{x}} = |\vec{\boldsymbol{x}}|^2 = 3^2 + (-4^2) = 9 + 16 = 25$$

2.85: Commutative Property

$$\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}} = \vec{\boldsymbol{b}} \cdot \vec{\boldsymbol{a}}$$

$$\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}} = |\vec{\boldsymbol{a}}||\vec{\boldsymbol{b}}| \cos\theta = |\vec{\boldsymbol{b}}||\vec{\boldsymbol{a}}| \cos\theta = \vec{\boldsymbol{b}} \cdot \vec{\boldsymbol{a}}$$

2.86: Dot Product of Three Vectors

$$\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}} \cdot \vec{\boldsymbol{c}}$$

The dot product of three vectors shown above is not defined.

Let $\vec{\boldsymbol{b}} \cdot \vec{\boldsymbol{c}} = x$:

$$\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}} \cdot \vec{\boldsymbol{c}} = \vec{\boldsymbol{a}} \cdot x = \underbrace{\vec{\boldsymbol{a}}}_{Vector} \cdot \underbrace{x}_{Scalar} \Rightarrow Not\ Defined$$

Example 2.87

You will later learn about the "triple product" of vectors, where three vectors are multiplied. Explain why $\hat{\boldsymbol{\imath}} \cdot \hat{\boldsymbol{\jmath}} \cdot \hat{\boldsymbol{k}}$ is not a valid triple product.

If you take the dot product of any two vectors from $\hat{\boldsymbol{\imath}} \cdot \hat{\boldsymbol{\jmath}} \cdot \hat{\boldsymbol{k}}$, the answer will be a scalar, and a scalar cannot be used an input into a dot product.

2.88: Distributive Property

$$\vec{\boldsymbol{c}} \cdot (\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}}) = \vec{\boldsymbol{c}} \cdot \vec{\boldsymbol{a}} + \vec{\boldsymbol{c}} \cdot \vec{\boldsymbol{b}}$$

Let the angle between:

$$\vec{\boldsymbol{c}}\ and\ \vec{\boldsymbol{a}} = \theta_1, \quad \vec{\boldsymbol{c}}\ and\ \vec{\boldsymbol{b}} = \theta_2, \qquad \vec{\boldsymbol{c}}\ and\ (\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}}) = \theta_2$$

Start with the right-hand side:

$$RHS = \vec{\boldsymbol{c}} \cdot \vec{\boldsymbol{a}} + \vec{\boldsymbol{c}} \cdot \vec{\boldsymbol{b}}$$

Expand using the definition of the dot product:

$$|\vec{\boldsymbol{c}}||\vec{\boldsymbol{a}}| \cos\theta_1 + |\vec{\boldsymbol{c}}||\vec{\boldsymbol{b}}| \cos\theta_2$$

Factor $|\vec{\boldsymbol{c}}|$:

$$|\vec{\boldsymbol{c}}|[|\vec{\boldsymbol{a}}| \cos\theta_1 + |\vec{\boldsymbol{b}}| \cos\theta_2]$$

Note that the quantities inside the brackets are projections. Hence, substitute:

$$|\vec{\boldsymbol{c}}| \left[Proj_{|\vec{\boldsymbol{a}}|\ upon\ \vec{\boldsymbol{c}}} + Proj_{|\vec{\boldsymbol{b}}|\ upon\ \vec{\boldsymbol{c}}}\right]$$

Use the property that $Proj(\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}}) = Proj(\vec{\boldsymbol{a}}) + Proj(\vec{\boldsymbol{b}})$:

$$|\vec{\boldsymbol{c}}| \left[Proj_{|\vec{\boldsymbol{a}}+\vec{\boldsymbol{b}}|\ upon\ \vec{\boldsymbol{c}}}\right]$$

Convert the second term using the definition of projection:

$$|\vec{\boldsymbol{c}}|[|\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}}| \cos\theta_3]$$

But note the above is exactly a dot product, using the definition:

$$= \vec{\boldsymbol{c}} \cdot (\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}}) = LHS$$

Note

- We proved this for two dimensional vectors. However, it is true for vectors of any dimension.

Example 2.89

Two vectors are such that the magnitude of their sum is also the magnitude of their difference. What condition must these vectors meet? Justify.

As per the condition, we know that:

$$|\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}}| = |\vec{\boldsymbol{a}} - \vec{\boldsymbol{b}}|$$

Square both sides:

$$|\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}}|^2 = |\vec{\boldsymbol{a}} - \vec{\boldsymbol{b}}|^2$$

Dot product of a vector with itself is the square of its magnitude

$$(\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}}) \cdot (\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}}) = (\vec{\boldsymbol{a}} - \vec{\boldsymbol{b}}) \cdot (\vec{\boldsymbol{a}} - \vec{\boldsymbol{b}})$$

Using the distributive property of vectors:

$$\vec{a}\cdot\vec{a}+\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{a}+\vec{b}\cdot\vec{b}=\vec{a}\cdot\vec{a}-\vec{a}\cdot\vec{b}-\vec{b}\cdot\vec{a}+\vec{b}\cdot\vec{b}$$

Cancel common terms on both sides:

$$\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{a}=-\vec{a}\cdot\vec{b}-\vec{b}\cdot\vec{a}$$

Using the commutative property of dot product:

$$2(\vec{a}\cdot\vec{b})=-2(\vec{a}\cdot\vec{b})$$
$$\vec{a}\cdot\vec{b}=0$$

If dot product is zero, then the vectors are perpendicular

$$\vec{a}\perp\vec{b}$$

2.90: Binomial Product

Example 2.91

Let $\vec{a}$ and $\vec{a}$ be two unit vectors. If the vectors $\vec{c}=\vec{a}+2\vec{b}$ and $\vec{d}=5\vec{a}-4\vec{b}$ are perpendicular to each other, then the angle between $\vec{a}$ and $\vec{b}$ (in radians) is: (JEE Main 2014)

$$(\vec{a}+2\vec{b})\cdot(5\vec{a}-4\vec{b})=0$$
$$5|\vec{a}|^2+10\vec{a}\cdot\vec{b}-\mathbf{4}\vec{a}\cdot\vec{b}-\mathbf{8}\left|\vec{b}\right|^2=0$$
$$5|\vec{a}|^2+6\vec{a}\cdot\vec{b}-\mathbf{8}\left|\vec{b}\right|^2=0$$

Substitute $|\vec{a}|^2=\left|\vec{b}\right|^2=1$:

$$5+6\vec{a}\cdot\vec{b}-\mathbf{8}=0$$
$$6\vec{a}\cdot\vec{b}=3$$

Use the definition of the dot product:

$$|\vec{a}|\cdot\left|\vec{b}\right|\cos\theta=\frac{1}{2}$$

Substitute $|\vec{a}|^2=\left|\vec{b}\right|^2=1$

$$\cos\theta=\frac{1}{2}\Rightarrow\theta=\frac{\pi}{3}$$

Example 2.92

If $(\vec{a}+3\vec{b})$ is perpendicular to $(7\vec{a}-5\vec{b})$ and $(\vec{a}-4\vec{b})$ is perpendicular to $(7\vec{a}-2\vec{b})$, then the angle between $\vec{a}$ and $\vec{b}$ (in degrees) is: (JEE Main 2021, 25 July, Shift-II)

Use the first perpendicularity condition:

$$(\vec{a}+3\vec{b})\cdot(7\vec{a}-5\vec{b})=0$$
$$7|\vec{a}|^2+21\vec{a}\cdot\vec{b}-5\vec{a}\cdot\vec{b}-15\left|\vec{b}\right|^2=0$$
$$\underbrace{7|\vec{a}|^2+16\vec{a}\cdot\vec{b}-15\left|\vec{b}\right|^2=0}_{\textit{Equation I}}$$

Use the second perpendicularity condition:

$$(\vec{a} - 4\vec{b}) \cdot (7\vec{a} - 2\vec{b}) = 0$$
$$7|\vec{a}|^2 - 2\vec{a} \cdot \vec{b} - 28\vec{a} \cdot \vec{b} + 8|\vec{b}|^2 = 0$$
$$\underbrace{7|\vec{a}|^2 - 30\vec{a} \cdot \vec{b} + 8|\vec{b}|^2 = 0}_{\textit{Equation II}}$$

There are more than two variables here. So, first eliminate the dot product $\vec{a} \cdot \vec{b}$
Multiply Equation I by 15, and Equation II by 8:

$$105|\vec{a}|^2 + 240\vec{a} \cdot \vec{b} - 225|\vec{b}|^2 = 0$$
$$56|\vec{a}|^2 - 240\vec{a} \cdot \vec{b} + 64|\vec{b}|^2 = 0$$

Add the two equations above:

$$161|\vec{a}|^2 - 161|\vec{b}|^2 = 0 \Rightarrow |\vec{a}|^2 = |\vec{b}|^2 \Rightarrow |\vec{a}| = |\vec{b}|$$

Substitute $|\vec{a}| = |\vec{b}|$ into Equation I:

$$7|\vec{a}|^2 + 16\vec{a} \cdot \vec{b} - 15|\vec{a}|^2 = 0 \Rightarrow 16\vec{a} \cdot \vec{b} = 8|\vec{a}|^2 \Rightarrow 2\vec{a} \cdot \vec{b} = |\vec{a}|^2$$

Use the definition of the dot product:

$$2|\vec{a}| \cdot |\vec{b}| \cos\theta = |\vec{a}|^2$$

Substitute $|\vec{a}| = |\vec{b}|$ into the above:

$$2|\vec{a}| \cdot |\vec{a}| \cos\theta = |\vec{a}|^2 \Rightarrow 2\cos\theta = 1 \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = 60°$$

B. Square of a Sum

2.93: Square of a Sum

Just as know $(a+b)^2 = a^2 + 2ab + b^2$, we have a similar property in vectors:

$$|\vec{a} + \vec{b}|^2 = (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) = |\vec{a}|^2 + 2\vec{a} \cdot \vec{b} + |\vec{b}|^2$$

Dot product of a vector with itself is the square of its magnitude. That is $\vec{x} \cdot \vec{x} = |\vec{x}|^2$. Apply the property to the LHS of the above:

$$\boldsymbol{LHS} = |\vec{a} + \vec{b}|^2 = (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b})$$

Using the distributive property of vectors:

$$\vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b}$$

Substitute $\vec{a} \cdot \vec{a} = |\vec{a}|^2$:

$$|\vec{a}|^2 + \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{a} + |\vec{b}|^2$$

Using the commutative property of dot product:

$$|\vec{a}|^2 + 2\vec{a} \cdot \vec{b} + |\vec{b}|^2$$

Example 2.94

If $\vec{a}$ and $\vec{b}$ are perpendicular vectors, $|\vec{a} + \vec{b}| = 13$ and $|\vec{a}| = 5$, then find the value of $|\vec{b}|$. (CBSE 2014, ISC 2019)

Square both sides of $|\vec{a} + \vec{b}| = 13$:

$$|\vec{a} + \vec{b}|^2 = 169$$

Use the property that $\left|\vec{a}+\vec{b}\right|^2 = |\vec{a}|^2 + 2\vec{a}\cdot\vec{b} + \left|\vec{b}\right|^2$:

$$|\vec{a}|^2 + 2\vec{a}\cdot\vec{b} + \left|\vec{b}\right|^2 = 169$$

Substitute $|\vec{a}|^2 = 5^2 = 25, \vec{a}\cdot\vec{b} = 0$:

$$25 + \left|\vec{b}\right|^2 = 169$$
$$\left|\vec{b}\right|^2 = 144$$
$$\left|\vec{b}\right| = 12$$

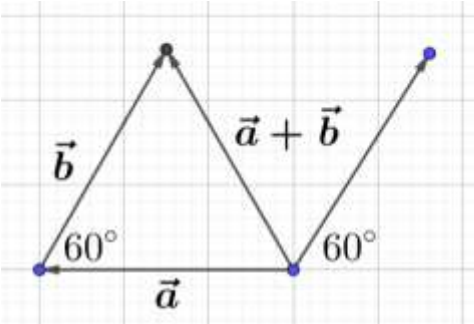
Example 2.95

If $\vec{a}$ and $\vec{b}$ are two unit vectors such that $\left|\vec{a}+\vec{b}\right|$ is also a unit vector, then find the angle between $\vec{a}$ and $\vec{b}$. (CBSE 2014)

Geometric Method

We get an equilateral triangle, as shown alongside.
To find the angle between the vectors, we arrange the vectors tail to tail, getting an angle of

$$180 - 60 = 120°$$



Algebraic Method

Square both sides of $\left|\vec{a}+\vec{b}\right| = 1$:

$$|\vec{a}|^2 + 2\vec{a}\cdot\vec{b} + \left|\vec{b}\right|^2 = 1$$
$$1 + 2\vec{a}\cdot\vec{b} + 1 = 1$$
$$\vec{a}\cdot\vec{b} = -\frac{1}{2}$$
$$|\vec{a}|\left|\vec{b}\right|\cos\theta = -\frac{1}{2}$$
$$\cos\theta = -\frac{1}{2}$$
$$\theta = 120°$$

Example 2.96

If $\vec{a}$ and $\vec{b}$ are two vectors such that $\left|\vec{a}+\vec{b}\right| = |\vec{a}|$, then prove that vector $2\vec{a}+\vec{b}$ is perpendicular to vector $\vec{b}$. (CBSE 2013)

Square both sides of $\left|\vec{a}+\vec{b}\right| = |\vec{a}|$:

$$|\vec{a}|^2 + 2\vec{a}\cdot\vec{b} + \left|\vec{b}\right|^2 = |\vec{a}|^2$$

Subtract $|\vec{a}|^2$ from both sides and substitute $\left|\vec{b}\right|^2 = \vec{b}\cdot\vec{b}$

$$2\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{b} = 0$$
$$\vec{b}\cdot\left(2\vec{a}+\vec{b}\right) = 0$$

Since the dot product is zero, the two vectors are perpendicular.

Example 2.97

If $\vec{a}+\vec{b}+\vec{c} = 0$, and $|\vec{a}| = 5$, $\left|\vec{b}\right| = 6$, and $|\vec{c}| = 9$, then find the angle between $\vec{a}$ and $\vec{b}$. (CBSE 2018)

$$\vec{a}+\vec{b} = -\vec{c}$$

Square both sides of the above:

$$(\vec{a}+\vec{b})\cdot(\vec{a}+\vec{b})=(-\vec{c})\cdot(-\vec{c})$$

Expand:

$$|\vec{a}|^2+2\vec{a}\cdot\vec{b}+|\vec{b}|^2=|\vec{c}|^2$$
$$25+2\vec{a}\cdot\vec{b}+36=81$$
$$2\vec{a}\cdot\vec{b}=20$$
$$\vec{a}\cdot\vec{b}=10$$
$$|\vec{a}||\vec{b}|\cos\theta=10$$
$$(5)(6)\cos\theta=10$$
$$\cos\theta=\frac{1}{3}$$
$$\theta=\cos^{-1}\left(\frac{1}{3}\right)$$

Example 2.98

If $\vec{x}$ and $\vec{y}$ be two non-zero vectors such that $|\vec{x}+\vec{y}|=|\vec{x}|$ and $2\vec{x}+\lambda\vec{y}$ is perpendicular to $\vec{y}$, then the value of λ is: **(JEE Main 2020, 6 Sep, Shift-II)**

Square both sides of $|\vec{x}+\vec{y}|=|\vec{x}|$

$$|\vec{x}|^2+2\vec{x}\cdot\vec{y}+|\vec{y}|^2=|\vec{x}|^2$$
$$\underbrace{2\vec{x}\cdot\vec{y}+|\vec{y}|^2=0}_{\textit{Equation I}}$$

Use the perpendicularity condition:

$$(2\vec{x}+\lambda\vec{y})\cdot(\vec{y})=0$$
$$\underbrace{2\vec{x}\cdot\vec{y}+\lambda|\vec{y}|^2=0}_{\textit{Equation II}}$$

Equate the LHS of Equations I and II:

$$2\vec{x}\cdot\vec{y}+|\vec{y}|^2=2\vec{x}\cdot\vec{y}+\lambda|\vec{y}|^2$$
$$|\vec{y}|^2=\lambda|\vec{y}|^2$$
$$1=\lambda$$

Example 2.99

Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|2\vec{a}+3\vec{b}|=|3\vec{a}+\vec{b}|$ and the angle between $\vec{a}$ and $\vec{b}$ is 60°. If $\frac{1}{8}\vec{a}$ is a unit vector, then $|\vec{b}|$ is equal to: **(JEE Main 2021, 31 Aug, Shift-I)**

$$|2\vec{a}+3\vec{b}|=|3\vec{a}+\vec{b}|$$

Square both sides:

$$4|\vec{a}|^2+12\vec{a}\cdot\vec{b}+9|\vec{b}|^2=9|\vec{a}|^2+6\vec{a}\cdot\vec{b}+|\vec{b}|^2$$

Collate all terms on the RHS:

$$5|\vec{a}|^2-6\vec{a}\cdot\vec{b}-8|\vec{b}|^2=0$$

Note that

$$\left|\frac{1}{8}\vec{a}\right|=1\Rightarrow\frac{1}{8}|\vec{a}|=1\Rightarrow|\vec{a}|=8\Rightarrow|\vec{a}|^2=64\Rightarrow 5|\vec{a}|^2=320$$

$$6\vec{a}\cdot\vec{b} = 6|\vec{a}||\vec{b}|\cos\theta = 6\cdot 8|\vec{b}|\cos 60° = 6\cdot 8|\vec{b}|\left(\frac{1}{2}\right) = 24|\vec{b}|$$

Making the above substitutions:

$$320 - 24|\vec{b}| - \mathbf{8}|\vec{b}|^2 = 0$$
$$|\vec{b}|^2 + 3|\vec{b}| - 40 = 0$$

Use a change of variable. Let $y = |\vec{b}|$:

$$y^2 + 3y - 40 = 0$$
$$(y+8)(y-5) = 0$$
$$y \in \{-8,5\}$$

Reject the negative value:

$$y = |\vec{b}| = 5$$

C. Square of a Difference

2.100: Square of a Difference

Similarly:

$$|\vec{a}-\vec{b}|^2 = (\vec{a}-\vec{b})\cdot(\vec{a}-\vec{b}) = |\vec{a}|^2 - 2\vec{a}\cdot\vec{b} + |\vec{b}|^2$$

Dot product of a vector with itself is the square of its magnitude. That is $\vec{x}\cdot\vec{x} = |\vec{x}|^2$. Apply the property to the LHS of the above:

$$\boldsymbol{LHS} = |\vec{a}-\vec{b}|^2 = (\vec{a}-\vec{b})\cdot(\vec{a}-\vec{b})$$

Using the distributive property of vectors:

$$\vec{a}\cdot\vec{a} - \vec{a}\cdot\vec{b} - \vec{b}\cdot\vec{a} + \vec{b}\cdot\vec{b}$$

Cancel common terms on both sides:

$$|\vec{a}|^2 - \vec{a}\cdot\vec{b} - \vec{b}\cdot\vec{a} + |\vec{b}|^2$$

Using the commutative property of dot product:

$$|\vec{a}|^2 - 2\vec{a}\cdot\vec{b} + |\vec{b}|^2$$

Example 2.101

If $\vec{a}$ and $\vec{b}$ are unit vectors, then find the angle between $\vec{a}$ and $\vec{b}$ given that $(\sqrt{3}\vec{a}-\vec{b})$ is a unit vector.

$$|\sqrt{3}\vec{a}-\vec{b}| = 1$$

Square both sides:

$$|\sqrt{3}\vec{a}-\vec{b}| = 1$$

Use the property that $|\vec{a}-\vec{b}|^2 = |\vec{a}|^2 - 2\vec{a}\cdot\vec{b} + |\vec{b}|^2$:

$$(\sqrt{3})^2|(\vec{a})|^2 - (2)(\sqrt{3}\vec{a})\cdot(\vec{b}) + |\vec{b}|^2 = 1$$

Substitute $|\vec{a}| = |\vec{b}| = 1$:

$$3 - (2\sqrt{3})(\vec{a}\cdot\vec{b}) + 1 = 1$$

Isolate the dot product:

$$(2\sqrt{3})(\vec{a}\cdot\vec{b}) = 3$$

$$\vec{a}\cdot\vec{b}=\frac{\sqrt{3}}{2}$$
$$|\vec{a}||\vec{b}|\cos\theta=\frac{\sqrt{3}}{2}$$
$$\cos\theta=\frac{\sqrt{3}}{2}$$
$$\theta=\cos^{-1}\frac{\sqrt{3}}{2}=\frac{\pi}{6}$$

Example 2.102

If $\vec{a}$ and $\vec{b}$ are unit vectors, then find the maximum and minimum value of $\vec{a}\cdot\vec{b}$:

A. Using the angle definition of the dot product
B. Using the property $|\vec{a}\pm\vec{b}|^2=|\vec{a}|^2\pm 2\vec{a}\cdot\vec{b}+|\vec{b}|^2$

Part A

$$\vec{a}\cdot\vec{b}=|\boldsymbol{a}||\boldsymbol{b}|\cos\theta=(1)(1)(\cos\theta)=\cos\theta$$
$$Max\ =1, Min=-1$$

Part B

Since the magnitude of a vector is never negative, we must have:

$$|\vec{a}\pm\vec{b}|\geq 0$$

Square both sides:

$$|\vec{a}\pm\vec{b}|^2\geq 0$$

Use the property $|\vec{a}\pm\vec{b}|^2=|\vec{a}|^2\pm 2\vec{a}\cdot\vec{b}+|\vec{b}|^2$:

$$|\vec{a}|^2\pm 2\vec{a}\cdot\vec{b}+|\vec{b}|^2\geq 0$$

Substitute $|\vec{a}|^2=|\vec{b}|^2=1$:

$$1\pm 2\vec{a}\cdot\vec{b}+1\geq 0$$
$$\pm 2\vec{a}\cdot\vec{b}\geq -2$$

Case I:

$$2\vec{a}\cdot\vec{b}\geq -2$$
$$\vec{a}\cdot\vec{b}\geq -1$$

Case II:

$$-2\vec{a}\cdot\vec{b}\geq -2$$

Divide by -2 both sides and flip the sign of the inequality

$$\vec{a}\cdot\vec{b}\leq 1$$

We get the same answer from Part A and from Part B:

$$\vec{a}\cdot\vec{b}\in[-1,1]$$

D. Square of a Difference

2.103: Square of a Difference

$$(\vec{a}+\vec{b})\cdot(\vec{a}-\vec{b})=|\boldsymbol{a}|^2-|\boldsymbol{b}|^2$$

Example 2.104

Find $|\vec{x}|$ if for a unit vector $\hat{a}$, $(\vec{x}-\vec{a})\cdot(\vec{x}+\vec{a})=15$. (CBSE 2013)

$$|\vec{x}|^2-|\vec{a}|^2=15$$
$$|\vec{x}|^2-1=15$$
$$|\vec{x}|^2=16$$
$$|\vec{x}|=4$$

Example 2.105

Two vectors $\vec{A}$ and $\vec{B}$ are such that $|\vec{A}+\vec{B}|=|\vec{A}-\vec{B}|$. The angle between the two vectors will be (BITSAT 2011)

Square both sides

$$|\vec{a}|^2 + 2\vec{a}\cdot\vec{b} + |\vec{b}|^2 = |\vec{a}|^2 - 2\vec{a}\cdot\vec{b} + |\vec{b}|^2$$
$$2\vec{a}\cdot\vec{b} = -2\vec{a}\cdot\vec{b}$$
$$4\vec{a}\cdot\vec{b} = 0$$
$$\vec{a}\cdot\vec{b} = 0$$
$$\vec{a}\perp\vec{b}$$
$$Angle\ between\ two\ vectors = 90°$$

Example 2.106

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three unit vectors such that $|\vec{a}-\vec{b}|^2 + |\vec{a}-\vec{c}|^2 = 8$. Then $|\vec{a}+2\vec{b}|^2 + |\vec{a}+2\vec{c}|^2$ is equal to
(JEE Main, 2020, 2 Sep, Shift-I)

Expand $|\vec{a}-\vec{b}|^2 + |\vec{a}-\vec{c}|^2 = 8$ to get:
$$|\vec{a}|^2 - 2\vec{a}\cdot\vec{b} + |\vec{b}|^2 + |\vec{a}|^2 - 2\vec{a}\cdot\vec{c} + |\vec{c}|^2 = 8$$
Substitute $|\vec{a}|^2 = |\vec{b}|^2 = |\vec{c}|^2 = 1$:
$$-2\vec{a}\cdot\vec{b} - 2\vec{a}\cdot\vec{c} = 4$$
$$\vec{a}\cdot\vec{b} + \vec{a}\cdot\vec{c} = -2$$

Expand $|\vec{a}+2\vec{b}|^2 + |\vec{a}+2\vec{c}|^2$ to get:
$$|\vec{a}|^2 + 4\vec{a}\cdot\vec{b} + 4|\vec{b}|^2 + |\vec{a}|^2 + 4\vec{a}\cdot\vec{c} + 4|\vec{c}|^2$$
Substitute $|\vec{a}|^2 = |\vec{b}|^2 = |\vec{c}|^2 = 1$:
$$10 + 4(\vec{a}\cdot\vec{b} + 4\vec{a}\cdot\vec{c})$$
Substitute $\vec{a}\cdot\vec{b} + \vec{a}\cdot\vec{c} = -2$:
$$10 + 4(-2) = 10 - 8 = 2$$

E. Square of Three Terms

2.107: Square of a sum of three terms

$$|\vec{a}+\vec{b}+\vec{c}|^2 = (\vec{a}+\vec{b}+\vec{c})\cdot(\vec{a}+\vec{b}+\vec{c}) = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a})$$

Applying the property $(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$ to the LHS of the given property immediately gives us the RHS.

Example 2.108

If $\vec{a}, \vec{b}, \vec{c}$ are three unit vectors such that $\vec{a}+\vec{b}+\vec{c} = \vec{0}$, where $\vec{0}$ is the null vector, then $\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a}$ is
(BITSAT 2013)

$$\vec{a}+\vec{b}+\vec{c} = \vec{0}$$
Square both sides:
$$|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a}) = 0$$

Substitute $|\vec{a}|^2 = |\vec{b}|^2 = |\vec{c}|^2 = 1$:

$$1 + 1 + 1 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0$$

$$\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -\frac{3}{2}$$

Example 2.109

If $\vec{a}, \vec{b},$ and $\vec{c}$ are unit vectors, then $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2$ does not exceed (JEE Advanced, 2001-Screening)

Expand the given expression:

$$\underbrace{|\vec{a}|^2 - 2\vec{a} \cdot \vec{b} + |\vec{b}|^2}_{|\vec{a}-\vec{b}|^2} + \underbrace{|\vec{b}|^2 - 2\vec{b} \cdot \vec{c} + |\vec{c}|^2}_{|\vec{b}-\vec{c}|^2} + \underbrace{|\vec{c}|^2 - 2\vec{c} \cdot \vec{a} + |\vec{a}|^2}_{|\vec{c}-\vec{a}|^2}$$

Simplify:

$$2\left(|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2\right) - 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$$

Substitute $|\vec{a}|^2 = |\vec{b}|^2 = |\vec{c}|^2 = 1$:

$$\underbrace{6 - 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})}_{Expression\ I}$$

To find the maximum value of the above, square both sides of $|\vec{a} + \vec{b} + \vec{c}| \geq 0$:

$$|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \geq 0$$

Substitute $|\vec{a}|^2 = |\vec{b}|^2 = |\vec{c}|^2 = 1$:

$$3 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \geq 0$$

Subtract three from both sides:

$$2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \geq -3$$

Multiply by -1 both sides and flip the inequality:

$$-2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \leq 3$$

Add six to both sides:

$$\underbrace{6 - 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})}_{Expression\ I} \leq 9$$

Since Expression I was what we wanted to find the maximum value of, its maximum value is:

$$9$$

2.110: Law of Cosines

We earlier established the law of cosines using trigonometry. We can arrive at the same result using the dot product.

2.111: Triangle Inequality (Revisited)

2.5 Components

A. Component Definition

2.112: Component

Given two vectors $\vec{\boldsymbol{a}} = (x_1, y_1, z_1), \vec{\boldsymbol{b}} = (x_2, y_2, z_2)$, the dot product of the vectors is given by:

$$\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}} = (x_1, y_1, z_1)\begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} = x_1x_2 + y_1y_2 + z_1z_2$$

Two-dimensional version

$$\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}}$$

Write each vector in component form:

$$(x_1\hat{\boldsymbol{\imath}} + y_1\hat{\boldsymbol{\jmath}}) \cdot (x_2\hat{\boldsymbol{\imath}} + y_2\hat{\boldsymbol{\jmath}})$$

Use the distributive property:

$$x_1x_2\hat{\boldsymbol{\imath}} \cdot \hat{\boldsymbol{\imath}} + y_1y_2\hat{\boldsymbol{\jmath}} \cdot \hat{\boldsymbol{\jmath}} + x_1y_2\hat{\boldsymbol{\imath}} \cdot \hat{\boldsymbol{\jmath}} + y_1x_2\hat{\boldsymbol{\jmath}} \cdot \hat{\boldsymbol{\imath}}$$

Substitute $\hat{\imath} \cdot \hat{\imath} = \hat{\jmath} \cdot \hat{\jmath} = 1$ and $\hat{\imath} \cdot \hat{\jmath} = \hat{\jmath} \cdot \hat{\imath} = 0$

$$x_1x_2(1) + y_1y_2(1) + x_1y_2(0) + y_1x_2(0)$$

Simplify:

$$x_1x_2 + y_1y_2$$

Three-dimensional version

$$\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}}$$

Write each vector in component form:

$$\left(x_1\hat{\boldsymbol{\imath}} + y_1\hat{\boldsymbol{\jmath}} + z_1\hat{\boldsymbol{k}}\right) \cdot \left(x_2\hat{\boldsymbol{\imath}} + y_2\hat{\boldsymbol{\jmath}} + +z_2\hat{\boldsymbol{k}}\right)$$

Use the distributive property to get nine terms. Note that we substitute $\hat{\boldsymbol{\imath}} \cdot \hat{\boldsymbol{\imath}} = \hat{\boldsymbol{\jmath}} \cdot \hat{\boldsymbol{\jmath}} = \hat{\boldsymbol{k}} \cdot \hat{\boldsymbol{k}} = 1$ and every other dot product is zero, we are left with

$$x_1x_2(1) + y_1y_2(1) + z_1z_2(1)$$

Simplify:

$$x_1x_2 + y_1y_2 + z_1z_2$$

Example 2.113

A. Find the dot product of $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$ given that $\vec{\boldsymbol{a}} = (2,5,-2), \vec{\boldsymbol{b}} = \left(-3, \frac{1}{2}, -4\right)$

$$\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}} = (2)(-3) + (5)\left(\frac{1}{2}\right) + (-2)(-4) = -6 + \frac{5}{2} + 8 = 4.5$$

Example 2.114

The scalar product of the vector $\vec{\boldsymbol{a}} = \hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}} + \hat{\boldsymbol{k}}$ with a unit vector along the sum of the vectors $\boldsymbol{b} = 2\hat{\boldsymbol{\imath}} + 4\hat{\boldsymbol{\jmath}} - 5\hat{\boldsymbol{k}}$ and $\boldsymbol{c} = \lambda\hat{\boldsymbol{\imath}} + 2\hat{\boldsymbol{\jmath}} + 3\hat{\boldsymbol{k}}$ is equal to 1. Find the value of λ and hence find the unit vector along $\boldsymbol{b} + \boldsymbol{c}$. (CBSE 2008,2009,2014,2019)

Unit vector in the direction of $\vec{\boldsymbol{b}} + \vec{\boldsymbol{c}}$ is

$$\vec{\boldsymbol{v}} = \frac{\vec{\boldsymbol{b}} + \vec{\boldsymbol{c}}}{\left|\vec{\boldsymbol{b}} + \vec{\boldsymbol{c}}\right|} = \frac{(2+\lambda, 6, -2)}{\sqrt{(2+\lambda)^2 + 6^2 + (-2)^2}} = \frac{(2+\lambda, 6, -2)}{\sqrt{(2+\lambda)^2 + 40}}$$

As per the given condition, $\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{v}} = 1$, which means:

$$(1,1,1) \cdot \frac{(2+\lambda, 6, -2)}{\sqrt{(2+\lambda)^2 + 40}} = 1$$

$$(1,1,1) \cdot (2+\lambda, 6, -2) = \sqrt{(2+\lambda)^2 + 40}$$

$$(1)(2+\lambda) + (1)(6) + (1)(-2) = \sqrt{(2+\lambda)^2 + 40}$$

$$(6+\lambda) = \sqrt{(2+\lambda)^2 + 40}$$

Square both sides:

$$(6+\lambda)^2 = (2+\lambda)^2 + 40$$
$$36 + 12\lambda + \lambda^2 = 4 + 4\lambda + \lambda^2 + 40$$
$$8\lambda = 8$$
$$\lambda = 1$$

$$\vec{\boldsymbol{v}} = \frac{(2+\lambda, 6, -2)}{\sqrt{(2+\lambda)^2 + 40}} = \frac{(3,6,-2)}{\sqrt{9+40}} = \frac{(3,6,-2)}{7} = \left(\frac{3}{7}, \frac{6}{7}, -\frac{2}{7}\right)$$

Example 2.115

Using the component form of the dot product, prove the

A. The commutative property
B. The distributive property
C. That the dot product of a vector with itself is the square of its magnitude

Part A

$$\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}} = \vec{\boldsymbol{b}} \cdot \vec{\boldsymbol{a}}$$
$$LHS = \vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}} = x_1x_2 + y_1y_2 + z_1z_2 = x_2x_1 + y_2y_1 + z_2z_1 = \vec{\boldsymbol{b}} \cdot \vec{\boldsymbol{a}} = RHS$$

Part B

Let $\vec{\boldsymbol{a}} = (x, y, z)$, $\vec{\boldsymbol{b}} = (x_1, y_1, z_1)$, $\vec{\boldsymbol{c}} = (x_2, y_2, z_2)$. Then:

$$\vec{\boldsymbol{a}} \cdot \left(\vec{\boldsymbol{b}} + \vec{\boldsymbol{c}}\right) = (x, y, z) \cdot (x_1 + x_2, y_1 + y_2, z_1 + z_2)$$

Expand

$$= xx_1 + xx_2 + yy_1 + yy_2 + zz_1 + zz_2$$

Rearrange:

$$= \underbrace{xx_1 + yy_1 + zz_1}_{\vec{a}\cdot\vec{b}} + \underbrace{xx_2 + yy_2 + zz_2}_{\vec{a}\cdot\vec{c}}$$
$$= \vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}} + \vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{c}}$$

Part C

Let $\vec{\boldsymbol{a}} = (x, y, z)$. Then:

$$LHS = \vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{a}} = (x, y, z) \cdot (x, y, z) = x^2 + y^2 + z^2 = |\vec{\boldsymbol{a}}|^2$$

$$\vec{\boldsymbol{x}} \cdot \vec{\boldsymbol{x}} = |\vec{\boldsymbol{x}}|^2$$

2.116: Perpendicular Vectors

Given two non-zero vectors, $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$ their dot product $\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}}$ is zero *if and only if* they are perpendicular

Example 2.117

Find a unit vector perpendicular to $\vec{\boldsymbol{v}} = (1, -3)$

Let the vector to be found be $\vec{\boldsymbol{u}} = (x, y)$

$$|\vec{\boldsymbol{u}}| = 1$$
$$\sqrt{x^2 + y^2} = 1$$
$$\underbrace{x^2 + y^2 = 1}_{Equation\ I}$$

Since the two vectors are perpendicular, their dot product is zero:

$$\vec{u}\cdot\vec{v}=0$$
$$(x,y)\cdot(1,-3)=0$$
$$x-3y=0$$
$$x=3y$$

$$(3y)^2+y^2=1$$
$$10y^2=1$$
$$y^2=\frac{1}{10}$$
$$y=\pm\frac{1}{\sqrt{10}}$$
$$x=\pm\frac{1}{\sqrt{10}}$$

$$\vec{u}=(x,y)=\left(\frac{3}{\sqrt{10}},\frac{1}{\sqrt{10}}\right)\ or\ \left(-\frac{3}{\sqrt{10}},\frac{1}{\sqrt{10}}\right)$$

Example 2.118

A. Write the value of λ, so that the vectors $\vec{a}=2\hat{\imath}+\lambda\boldsymbol{j}+\boldsymbol{k}$ and $\boldsymbol{b}=\boldsymbol{i}-2\boldsymbol{j}+2\boldsymbol{k}$ are perpendicular to each other. (CBSE 2008,2013)
B. For what value of λ are the vectors $\boldsymbol{i}+2\lambda\boldsymbol{j}+\boldsymbol{k}$ and $2\boldsymbol{i}+\boldsymbol{j}-3\boldsymbol{k}$ perpendicular. (CBSE 2011)
C. Find the value of λ, if the vectors $2\boldsymbol{i}+\lambda\boldsymbol{j}+3\boldsymbol{k}$ and $3\boldsymbol{i}+2\boldsymbol{j}-4\boldsymbol{k}$ are perpendicular to each other. (CBSE 2010)

Part A

$$(2,\lambda,1)\cdot(1,-2,2)=0$$
$$2-2\lambda+2=0$$
$$4=2\lambda$$
$$\lambda=2$$

Part B

$$(1,2\lambda,1)\cdot(2,1,-3)=0$$
$$2+2\lambda-3=0$$
$$2\lambda=1$$
$$\lambda=\frac{1}{2}$$

Part C

$$(2,\lambda,3)\cdot(3,2,-4)=0$$
$$6+2\lambda-12=0$$
$$2\lambda=6$$
$$\lambda=3$$

Explain why the formula $\vec{a}\cdot\vec{b}=|\vec{a}||\vec{b}|\cos\theta$ is not useful in solving the above questions.

$$\vec{a}\cdot\vec{b}=|\vec{a}||\vec{b}|\cos\theta=0$$
$$|\vec{a}||\vec{b}|(0)=0$$

We cannot conclude anything about either $|\vec{a}|$ or $|\vec{b}|$

Example 2.119

If the vectors $\vec{a}=\hat{\imath}-\hat{\jmath}+2\hat{k},\vec{b}=2\hat{\imath}+4\hat{\jmath}+\hat{k}$, and $\vec{c}=\lambda\hat{\imath}+\hat{\jmath}+\mu\hat{k}$ are mutually orthogonal, then (λ,μ) is equal to: (JEE Main 2010)

$$0=\vec{a}\cdot\vec{c}=(1,-1,2)\cdot(\lambda,1,\mu)=\lambda-1+2\mu=2\lambda-2+4\mu$$
$$0=\vec{b}\cdot\vec{c}=(2,4,1)\cdot(\lambda,1,\mu)=2\lambda+4+\mu$$

$$2\lambda - 2 + 4\mu = 0$$
$$2\lambda + 4 + \mu = 0$$
$$(\lambda, \mu) = (-3,2)$$

B. Angle between Vectors

2.120: Angle between Vectors

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta \Leftrightarrow \theta = \cos^{-1}\left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}\right)$$

Example 2.121

If θ be the angle between vectors $\vec{a} = \hat{\imath} + 2\hat{\jmath} + 3\hat{k}$ and $\vec{b} = 3\boldsymbol{i} + 2\boldsymbol{j} + \boldsymbol{k}$, then $cos\ \theta =$ (BITSAT 2007)

$$\cos\theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \frac{1 \times 3 + 2 \times 2 + 3 \times 1}{\sqrt{1^2 + 2^2 + 3^2}\sqrt{3^2 + 2^2 + 1^2}} = \frac{3 + 4 + 3}{\sqrt{14}\sqrt{14}} = \frac{10}{14} = \frac{5}{7}$$

Example 2.122

Find the angle between the vectors $\vec{A} = \hat{\imath} + \hat{\jmath} - 2\hat{k}$ and $\vec{B} = -\hat{\imath} + 2\hat{\jmath} - \hat{k}$. (BITSAT 2007)

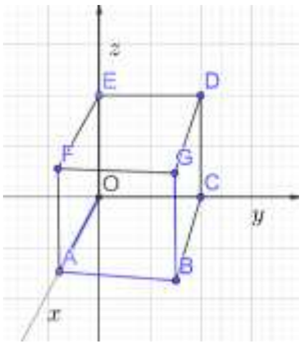
$$\theta = \cos^{-1}\left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}\right) = \cos^{-1}\left(\frac{-1 + 2 + 2}{\sqrt{1^2 + 1^2 + 2^2}\sqrt{1^2 + 2^2 + 1^2}}\right) = \cos^{-1}\left(\frac{3}{6}\right) = \cos^{-1}\left(\frac{1}{2}\right) = 60°$$

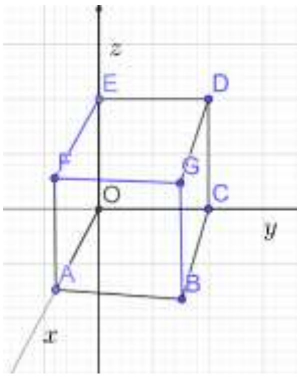
Example 2.123

The angle between any two diagonals of a cube can be written in the form $\tan^{-1} x$. Find the value of x. (BITSAT 2014, Adapted)

Note: There are two kinds of diagonals of a cube: space diagonals and face diagonals. From the options (which are removed here), we know that the diagonals referred to are space diagonals, since the angle between two face diagonals is a right angle.

By similarity, all cubes will have the same angle. Hence, we consider the cube with side length 1, and bottom left corner at the origin.





Consider the space diagonals

$$\overrightarrow{\boldsymbol{OG}}, and\ \overrightarrow{\boldsymbol{EB}}$$

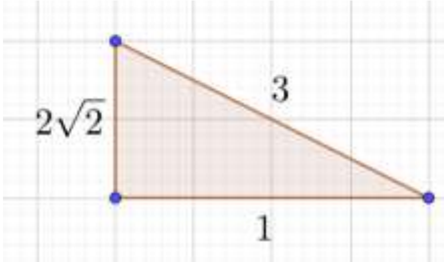
Write them in terms of unit vectors:

$$\overrightarrow{\boldsymbol{OG}} = \hat{\imath} + \hat{\jmath} + \hat{k}, \overrightarrow{\boldsymbol{EB}} = \hat{\imath} + \hat{\jmath} - \hat{k}$$

The angle between the two vectors is given by:

$$\cos\theta = \frac{\overrightarrow{\boldsymbol{OG}} \cdot \overrightarrow{\boldsymbol{EB}}}{|\overrightarrow{\boldsymbol{OG}}||\overrightarrow{\boldsymbol{EB}}|} = \frac{1 + 1 - 1}{\sqrt{3}\sqrt{3}} = \frac{1}{3}$$

We have $\cos\theta$, and we need to find $\tan\theta$. Draw a reference triangle with $adj = 1, hyp = 3$ and find $opp = 2\sqrt{2}$.



Then, from the reference triangle:

$$\tan\theta = 2\sqrt{2}$$
$$\theta = \tan^{-1} 2\sqrt{2}$$
$$x = 2\sqrt{2}$$

Example 2.124

If $\hat{\imath} + \hat{\jmath} + \hat{k}, 2\hat{\imath} + 5\hat{\jmath}, 3\hat{\imath} + 2\hat{\jmath} - 3\hat{k}$ and $\hat{\imath} - 6\hat{\jmath} - \hat{k}$ respectively are the position vectors of point A, B, C and D, then find:

A. the angle between the straight lines AB and CD.
B. whether $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are collinear or not. **(CBSE 2019)**

Part A

If the angle between the lines is θ, then:

$$\cos\theta = \frac{\overrightarrow{AB} \cdot \overrightarrow{CD}}{|\overrightarrow{AB}||\overrightarrow{CD}|}$$

Substitute using the definition:

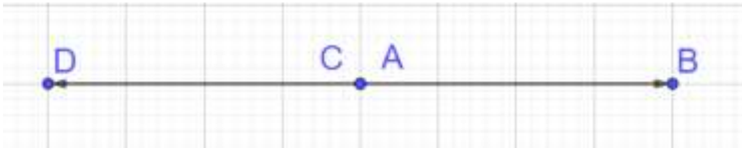
$$= \frac{(2-1, 5-1, 0-1) \cdot (1-3, -6-2, -1+3)}{\sqrt{1^2+4^2+(-1)^2}\sqrt{2^2+(-8)^2+2^2}} = \frac{(1,4,-1)\cdot(-2,-8,2)}{\sqrt{18}\sqrt{72}} = \frac{-2-32-2}{(3\sqrt{2})(6\sqrt{2})} = -\frac{36}{36} = -1$$

Since

$$\cos\theta = -1 \Rightarrow \theta = \cos^{-1}(-1) = \pi$$

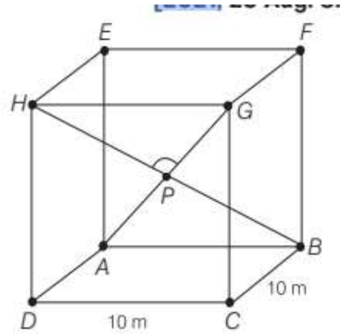
Part B

Since the angle between the lines is π, they are in opposite directions.
They are collinear.



Example 2.125

17: A hall has a square floor of dimension $10m \times 10m$ (see the figure) and vertical walls. If the $\angle GPH$ between the diagonals AG and BH is $\cos^{-1}\frac{1}{5}$, then the height of the hall (in m) is: **(JEE Main 2021, 26 Aug, Shift-II)**



The dot product of the diagonals $\overrightarrow{AG}$ and $\overrightarrow{BH}$ is:

$$\overrightarrow{AG} \cdot \overrightarrow{BH} = |\overrightarrow{AG}||\overrightarrow{BH}|\cos\theta$$

Introduce an origin at A. Substitute $\overrightarrow{AG} = (10,10,h), \overrightarrow{BH} = (-10,10,h)$:

$$(10,10,h)\cdot(-10,10,h) = \sqrt{10^2+10^2+h^2}\sqrt{(-10)^2+10^2+h^2}\cos\left(cos^{-1}\left(\frac{1}{5}\right)\right)$$

$$-100+100+h^2 = (h^2+200)\left(\frac{1}{5}\right)$$
$$5h^2 = h^2 + 200$$
$$4h^2 = 200$$
$$h^2 = 50$$

$$h = 5\sqrt{2}$$

2.126: Sum to Product Identities

$$\cos\alpha + \cos\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$

Challenge 2.127

Let $a, b, c \in \mathbb{R}$ such that $a^2 + b^2 + c^2 = 1$. If $a\cos\theta = b\cos\left(\theta + \frac{2\pi}{3}\right) = c\cos\left(\theta + \frac{4\pi}{3}\right)$, where $\theta = \frac{\pi}{9}$, then the angle between the vectors $a\hat{\boldsymbol{\imath}} + b\hat{\boldsymbol{\jmath}} + c\hat{\boldsymbol{k}}$, and $b\hat{\boldsymbol{\imath}} + c\hat{\boldsymbol{\jmath}} + a\hat{\boldsymbol{k}}$ is: **(JEE Main 2020, 3 Sep, Shift-II)**

If the angle between the vectors is α, then:

$$\cos\alpha = \frac{(a,b,c)\cdot(b,c,a)}{|(a,b,c)||(b,c,a)|} = \frac{ab+bc+ca}{\sqrt{a^2+b^2+c^2}\sqrt{a^2+b^2+c^2}} = ab+bc+ca$$

(Where in the last step we used $a^2 + b^2 + c^2 = 1$)

Since we have a tripartite equality, use the k method. Let:

$$a\cos\theta = b\cos\left(\theta+\frac{2\pi}{3}\right) = c\cos\left(\theta+\frac{4\pi}{3}\right) = k$$

Find a, b, c in terms of k:

$$\underbrace{a = \frac{k}{\cos\theta}, b = \frac{k}{\cos\left(\theta+\frac{2\pi}{3}\right)}, c = \frac{k}{\cos\left(\theta+\frac{4\pi}{3}\right)}}_{System\ I}$$

Substitute values from $System\ I$ in the angle between the vectors to get:

$$ab+bc+ca = \left(\frac{k}{\cos\theta}\right)\left(\frac{k}{\cos\left(\theta+\frac{2\pi}{3}\right)}\right) + \left(\frac{k}{\cos\left(\theta+\frac{2\pi}{3}\right)}\right)\left(\frac{k}{\cos\left(\theta+\frac{4\pi}{3}\right)}\right) + \left(\frac{k}{\cos\left(\theta+\frac{4\pi}{3}\right)}\right)\left(\frac{k}{\cos\theta}\right)$$

Add the fractions, and factor k^2:

$$= k^2\left[\frac{\cos\left(\theta+\frac{4\pi}{3}\right) + \cos\theta + \cos\left(\theta+\frac{2\pi}{3}\right)}{\cos\theta\cos\left(\theta+\frac{2\pi}{3}\right)\cos\left(\theta+\frac{4\pi}{3}\right)}\right]$$

Consider only the numerator.

Use the sum to product formula $\cos\alpha + \cos\beta = 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$ in the first and the last term to get:

$$= 2\cos\left(\frac{1}{2}\left[\left(\theta+\frac{4\pi}{3}\right)+\left(\theta+\frac{2\pi}{3}\right)\right]\right)\cos\left(\frac{1}{2}\left[\left(\theta+\frac{4\pi}{3}\right)-\left(\theta+\frac{2\pi}{3}\right)\right]\right) + \cos\theta$$

Simplify:

$$= 2\cos(\theta+\pi)\cos\left(\frac{\pi}{3}\right) + \cos\theta$$

Substitute $\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}, \cos(\theta+\pi) = -\cos\theta$

$$= 2(-\cos\theta)\left(\frac{1}{2}\right) + \cos\theta = -\cos\theta + \cos\theta = 0$$

Hence, the expression becomes:

$$\cos\alpha = k^2\left[\frac{0}{\cos\theta\cos\left(\theta+\frac{2\pi}{3}\right)\cos\left(\theta+\frac{4\pi}{3}\right)}\right]$$
$$\cos\alpha = 0$$

Take the *cos* inverse of both sides:

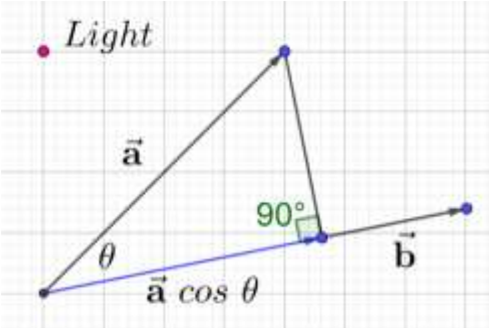
$$\alpha = \frac{\pi}{2}$$

C. Projection

2.128: Restating the projection definition

The projection of $\vec{a}$ along $\vec{b}$ is

$$\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|} = |\vec{a}|\cos\theta$$



$$LHS = \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|} = \frac{|\vec{a}||\vec{b}|\cos\theta}{|\vec{b}|} = |\vec{a}|\cos\theta = RHS$$

Example 2.129

If $|\vec{a}| = 2$, $|\vec{b}| = 3$, and $\vec{a}\cdot\vec{b} = 3$, find the projection of $|\vec{b}|$ on $|\vec{a}|$. (CBSE 2010)
Note: By projection here is meant the length of the projection, not the projection vector.

$$\frac{\vec{a}\cdot\vec{b}}{|\vec{a}|} = \frac{3}{2}$$

Example 2.130

A. If $\vec{a} = 7\hat{\imath} + \hat{\jmath} - 4\hat{k}$ and $\vec{b} = 2\hat{\imath} + 6\hat{\jmath} + 3\hat{k}$, then find the projection of $\vec{a}$ on $\vec{b}$. (CBSE 2013,2015)
B. Write the projection of the vector $\vec{a} = 2\hat{\imath} - \hat{\jmath} + \hat{k}$ on the vector $\vec{b} = \hat{\imath} + 2\hat{\jmath} + 2\hat{k}$. (CBSE 2014)
C. Write the projection of $(\vec{b} + \vec{c})$ on $\vec{a}$, where $\vec{a} = 2\hat{\imath} - 2\hat{\jmath} + \hat{k}$, $\vec{b} = \hat{\imath} + 2\hat{\jmath} - 2\hat{k}$ and $\vec{c} = 2\hat{\imath} - \hat{\jmath} + 4\boldsymbol{k}$ (CBSE 2013)

Part A

$$\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|} = \frac{(7,1,-4)\cdot(2,6,3)}{\sqrt{2^2+6^2+3^2}} = \frac{14+6-12}{\sqrt{49}} = \frac{8}{7}$$

Part B

$$\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|} = \frac{(2,-1,1)\cdot(1,2,2)}{\sqrt{1^2+2^2+2^2}} = \frac{2-2+2}{\sqrt{9}} = \frac{2}{3}$$

Part C

$$\frac{(\vec{b}+\vec{c})\cdot\vec{a}}{|\vec{a}|} = \frac{(3,1,2)\cdot(2,-2,1)}{\sqrt{2^2+(-2)^2+1^2}} = \frac{6-2+2}{\sqrt{9}} = \frac{6}{3} = 2$$

Example 2.131

$$\vec{v} = a\hat{\imath} + b\hat{\jmath} + c\hat{k}$$

Find the projection of $\vec{v}$ along

A. $\hat{\boldsymbol{\imath}}$
B. $\hat{\boldsymbol{\jmath}}$
C. $\hat{\boldsymbol{k}}$

Part A

$$\frac{\vec{v}\cdot\hat{\imath}}{|\hat{\imath}|}=\frac{(a,b,c)\cdot(1,0,0)}{1}=\frac{a}{1}=a$$

Part B

$$\frac{\vec{v}\cdot\hat{\jmath}}{|\hat{\jmath}|}=\frac{(a,b,c)\cdot(0,1,0)}{1}=\frac{b}{1}=b$$

Part C

$$\frac{\vec{v}\cdot\hat{k}}{|\hat{k}|}=\frac{(a,b,c)\cdot(0,0,1)}{1}=\frac{c}{1}=c$$

Example 2.132: Back Calculations

A. Find λ when projection of $\vec{a}=\lambda\hat{\imath}+\hat{\jmath}+4\hat{k}$ on $\vec{b}=2\hat{\imath}+6\hat{\jmath}+3\hat{k}$ is 4 units. (CBSE 2012)
B. Find the number of values of λ such that the projection of $\vec{a}=\lambda\hat{\imath}+\hat{\jmath}+4\hat{k}$ on $\vec{b}=2\hat{\imath}+\lambda\hat{\jmath}+3\hat{k}$ is 4 units. (CBSE 2012)

Part A

$$\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}=\frac{(\lambda,1,4)\cdot(2,6,3)}{\sqrt{2^2+6^2+3^2}}=\frac{2\lambda+6+12}{\sqrt{49}}=\frac{2\lambda+18}{7}=\underbrace{4}_{Given}$$

$$2\lambda+18=28$$
$$2\lambda=10$$
$$\lambda=5$$

Part B

$$\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}=\frac{(\lambda,1,4)\cdot(2,\lambda,3)}{\sqrt{2^2+\lambda^2+3^2}}=\frac{2\lambda+\lambda+12}{\sqrt{\lambda^2+13}}=\frac{3\lambda+12}{\sqrt{\lambda^2+13}}=\underbrace{4}_{Given}$$

$$3\lambda+12=4\sqrt{\lambda^2+13}$$
$$9\lambda^2+72\lambda+144=16(\lambda^2+13)$$
$$7\lambda^2-72\lambda+64=0$$

$$b^2-4ac=5184-(4)(7)(64)=3392>0$$
$$\textit{Quadratic has two distinct roots}$$
$$2\ \textit{values of}\ \lambda$$

Example 2.133

18: If the projection of the vector $\hat{\imath}+2\hat{\jmath}+\hat{k}$ on the sum of the two vectors $2\hat{\imath}+4\hat{\jmath}-5\hat{k}$ and $-\lambda\hat{\imath}+2\hat{\jmath}+3\hat{k}$ is 1, then λ is equal to: (JEE Main 2021, 25 July, Shift-II)

The projection of $\vec{a}$ on $\vec{b}$ is

$$\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}=1$$

Substitute $\vec{a}=(1,2,1)$ and $\vec{b}=(2,4,-5)+(-\lambda,2,3)=(2-\lambda,6,-2)$ in the above:

$$\frac{(1,2,1)\cdot(2-\lambda,6,-2)}{|(2-\lambda,6,-2)|}=1$$

Carry the multiplication using the dot product:

$$2-\lambda+12-2=|(2-\lambda,6,-2)|$$

Use the definition of magnitude on the RHS:

$$12-\lambda=\sqrt{(2-\lambda)^2+6^2+(-2)^2}$$

Square both sides:

$$(12-\lambda)^2=(2-\lambda)^2+6^2+(-2)^2$$

Expand:

$$144-24\lambda+\lambda^2=4-4\lambda+\lambda^2+36+4$$

Collate all terms on side:

$$100=20\lambda\Rightarrow\lambda=5$$

D. Coplanar Vectors

Example 2.134

24: Let $\vec{x}$ be a vector in the plane containing vectors $\vec{a}=2\hat{\imath}-\hat{\jmath}+\hat{k}$ and $\vec{b}=\hat{\imath}+2\hat{\jmath}-\hat{k}$. If the vector $\vec{x}$ is perpendicular to $(3\hat{\imath}+2\hat{\jmath}-\hat{k})$ and its projection on $\vec{a}$ is $\frac{17\sqrt{6}}{2}$, then the value of $|\vec{x}|^2$ is equal to: (JEE Main 2021, 17 March, Shift-II)

Let

$$\vec{x}=\lambda\vec{a}+\mu\vec{b}=\lambda(2,-1,1)+\mu(1,2,-1)=(2\lambda+\mu,-\lambda+2\mu,\lambda-\mu)$$

Since the vector $\vec{x}$ is perpendicular to $(3\hat{\imath}+2\hat{\jmath}-\hat{k})$

$$\vec{x}\cdot(3\hat{\imath}+2\hat{\jmath}-\hat{k})=0$$

Substitute $\vec{x}=(2\lambda+\mu,-\lambda+2\mu,\lambda-\mu)$

$$(2\lambda+\mu,-\lambda+2\mu,\lambda-\mu)\cdot(3\hat{\imath}+2\hat{\jmath}-\hat{k})=0$$

Multiply using the dot product:

$$(6\lambda+3\mu)+(-2\lambda+4\mu)+(\mu-\lambda)=0$$

Simplify:

$$\underbrace{3\lambda+8\mu=0}_{Equation\ I}$$

Projection of $\vec{x}$ on $\vec{a}$ is $\frac{17\sqrt{6}}{2}$:

$$\frac{\vec{x}\cdot\vec{a}}{|\vec{a}|}=\frac{17\sqrt{6}}{2}$$

Substitute $\vec{x}=(2\lambda+\mu,-\lambda+2\mu,\lambda-\mu),\vec{a}=2\hat{\imath}-\hat{\jmath}+\hat{k}$

$$\frac{(2\lambda+\mu,-\lambda+2\mu,\lambda-\mu)\cdot(2,-1,1)}{\sqrt{2^2+(-1)^2+1^2}}=\frac{17\sqrt{6}}{2}$$

Multiply using the dot product:

$$\frac{(4\lambda + 2\mu) + (\lambda - 2\mu) + (\lambda - \mu)}{\sqrt{6}} = \frac{17\sqrt{6}}{2}$$

Cross multiply, and simplify:

$$\underbrace{6\lambda - \mu = 51}_{Equation\ II}$$

Multiply Equation I by II:

$$\underbrace{6\lambda + 16\mu = 0}_{Equation\ III}$$

Subtract Equation II from Equation III:

$$17\mu = -51$$
$$\mu = -3$$
$$\lambda = 8$$

Substitute $\mu = -3, \lambda = 8$

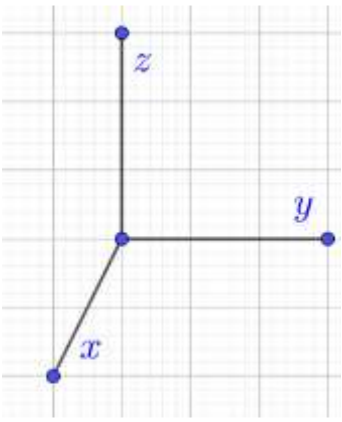
$$\vec{x} = (2\lambda + \mu, -\lambda + 2\mu, \lambda - \mu) = (13, -14, 11)$$

Finally, calculate the square of the magnitude of $\vec{x}$:

$$|\vec{x}|^2 = \sqrt{13^2 + (-14)^2 + 11^2} = 486$$

E. Vector ⊥ to Two Vectors

In the adjoining diagram, if you want a vector perpendicular to the x and the y axes, you can draw the z axis. However, you could also have drawn the z axis pointing downward, and that would also be perpendicular.
To eliminate the ambiguity, we define the z axis in such a way that the positive direction points upwards.



Example 2.135

A. Write the number of vectors of unit length perpendicular to both the vectors $\vec{a} = 2\hat{\imath} + \hat{\jmath} + 2\hat{k}$, and $\vec{b} = \hat{\jmath} + \hat{k}$. (CBSE 2016)
B. When will two vectors not have exactly two vectors perpendicular to them?

Part A
Two vectors
Part B
When these two vectors are parallel.

Example 2.136

Find the unit vector perpendicular to both of the vectors $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ where $\vec{a} = \hat{\imath} + \hat{\jmath} + \hat{k}$ and $\vec{b} = \hat{\imath} + 2\hat{\jmath} + 3\hat{k}$. (CBSE 2014)

Let the unit vector that we wish to find be

$$\hat{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$$

Since $\hat{r} \perp (\vec{a} + \vec{b})$, we must have

$$\hat{r} \cdot (\vec{a} + \vec{b}) = 0$$
$$(x\hat{\imath} + y\hat{\jmath} + z\hat{k}) \cdot (2\hat{\imath} + 3\hat{\jmath} + 4\hat{k}) = 0$$

Expand using the dot product:

$$\underbrace{2x + 3y + 4z = 0}_{\textit{Equation I}}$$

Similarly, $\hat{\boldsymbol{r}} \perp (\vec{\boldsymbol{a}} - \vec{\boldsymbol{b}})$, and hence we must have:

$$\hat{\boldsymbol{r}} \cdot (\vec{\boldsymbol{a}} - \vec{\boldsymbol{b}}) = \boldsymbol{0}$$

$$(x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}} + z\hat{\boldsymbol{k}}) \cdot (-\hat{\boldsymbol{\jmath}} - 2\hat{\boldsymbol{k}}) = 0$$

Expand using the dot product:

$$-y - 2z = 0$$

$$\underbrace{y = -2z}_{\textit{Equation II}}$$

Substitute the value of y from Equation II into Equation I:

$$2x + 3(-2z) + 4z = 0$$

$$2x - 2z = 0$$

$$\underbrace{x = z}_{\textit{Equation III}}$$

We need a third equation. Since we wish to find a unit vector:

$$\hat{\boldsymbol{r}} = 1$$

$$|x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}} + z\hat{\boldsymbol{k}}| = 1$$

$$\sqrt{x^2 + y^2 + z^2} = 1$$

$$\underbrace{x^2 + y^2 + z^2 = 1}_{\textit{Equation IV}}$$

Substitute the values from Equation II and Equation III into Equation IV:

$$z^2 + (-2z)^2 + z^2 = 1$$

$$6z^2 = 1$$

$$x = z = \pm\frac{1}{\sqrt{6}}$$

$$y = -2z = \pm\frac{2}{\sqrt{6}}$$

The vectors that we want are:

$$\frac{1}{\sqrt{6}}\hat{\boldsymbol{\imath}} - \frac{2}{\sqrt{6}}\hat{\boldsymbol{\jmath}} + \frac{1}{\sqrt{6}}\hat{\boldsymbol{k}}$$

$$-\frac{1}{\sqrt{6}}\hat{\boldsymbol{\imath}} + \frac{2}{\sqrt{6}}\hat{\boldsymbol{\jmath}} - \frac{1}{\sqrt{6}}\hat{\boldsymbol{k}}$$

2.6 Revision

A. Midpoint

HW 1

Example 2.137

31: A vector $\vec{\boldsymbol{a}} = \alpha\hat{\boldsymbol{\imath}} + 2\hat{\boldsymbol{\jmath}} + \beta\hat{\boldsymbol{k}}$ $(\alpha, \beta \in \mathbb{R})$ lies in the plane of the vectors $\vec{\boldsymbol{b}} = \hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}}$ and $\vec{\boldsymbol{c}} = \hat{\boldsymbol{\imath}} - \hat{\boldsymbol{\jmath}} + 4\hat{\boldsymbol{k}}$. If $\vec{\boldsymbol{a}}$ bisects the angle between $\vec{\boldsymbol{b}}$ and $\vec{\boldsymbol{c}}$, then:

A. $a.\hat{\boldsymbol{\imath}} + 3 = 0$
B. $a.k + 2 = 0$
C. $a.\hat{\boldsymbol{\imath}} + 1 = 0$
D. $a.k + 4 = 0$ (JEE Main 2020, 7 Jan, Shift-I)

Angle Bisector Theorem OR Basic Geometry

B. Projections/Dot Product

Example 2.138

37: Let $ABCD$ be a parallelogram such that $\overrightarrow{\boldsymbol{AB}} = \vec{\boldsymbol{q}}, \overrightarrow{\boldsymbol{AD}} = \vec{\boldsymbol{p}}$ and $\angle BAD$ be an acute angle. If $\vec{\boldsymbol{r}}$ is the vector that coincides with the altitude directed from the vertex B to the side AD, then $\vec{\boldsymbol{r}}$ is given by:

A. $r = 3q\frac{3(p \cdot q)}{p \cdot p}p$
B. r=-q+(p\bulletq)/(p\bulletp) p
C. r=q-(p\bulletq)/(p\bulletp) p
D. r=-3q+3(p\bulletq)/(p\bulletp) p

C. Square of a Difference

2.139: Maximum of a Trig Expression

Maximum of $a\cos\theta + b\sin\theta$ is

$$\sqrt{a^2+b^2}$$

Example 2.140

29: If $\vec{a}$ and $\vec{b}$ are unit vectors, then the greatest value of $\sqrt{3}|\vec{a}+\vec{b}|+|\vec{a}-\vec{b}|$ is: (JEE Main 2020, 6 Sep, Shift-I)

D. Square of Three Terms

Example 2.141

28: Let the vectors $\vec{a}, \vec{b}, \vec{c}$ be such that $|\vec{a}|=2, |\vec{b}|=4$, and $|\vec{c}|=4$. If the projection of $\vec{b}$ on $\vec{a}$ is equal to the projection of $\vec{c}$ on $\vec{a}$ and $\vec{b}$ is perpendicular to $\vec{c}$, then the value of $|\vec{a}+\vec{b}-\vec{c}|$ is: (JEE Main 2020, 5 Sep, Shift-II)

Example 2.142

41: Let $\vec{u}, \vec{v}, \vec{w}$ be such that $|\vec{u}|=1, |\vec{v}|=2, |\vec{w}|=3$. If the projection $\vec{v}$ along $\vec{u}$ is equal to that of $\vec{w}$ along $\vec{u}$ and $\vec{v}, \vec{w}$ are perpendicular to each other, then $|\vec{u}-\vec{v}+\vec{w}|$ is equal to: (JEE Main 2004)

Example 2.143

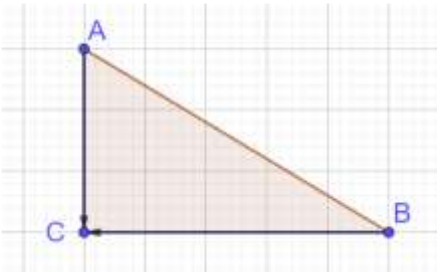
42: $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that $\vec{a}+\vec{b}+\vec{c}=0, |\vec{a}|=1, |\vec{b}|=2, |\vec{c}|=3$, then $\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a}$ is equal to: (JEE Main 2003)

E. Perpendicularity

HW

Example 2.144

The value(s) of $\vec{a}$, for which the points A, B, C with position vectors $2\hat{\imath}-\hat{\jmath}+\hat{k}, \hat{\imath}-3\hat{\jmath}-5\hat{k}$ and $a\hat{\imath}-3\hat{\jmath}+\hat{k}$ respectively are the vertices of a right-angled triangle with $C=\frac{\pi}{2}$ are: (JEE Main 2006)



Draw a triangle with the points A, B and C.

Since $\angle ACB = 90°$:

$$\overrightarrow{AC}\cdot\overrightarrow{BC}=0$$

Use the definition of vector as displacement:

$$[(a,-3,1)-(2,-1,1)]\cdot[(a,-3,1)-(1,-3,-5)]=0$$

Simplify:

$$(a-2,-4,0)\cdot(a-1,0,-6)=0$$

Use the definition of the dot product:

$$(a-2)(a-1)=0 \Rightarrow a\in\{1,2\}$$

2.145: Coplanar Vector

A coplanar vector is a linear combination of two non-parallel vectors in that plane.

Example 2.146

Given two vectors are $\hat{\boldsymbol{\imath}} - \hat{\boldsymbol{\jmath}}$ and $\hat{\boldsymbol{\imath}} + 2\hat{\boldsymbol{\jmath}}$, the unit vector coplanar with the two vectors and perpendicular to the first is: (JEE Main 2002)

A vector coplanar to $\hat{\boldsymbol{\imath}} - \hat{\boldsymbol{\jmath}}$ and $\hat{\boldsymbol{\imath}} + 2\hat{\boldsymbol{\jmath}}$ is:

$$\vec{\boldsymbol{a}} = (x, y)$$

From the perpendicularity condition:

$$(x, y) \cdot (1, -1) = 0 \Rightarrow x - y = 0 \Rightarrow x = y$$

$$|\vec{\boldsymbol{a}}| = \sqrt{x^2 + y^2} = \sqrt{x^2 + x^2} = \sqrt{2x^2} = \sqrt{2}x$$

Hence, the unit vector in the direction of $\vec{\boldsymbol{a}}$ is given by:

$$\frac{\vec{\boldsymbol{a}}}{|\vec{\boldsymbol{a}}|} = \frac{(x, x)}{\sqrt{2}x} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

HW 2

Example 2.147

35: Let $\vec{\boldsymbol{u}}$ be a vector coplanar with the vectors $\vec{\boldsymbol{a}} = 2\hat{\boldsymbol{\imath}} + 3\hat{\boldsymbol{\jmath}} - \hat{\boldsymbol{k}}$ and $\vec{\boldsymbol{b}} = \hat{\boldsymbol{\jmath}} + \hat{\boldsymbol{k}}$. If $\vec{\boldsymbol{u}}$ is perpendicular to $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{u}} \cdot \vec{\boldsymbol{b}} = 24$, then $|\vec{\boldsymbol{u}}|^2$ is equal to: (JEE Main 2018)

Using the perpendicularity condition:

$$\vec{\boldsymbol{u}} \cdot \vec{\boldsymbol{a}} = 0$$

Since $\vec{\boldsymbol{u}}$ is coplanar to $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$**,** substitute $\vec{\boldsymbol{u}} = \lambda\vec{\boldsymbol{a}} + \mu\vec{\boldsymbol{b}}$

$$\left(\lambda\vec{\boldsymbol{a}} + \mu\vec{\boldsymbol{b}}\right) \cdot \vec{\boldsymbol{a}} = 0$$

Using the distributive property:

$$\underbrace{\lambda|\vec{\boldsymbol{a}}|^2 + \mu\vec{\boldsymbol{b}} \cdot \vec{\boldsymbol{a}} = 0}_{Equation\ I}$$

Note that

$$|\vec{\boldsymbol{a}}|^2 = 2^2 + 3^2 + (-1)^2 = 4 + 9 + 1 = 14$$
$$\vec{\boldsymbol{b}} \cdot \vec{\boldsymbol{a}} = (0,1,1) \cdot (2,3,-1) = 0 + 3 - 1 = 2$$

Substitute the above into Equation I:

$$14\lambda + 2\mu = 0$$

Use the

$$\left(\lambda\vec{\boldsymbol{a}} + \mu\vec{\boldsymbol{b}}\right) \cdot \vec{\boldsymbol{b}} = 24$$
$$\lambda\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}} + \mu\left|\vec{\boldsymbol{b}}\right|^2 = 24$$

HW 3

Example 2.148

33: Let $\vec{a} = \hat{\imath} + \hat{\jmath} + \sqrt{2}\hat{k}$,$\vec{b} = b_1\hat{\imath} + b_2\hat{\jmath} + \sqrt{2}\hat{k}$ and $\vec{c} = 5\hat{\imath} + \hat{\jmath} + \sqrt{2}\hat{k}$ be three vectors such that the projection vector of $\vec{b}$ on $\vec{a}$ is $\vec{a}$. If $\vec{a} + \vec{b}$ is perpendicular to $\vec{c}$, then $|\vec{c}|$ is equal to: (JEE Main 2019, 9 Jan, Shift-II)

Example 2.149

34: Let $\vec{a} = 2\hat{\imath} + \lambda_1\hat{\jmath} + 3\hat{k}$, $\vec{b} = 4\hat{\imath} + (3 - \lambda_2)\hat{\jmath} + 6\hat{k}$ and $\vec{c} = 3\hat{\imath} + 6\hat{\jmath} + (\lambda_3 - 1)\hat{k}$ be three vectors such that $\vec{b} = 2\vec{a}$ and $\vec{a}$ is perpendicular to $\vec{c}$. Then a possible value of $\lambda_1, \lambda_2, \lambda\_3$ is:
A. (1,3,1)
B. (1,5,1)
C. (-1/2, 4,0)
D. (1/2,4,-2) (JEE Main 2019, 10 Jan, Shift-I)

F. Applications

Example 2.150

20: Let $\vec{a}, \vec{b}, \vec{c}$ be three mutually perpendicular vectors of the same magnitude and equally inclined at an angle θ with the vector $\vec{a} + \vec{b} + \vec{c}$. Then, $36\cos^2\theta$ is equal to: (JEE Main 2021, 20 July, Shift-II)

Example 2.151

21: For $p > 0$, a vector $V_2 = 2\hat{\imath} + (p+1)\hat{\jmath}$ is obtained by rotating the vector $v_1 = rt(3p)\hat{\imath} + \hat{\jmath}$ by an angle θ about the origin in counter clockwise direction. If $\tan\theta = \frac{\alpha\sqrt{3}-2}{4\sqrt{3}+3}$, then the value of α is equal to: (JEE Main 2021, 20 July, Shift-II)

G. Section Formula

Example 2.152

32: Let $A(3,0,-1)$, $B(2,10,6)$ and $C(1,2,1)$ be the vertices of a triangle and M be the mid-point of AC. If G divides BM in the ratio $2:1$, then $\cos\angle GOA$ (O being the origin) is equal to: (JEE Main 2019, 10 April, Shift-I)

A. Geometry with Vectors

Example 2.153

Show that the line drawn from the center of a circle, bisecting the chord, is perpendicular to the chord.

Consider circle with center O and chord AB with midpoint X.

Note that

$$\overrightarrow{OX} = \frac{1}{2}\left(\overrightarrow{OB} + \overrightarrow{OA}\right)\ (By\ midpoint\ theorem)$$
$$\overrightarrow{AB} = \overrightarrow{OB} + \overrightarrow{AO} = \overrightarrow{OB} - \overrightarrow{OA}$$

Then, using the above:

$$\overrightarrow{OX} \cdot \overrightarrow{AB} = \frac{1}{2}\left(\overrightarrow{OB} + \overrightarrow{OA}\right)\left(\overrightarrow{OB} - \overrightarrow{OA}\right)$$

Using the difference of squares for dot product of vectors:

$$= \frac{1}{2}\left(\left|\overrightarrow{OB}\right|^2 - \left|\overrightarrow{OA}\right|^2\right)$$

But since both $\overrightarrow{OB}$ and $\left|\overrightarrow{OA}\right|$ are radii of a circle, their magnitude must be the same.
Hence, the expression is equal to zero.

And since

$$\overrightarrow{OX} \cdot \overrightarrow{AB} = 0 \Rightarrow \overrightarrow{OX} \perp \overrightarrow{AB}$$

Example 2.154
Orthocenter of a triangle using vectors
Uses dot product for perpendicularity

3. Cross Product

3.1 Angle Definition

A. Magnitude of Cross Product

We have looked at the product of a vector with a scalar, which is a vector with the same direction but different magnitude. We also looked at the dot product, which is the product of two vectors, which has a scalar output. The last product that we at is the cross product, which is the product of a vector with a vector, and has a vector perpendicular to the original two vectors for output.

	Input	Output	Example
Scalar Multiple	A scalar and a vector	Vector	$\frac{1}{\sqrt{6}}\hat{\boldsymbol{\imath}}$
Dot Product	Two vectors	Scalar	
Cross Product	Two vectors	Vector	

3.1: Magnitude of Cross Product

Given two vectors $\vec{a}$ and $\vec{b}$, with angle θ, between them, the magnitude of their cross product is:

$$|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}|\sin\theta$$

Example 3.2

A. $(Definition)$ Find $|\vec{a} \times \vec{b}|^2$ if $|\vec{a}| = 2, |\vec{b}| = 3$, and $(\vec{a}, \vec{b}) = \frac{\pi}{6}$. (AP EAPCET, 17 Sep 2020, Shift-I)

B. $(Back\ calculation)$ If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a} \cdot \vec{b}| = |\vec{a} \times \vec{b}|$, then find the angle between $\vec{a}$ and $\vec{b}$. (CBSE 2010)

Part A

$$|\vec{a} \times \vec{b}|^2 = (|\vec{a}||\vec{b}|\sin\theta)^2 = \left(2 \cdot 3 \cdot \sin\frac{\pi}{6}\right)^2 = \left(2 \cdot 3 \cdot \frac{1}{2}\right)^2 = 3^2 = 9$$

Part B

$$|\vec{a}||\vec{b}|\cos\theta = |\vec{a}||\vec{b}|\sin\theta \Rightarrow \cos\theta = \sin\theta \Rightarrow \theta = \frac{\pi}{4}$$

Example 3.3: Correct the proof

Identify the first step, in the following "proof" where a mistake is made, or incorrect logic is used, explain the issue, and fix it.

Step I: The magnitude of the cross product $|\vec{a} \times \vec{b}|$ is $|\vec{a}||\vec{b}|\sin\theta$. Since both $|\vec{a}|$ and $|\vec{b}|$ are nonnegative, the sign of $|\vec{a} \times \vec{b}|$ is controlled by $\sin\theta$.
Step II: It is a standard property that $-1 \le \sin x \le 1$
Step III: Multiply $-1 \le \sin\theta \le 1$ throughout by $|\vec{a}||\vec{b}|$ to get $-|\vec{a}||\vec{b}| \le |\vec{a}||\vec{b}|\sin\theta \le |\vec{a}||\vec{b}|$
Step IV: Substitute $|\vec{a}||\vec{b}|\sin\theta = |\vec{a} \times \vec{b}|$ to get $-|\vec{a}||\vec{b}| \le |\vec{a} \times \vec{b}| \le |\vec{a}||\vec{b}|$

The mistake is in Step III.
Since the angle between two vectors θ is always the smaller angle we have

$$0 \le \theta \le \pi \Rightarrow 0 \le \sin\theta \le 1$$

And multiplying the inequality above with $|\vec{a}||\vec{b}|\sin\theta$ gives us the correct version

$$0 \le |\vec{a}||\vec{b}| \sin\theta \le |\vec{a}||\vec{b}|$$

Where we can substitute $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin\theta$ to get

$$\underbrace{0 \le |\vec{a} \times \vec{b}| \le |\vec{a}||\vec{b}|}_{Correct\ Version}$$

3.4: Range of Magnitude

$$0 \le |\vec{a} \times \vec{b}| \le |\vec{a}||\vec{b}|$$

Refer to the previous example to see the proof of this property.

Example 3.5

A. If $|\vec{a}| = \frac{1}{2}, |\vec{b}| = \frac{3}{5}, x = |\vec{a} \times \vec{b}|, M = Max(x),\ m = Min(x)$, then find $M - \alpha m$, where $\alpha \in \mathbb{R}$.

$$M - 2m = \left(\frac{1}{2}\right)\left(\frac{3}{5}\right) - \alpha(0) = 0.3$$

MCQ 3.6

Mark the Correct Option

Let $\vec{u}$ and $\vec{v}$ be two non-zero vectors. Then the magnitude of the cross product $|\vec{u} \times \vec{v}|$is always: (AP EAPCET, 18 Sep 2020, Shift-I)

A. $\le |\vec{u}||\vec{v}|$
B. $= |\vec{u}||\vec{v}|$
C. $\ge |\vec{u}||\vec{v}|$
D. $= 0$

Using the property $\mathbf{0} \le |\vec{a} \times \vec{b}| \le |\vec{a}||\vec{b}|$

Option A is correct

Example 3.7: Justify the proof

Consider vectors $\vec{a}$ and $\vec{b}$ such that $|\vec{a} \times \vec{b}| > 0$

Step I: $|\vec{a}||\vec{b}| \sin\theta > 0$

Step II: $\sin\theta > 0$

Step III: $0 < \theta < \pi$

A. Justify Step I.
B. Justify Step II. Explain why we do not need to reverse the sign of the inequality.
C. Justify Step III. Explain why θ cannot be in Quadrant III.

Step I: Substitute the definition of the magnitude of the cross product: $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin\theta$

Step II: Divide both sides $|\vec{a}||\vec{b}|$, and note that $|\vec{a}||\vec{b}| > 0$ since they are both magnitudes and greater than zero.

Step III: $\sin\theta$ is positive in the first and second quadrants.

MCQ 3.8: Positive Magnitude of Cross Product

Mark the correct option

The cross product of two vectors with angle θ between them is positive. In what range must θ lie?

A. $0 < \theta < \pi$

B. $0 \le \theta \le \pi$
C. $0 < \theta < \frac{\pi}{2}$
D. $\frac{\pi}{2} < \theta < \pi$
E. $\frac{\pi}{2} \le \theta \le \pi$

Option A

Option B is not correct since:

$$\theta = \{0, \pi\} \Rightarrow \sin\theta = 0 \Rightarrow |\vec{\boldsymbol{a}}||\vec{\boldsymbol{b}}|\sin\theta = 0 \Rightarrow Cross\ product = 0 \neq positive$$

MCQ 3.9: Negative Magnitude of Cross Product

Mark the correct option

The cross product of two vectors with angle θ between them is calculated to be negative. This indicates:

A. That one of the vectors is a zero vector.
B. That both of the vectors are zero vectors.
C. That the calculations were incorrect.
D. The angle between the vectors is acute.
E. The angle between the vectors is obtuse.
F. The angle between the vectors is reflex.

Option C

Example 3.10

What can you conclude about the angle θ between two non-zero vectors if their cross product is zero?

The angle between the two vectors is zero or π:

$$\theta \in \{0, \pi\}$$

That is, the vectors are collinear.

3.11: Zero Cross Product

Show that:

A. The cross product of vectors in the same direction or opposite directions is the zero vector.
B. The cross product of a vector with itself is zero.

Part A

If the two vectors are in the same direction, then the angle between them is zero.
If the two vectors are in opposite directions, then the angle between them is π.

$$\theta \in \{0, \pi\} \Rightarrow \sin\theta = 0 \Rightarrow \vec{\boldsymbol{a}} \times \vec{\boldsymbol{b}} = |\vec{\boldsymbol{a}}||\vec{\boldsymbol{b}}|\sin\theta\,\hat{\boldsymbol{n}} = \mathbf{0}$$

Part B

A vector is parallel to itself. And the angle between two parallel vectors is zero. Hence, the cross product is zero.

Example 3.12

If the cross product of two vectors is zero, what can you conclude about the vectors.

Let the vectors be $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$.

$$\vec{\boldsymbol{a}} \times \vec{\boldsymbol{b}} = 0$$

Take the magnitude on both sides:

$$|\vec{\boldsymbol{a}} \times \vec{\boldsymbol{b}}| = 0$$

Use the definition of magnitude:

$$|\vec{a}||\vec{b}|\sin\theta = 0$$

Use the zero-product property:

$$|\vec{a}| = 0\ OR\ |\vec{b}| = 0\ OR\ \sin\theta = 0 \Rightarrow \theta \in \{0, \pi\}$$

3.13: Zero Property for Cross Product

$$\vec{a}\times\vec{b} = 0 \Rightarrow |\vec{a}| = 0\ OR\ |\vec{b}| = 0\ OR\ \sin\theta = 0 \Rightarrow \theta \in \{0, \pi\}$$

If the cross product of two vectors is zero, then one of the following holds:

- one of the vectors is zero
- both of the vectors are zero
- the vectors are collinear.

Note: The collinearity condition is equivalent to saying that one vector is a scalar multiple of the other.

3.14: Properties

Show that:

A. The magnitude of the cross product of two perpendicular vectors is the product of their magnitudes.
B. The magnitude of the cross product of two vectors is maximum when the angle between them is a right angle. (Assume that their lengths are given, and cannot be changed).

Part A

$$|\vec{a}\times\vec{b}| = |\vec{a}||\vec{b}|\sin\theta$$

If the two vectors are perpendicular, then the angle between them is a right angle:

$$\theta = \frac{\pi}{2} \Rightarrow \sin\theta = 1$$

$$|\vec{a}\times\vec{b}| = |\vec{a}||\vec{b}|$$

Part B

$$|\vec{a}\times\vec{b}| = |\vec{a}||\vec{b}|\sin\theta$$

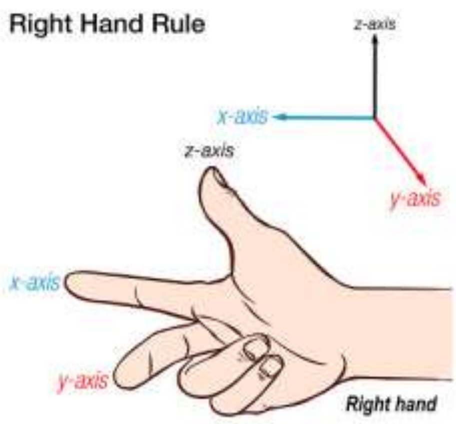
The maximum value of the RHS is obtained when

$$\sin\theta = 1 \Rightarrow \theta = \frac{\pi}{2} \Rightarrow Right\ Angle$$

B. Right-Handed Coordinate System

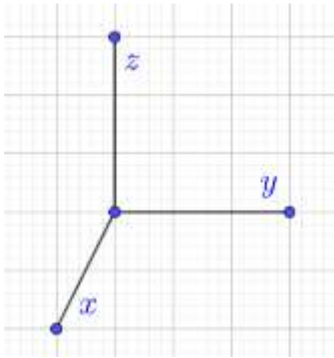
3.15: Right-Handed Co-ordinate System

- In a right-handed coordinate system, the Right-Hand Thumb Rule is applicable.
- $x - axis$ is the first vector (given by direction of index finger). $y - axis$ is the second vector (given by direction of middle finger).
- Then, the direction of cross product is given by the thumb of the right hand.



3.16: Properties of Handedness

- The order of vectors matters. Interchanging the labels x and y changes the handedness.
- Reversing the direction of an axis changes the handedness.
- Reversing the direction of all three axes also changes the handedness.



Example 3.17

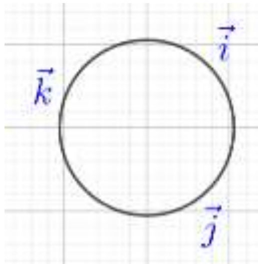
C. Cross Product

3.18: Cross Product

Given two vectors $\vec{a}$ and $\vec{b}$, with angle θ, between them, their cross product is:

$$\vec{a}\times\vec{b}=|\vec{a}||\vec{b}|\sin\theta\,\hat{n}$$

- When taking the angle between vectors, we take the smaller angle when they are arranged tip to tip, or tail to tail. That is $0\le\theta\le\pi$.
- $\hat{n}$ is a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$, such that $\vec{a},\vec{b}$ and $\hat{n}$ form a right-handed coordinate system.



3.19: Cross Product of Unit Vectors

$$\hat{\imath}\times\hat{\jmath}=\hat{k},\qquad \hat{\jmath}\times\hat{k}=\hat{\imath},\qquad \hat{k}\times\hat{\imath}=\hat{\jmath}$$
$$\hat{\jmath}\times\hat{\imath}=-\hat{k},\qquad \hat{k}\times\hat{\jmath}=-\hat{\imath},\qquad \hat{\imath}\times\hat{k}=-\hat{\jmath}$$

Example 3.20

Write the value of $(\hat{k}\times\hat{\jmath})\cdot\hat{\imath}+\hat{\jmath}\cdot\hat{k}$ (CBSE 2012)

$$(-\hat{\imath})\cdot\hat{\imath}+0=-1+0=-1$$

D. Distributive Property

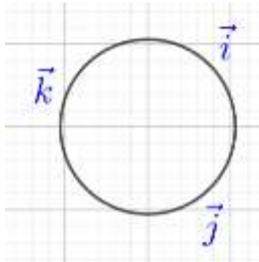
3.21: Distributive Over Addition and Subtraction

$$\vec{a}\times(\vec{b}+\vec{c})=\vec{a}\times\vec{b}+\vec{a}\times\vec{c}$$
$$\vec{a}\times(\vec{b}-\vec{c})=\vec{a}\times\vec{b}-\vec{a}\times\vec{c}$$

Example 3.22

A. Write the value of the following $\hat{\imath}\times(\hat{\jmath}+\hat{k})+\hat{\jmath}\times(\hat{k}+\hat{\imath})+\hat{k}\times(\hat{\imath}+\hat{\jmath})$ (CBSE 2014)
B. If $\vec{r}=x\hat{\imath}+y\hat{\jmath}+z\hat{k}$ find $(\vec{r}\times\hat{\imath})\cdot(\vec{r}\times\hat{\jmath})+xy$

Part A



Use the distributive property:
$$\left(\hat{\boldsymbol{\imath}}\times\hat{\boldsymbol{\jmath}}+\hat{\boldsymbol{\imath}}\times\hat{\boldsymbol{k}}\right)+\left(\hat{\boldsymbol{\jmath}}\times\hat{\boldsymbol{k}}+\hat{\boldsymbol{\jmath}}\times\hat{\boldsymbol{\imath}}\right)+\left(\hat{\boldsymbol{k}}\times\hat{\boldsymbol{\imath}}+\hat{\boldsymbol{k}}\times\hat{\boldsymbol{\jmath}}\right)$$
Use the cross products of the unit vectors:
$$=\left(\hat{\boldsymbol{k}}-\hat{\boldsymbol{k}}\right)+\left(\hat{\boldsymbol{\imath}}-\hat{\boldsymbol{k}}\right)+\left(\hat{\boldsymbol{\jmath}}-\hat{\boldsymbol{\imath}}\right)=0$$

Part B

Find the value of each term:
$$\vec{\boldsymbol{r}}\times\hat{\boldsymbol{\imath}}=\left(x\hat{\boldsymbol{\imath}}+y\hat{\boldsymbol{\jmath}}+z\hat{\boldsymbol{k}}\right)\times\hat{\boldsymbol{\imath}}=x(\hat{\boldsymbol{\imath}}\times\hat{\boldsymbol{\imath}})+y(\hat{\boldsymbol{\jmath}}\times\hat{\boldsymbol{\imath}})+z\left(\hat{\boldsymbol{k}}\times\hat{\boldsymbol{\imath}}\right)=-y\hat{\boldsymbol{k}}+z\hat{\boldsymbol{\jmath}}$$
$$\vec{\boldsymbol{r}}\times\hat{\boldsymbol{\jmath}}=\left(x\hat{\boldsymbol{\imath}}+y\hat{\boldsymbol{\jmath}}+z\hat{\boldsymbol{k}}\right)\times\hat{\boldsymbol{\jmath}}=x(\hat{\boldsymbol{\imath}}\times\hat{\boldsymbol{\jmath}})+y(\hat{\boldsymbol{\jmath}}\times\hat{\boldsymbol{\jmath}})+z\left(\hat{\boldsymbol{k}}\times\hat{\boldsymbol{\jmath}}\right)=x\hat{\boldsymbol{k}}-z\hat{\boldsymbol{\imath}}$$

Combine the above:
$$(\vec{\boldsymbol{r}}\times\hat{\boldsymbol{\imath}})\cdot(\vec{r}\times\hat{\boldsymbol{\jmath}})=\left(-y\hat{\boldsymbol{k}}+z\hat{\boldsymbol{\jmath}}\right)\cdot\left(x\hat{\boldsymbol{k}}-z\hat{\boldsymbol{\imath}}\right)=-yx+yz\cdot 0+zx\cdot 0+z^2\cdot 0=-yx$$

And finally:
$$(\vec{\boldsymbol{r}}\times\hat{\boldsymbol{\imath}})\cdot(\vec{r}\times\hat{\boldsymbol{\jmath}})+xy=-yx+xy=0$$

MCMC 3.23

Mark all correct options

Which condition ensures that $\vec{\boldsymbol{p}}\times\vec{\boldsymbol{q}}=\vec{\boldsymbol{p}}\times\vec{\boldsymbol{r}}$ and $\vec{\boldsymbol{p}}\cdot\vec{\boldsymbol{q}}=\vec{\boldsymbol{p}}\cdot\vec{\boldsymbol{r}}$:

A. $\vec{\boldsymbol{p}}=\vec{\boldsymbol{r}}$
B. $\vec{\boldsymbol{q}}=\vec{\boldsymbol{r}}$
C. $\vec{\boldsymbol{p}}=\vec{\boldsymbol{q}}$
D. $\vec{\boldsymbol{p}}+\vec{\boldsymbol{q}}=0$
E. $\vec{\boldsymbol{p}}=0$ (AP EAPCET, 22 Sep 2020, Shift-II, Adapted)

	Cross Product	Dot Product
Condition	$\vec{\boldsymbol{p}}\times\vec{\boldsymbol{q}}=\vec{\boldsymbol{p}}\times\vec{\boldsymbol{r}}$	$\vec{\boldsymbol{p}}\cdot\vec{\boldsymbol{q}}=\vec{\boldsymbol{p}}\cdot\vec{\boldsymbol{r}}$
Collate all terms on one side	$\vec{\boldsymbol{p}}\times\vec{\boldsymbol{q}}-\vec{\boldsymbol{p}}\times\vec{\boldsymbol{r}}=0$	$\vec{\boldsymbol{p}}\cdot\vec{\boldsymbol{q}}-\vec{\boldsymbol{p}}\cdot\vec{\boldsymbol{r}}=0$
Use the distributive property	$\vec{\boldsymbol{p}}\times(\vec{\boldsymbol{q}}-\vec{\boldsymbol{r}})=0$	$\vec{\boldsymbol{p}}\cdot(\vec{\boldsymbol{q}}-\vec{\boldsymbol{r}})=0$
Conclusion	Either the vectors are collinear, or one of the vectors is zero.	Either the vectors are perpendicular, or one of the vectors is zero.

The vectors cannot be collinear and perpendicular simultaneously. Hence, one of them must be zero.[3]
$$\vec{\boldsymbol{p}}=0\Rightarrow Option\ E$$
$$\vec{\boldsymbol{q}}-\vec{\boldsymbol{r}}=\mathbf{0}\Rightarrow\vec{\boldsymbol{q}}=\vec{\boldsymbol{r}}\Rightarrow Option\ B$$
Hence, the final answer is:
$$Options\ B,E$$

E. Anti-Commutative Property

3.24: Cross Product is anti-Commutative

$$\vec{\boldsymbol{a}}\times\vec{\boldsymbol{b}}=-\vec{\boldsymbol{b}}\times\vec{\boldsymbol{a}}$$

Example 3.25

A. Recall that the Cartesian product of two sets A and B, is the set of all ordered pairs (a,b) where a is in A and b is in B. Find the sum of the cross products of each element of the Cartesian product of the set $\left\{\vec{\boldsymbol{a}},\vec{\boldsymbol{b}},\vec{\boldsymbol{c}}\right\}$ with itself.

[3] This question is concept-intensive (distributive property) rather than calculation intensive.

B. Is the above result applicable to the set of n vectors $\{\vec{v_1}, \vec{v_2}, \dots, \vec{v_n}\}$. Explain why or why not.

Part A

$$(\vec{a}\times\vec{a}) + (\vec{b}\times\vec{b}) + (\vec{c}\times\vec{c}) + (\vec{a}\times\vec{b}) + (\vec{a}\times\vec{c}) + (\vec{b}\times\vec{c}) + (\vec{b}\times\vec{a}) + (\vec{c}\times\vec{a}) + (\vec{c}\times\vec{b})$$

Note that the first three terms all have the cross product of a vector with itself, and hence the output is zero:

$$0 + 0 + 0 + (\vec{a}\times\vec{b}) + (\vec{a}\times\vec{c}) + (\vec{b}\times\vec{c}) + (\vec{b}\times\vec{a}) + (\vec{c}\times\vec{a}) + (\vec{c}\times\vec{b})$$

Use the anti-commutative property in the last three terms:

$$= (\vec{a}\times\vec{b}) + (\vec{a}\times\vec{c}) + (\vec{b}\times\vec{c}) + [-(\vec{a}\times\vec{b})] + [-(\vec{a}\times\vec{c})] + [-(\vec{b}\times\vec{c})]$$

And these all add up to zero:

$$= 0$$

Part B

Since we are taking the Cartesian product of the set with itself, there are two cases:

Case I: You take the cross product of a vector with itself. This is zero.

Case II: You take the cross product of a vector with a vector that is not itself. In such a case, for every pair $(\vec{a}, \vec{b})$, there will be a pair $(\vec{b}, \vec{a})$ whose cross product will be the negative of the first pair.

Hence, the total must equal zero.

F. Scalar Multiplication

3.26: Scalar Multiplication

The cross product is commutative with scalar multiplication. In other words, you can "factor" any scalar out of a cross product.

$$\lambda(\vec{a}\times\vec{b}) = (\lambda\vec{a})\times\vec{b} = \vec{a}\times(\lambda\vec{b})$$

Use the definition $(\vec{a}\times\hat{b}) = |\vec{a}||\vec{b}|\sin\theta\,\hat{n}$ to expand each term and get:

$$\lambda(\vec{a}\times\vec{b}) = \lambda|\vec{a}||\vec{b}|\sin\theta\,\hat{n}$$
$$(\lambda\vec{a})\times\vec{b} = |\lambda\vec{a}||\vec{b}|\sin\theta\,\hat{n}$$
$$\vec{a}\times(\lambda\vec{b}) = |\vec{a}||\lambda\vec{b}|\sin\theta\,\hat{n}$$

By the properties of scalar multiplication, the second and third terms are equal to the first.

Example 3.27

Find the value of λ:

$$(3\vec{a})\times(2\vec{b}) = \lambda(\vec{a}\times\vec{b})$$

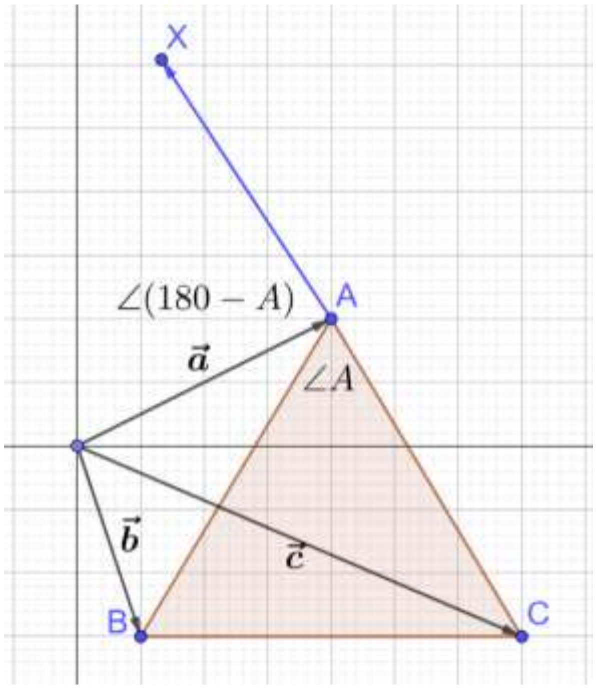
$$(3\vec{a})\times(2\vec{b}) = 6(\vec{a}\times\vec{b}) = \lambda(\vec{a}\times\vec{b}) \Rightarrow \lambda = 6$$

G. Triangles

Example 3.28

If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are position vectors of the vertices of ΔABC, then, in terms of the angles of the triangle (AP EAPCET, 22 Sep 2020, Shift-II, Adapted)

$$\frac{|(\vec{a}-\vec{c})\times(\vec{b}-\vec{a})|}{(\vec{b}-\vec{a})\cdot(\vec{c}-\vec{a})} =$$



Using vector addition and subtraction, the required expression is:
$$= \frac{|\overrightarrow{CA} \times \overrightarrow{AB}|}{\overrightarrow{AB} \cdot \overrightarrow{AC}}$$
Expand the numerator using the definition of the cross product. Note that $\overrightarrow{CA}$ and $\overrightarrow{AB}$ are not arranged tail to tail and hence move $\overrightarrow{CA}$ forward so as to arrange the vectors tail to tail. The angle between them is not $180 - A$, rather than A.
$$|\overrightarrow{CA} \times \overrightarrow{AB}| = |\overrightarrow{CA}||\overrightarrow{AB}|\sin(\pi - A)$$

Expand the denominator using the definition of the dot product:
$$\overrightarrow{AB} \cdot \overrightarrow{AC} = |\overrightarrow{AB}||\overrightarrow{AC}|\cos A$$

Combine the two:
$$\frac{|\overrightarrow{CA}||\overrightarrow{AB}|\sin(\pi - A)}{|\overrightarrow{AB}||\overrightarrow{AC}|\cos A}$$
Note that $|\overrightarrow{CA}| = |\overrightarrow{AC}|$ since the magnitudes are same (even though the vectors are different). Also, use the property that $\sin(\pi - \theta) = \sin\theta$ and simplify:
$$= \frac{\sin A}{\cos A} = \tan A$$

3.29: Area of Triangle

The area of a triangle with $\vec{a}$ and $\vec{b}$ for its sides is:
$$\frac{1}{2}|\vec{a} \times \vec{b}|$$

Consider a triangle with $\vec{a}$ and $\vec{b}$ for its sides. From trigonometry, we know that the area of the triangle is:
$$\frac{1}{2}|\vec{a}||\vec{b}|\sin\theta = \frac{1}{2}|\vec{a} \times \vec{b}|$$

Example 3.30

Vectors $\vec{x}$, $\vec{y}$, and $\vec{z}$ are the sides of a triangle.

A. Determine the magnitude of the cross product of $\vec{x}$ and $\vec{y}$ if the cross product of $\vec{y}$ and $\vec{z}$ is 7.
B. Determine the magnitude of the cross product of $\vec{x}$ and $\vec{y}$ if the area of the triangle formed by the vectors is 7.

Part A
$$2 \times Area\ of\ Triangle = |\vec{x} \times \vec{y}| = |\vec{y} \times \vec{z}| = 7$$
Part B
$$|\vec{x} \times \vec{y}| = 2 \times Area\ of\ Triangle = 2 \times 7 = 14$$

MCMC 3.31: Revision (Condition for Triangle)

Mark all correct options

If three vectors form the sides of a triangle, then:

A. A linear combination of two of them must be a scalar multiple of the third.
B. The sum of the three vectors must be zero.

C. The dot product of two of them must be non-zero.
D. The cross product of two of them must be zero.

The sum of the vectors that make a vector polygon is zero.

$$Option\ B \Rightarrow\ \vec{a}+\vec{b}+\vec{c}=0$$

$$Option\ A\ is\ correct$$

Example 3.32

Given three vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$, such that $\vec{a}+\vec{b}+\vec{c}=0$, show that:

A. $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}$
B. the magnitude of the cross product of any two of them is the same.

Part A

Setup:

The possible pairs are:

$$(\vec{a}\times\vec{b}),(\vec{a}\times\vec{c}),(\vec{b}\times\vec{c}),(\vec{b}\times\vec{a}),(\vec{c}\times\vec{a}),(\vec{c}\times\vec{b})$$

Prove $(\vec{b}\times\vec{c})=\vec{a}\times\vec{b}$

Substitute $\vec{a}+\vec{b}+\vec{c}=0\Rightarrow\vec{c}=-(\vec{a}+\vec{b})$:

$$\vec{b}\times\vec{c}=\vec{b}\times[-(\vec{a}+\vec{b})]$$

Move the minus sign outside (scalar multiple property), and distribute the cross product over the addition:

$$=-\vec{b}\times[(\vec{a}+\vec{b})]=-(\vec{b}\times\vec{a}+\vec{b}\times\vec{b})$$

Use the anti-commutative property (first term), and note that the product of a vector with itself is zero (for the second term):

$$=-(-\vec{a}\times\vec{b}+\vec{0})=\vec{a}\times\vec{b}$$

Prove $(\vec{c}\times\vec{a})=\vec{a}\times\vec{b}$

We can prove this using similar steps as above:

$$\vec{c}\times\vec{a}=\underbrace{[-(\vec{a}+\vec{b})]\times\vec{a}}_{Substitute}=\underbrace{-[(\vec{a}+\vec{b})]\times\vec{a}}_{\substack{Scalar\ Multiple\\ Property}}=\underbrace{-(\vec{a}\times\vec{a}+\vec{b}\times\vec{a})}_{\substack{Distributive\\ Property}}=\underbrace{-(-\vec{a}\times\vec{a}-\vec{a}\times\vec{b})}_{\substack{Anti\ Commutative\\ Property}}=\underbrace{0+\vec{a}\times\vec{b}}_{\substack{Cross\ Product\\ of\ \|vectors}}=\vec{a}\times\vec{b}$$

Part B

By the anti-commutative property:

$$\vec{b}\times\vec{a}=-(\vec{a}\times\vec{b})$$
$$\vec{a}\times\vec{c}=-(\vec{c}\times\vec{a})=-(\vec{a}\times\vec{b})$$
$$\vec{c}\times\vec{b}=-(\vec{b}\times\vec{c})=-(\vec{a}\times\vec{b})$$

Hence all six cross products have the same magnitude.

MCMC 3.33:

Mark all Correct Options

If the vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ form the sides BC, CA and AB respectively of a ΔABC, then:

A. $\vec{a}\cdot\vec{b}=\vec{b}\cdot\vec{c}=\vec{c}\cdot\vec{b}=0$
B. $\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}$
C. $\vec{a}\cdot\vec{b}=\vec{b}\cdot\vec{c}=\vec{c}\cdot\vec{a}=0$
D. $\vec{a}\times\vec{a}+\vec{b}\times\vec{b}+\vec{c}\times\vec{c}=0$

E. $\vec{a}\times\vec{a}+\vec{a}\times\vec{c}+\vec{a}\times\vec{a}=0$ (JEE Main 2002, Adapted)[4]

Since the vectors form the sides of the triangle, the sum of the vectors is zero.

$$\vec{a}+\vec{b}+\vec{c}=0\Rightarrow\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}\Rightarrow Option\ A$$

Since a vector is parallel to itself, the cross product of a vector with itself is zero:

$$\vec{a}\times\vec{a}+\vec{a}\times\vec{c}+\vec{a}\times\vec{a}=0+0+0=\mathbf{0}\Rightarrow Option\ D$$

Hence, the final answer is:

$$Options\ A, D$$

3.34: Cross Product when sum of three Vectors is zero

Given three vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$, such that $\vec{a}+\vec{b}+\vec{c}=0$, their cross products when taken in order are equal:

$$\vec{a}\times\vec{b}=\vec{b}\times\vec{c}=\vec{c}\times\vec{a}$$

This is an important property.

Correct the Proof 3.35:

Consider the following "proof" that $\vec{a}+\vec{b}+\vec{c}=0\Rightarrow|\vec{a}\times\vec{b}|=|\vec{b}\times\vec{c}|=|\vec{c}\times\vec{a}|$.

Step I: $\vec{a}+\vec{b}+\vec{c}=0$ means that the vectors form the sides of a triangle. Call it ΔA

Step II: From trigonometry, we know that:

$$Area\ of\ \Delta A=\frac{1}{2}|\vec{a}||\vec{b}|\sin\theta_{\vec{a}\vec{b}}=\frac{1}{2}|\vec{b}||\vec{c}|\sin\theta_{\vec{b}\vec{c}}=\frac{1}{2}|\vec{c}||\vec{a}|\sin\theta_{\vec{c}\vec{a}}$$

(where the angles are then angles between the vectors).

Step III: Note that the expressions in Step II are precisely the expressions for the magnitude of the cross product:

$$Area\ of\ \Delta A=|\vec{a}\times\vec{b}|=|\vec{b}\times\vec{c}|=|\vec{c}\times\vec{a}|$$

Is there any flaw in the proof? Is so, correct and complete it.

Fix the Problem with Step II

The trigonometry formula is incorrect. It has:

- A missing factor of $\frac{1}{2}$
- The angle between the vectors is when they are arranged tip to tip or tail to tail. Hence, we need to write:

$$Area\ of\ \Delta A=\frac{1}{2}|\vec{a}||\vec{b}|\sin(180-\theta_{\vec{a}\vec{b}})=\frac{1}{2}|\vec{b}||\vec{c}|\sin(180-\theta_{\vec{b}\vec{c}})=\frac{1}{2}|\vec{c}||\vec{a}|\sin(180-\theta_{\vec{c}\vec{a}})$$

Use $\sin(180-\theta)=\sin\theta$:

$$Area\ of\ \Delta A=\frac{1}{2}|\vec{a}||\vec{b}|\sin\theta_{\vec{a}\vec{b}}=\frac{1}{2}|\vec{b}||\vec{c}|\sin\theta_{\vec{b}\vec{c}}=\frac{1}{2}|\vec{c}||\vec{a}|\sin\theta_{\vec{c}\vec{a}}\ M$$

Multiply throughout by 2:

$$2\times Area\ of\ \Delta A=|\vec{a}||\vec{b}|\sin\theta_{\vec{a}\vec{b}}=|\vec{b}||\vec{c}|\sin\theta_{\vec{b}\vec{c}}=|\vec{c}||\vec{a}|\sin\theta_{\vec{c}\vec{a}}\ M$$

And now we can make the substitution to get:

$$2\times Area\ of\ \Delta A=|\vec{a}\times\vec{b}|=|\vec{b}\times\vec{c}|=|\vec{c}\times\vec{a}|$$

[4] The original question was easier. It was a single correct question (Option D was missing). This is a question directly based on your knowledge of properties.

The proof is correct when the vectors form the sides of a triangle. We need to consider the cases when the vectors do not form a triangle.

The vectors will not form a triangle if:
- One or more of the vectors is zero
- The vectors are collinear

Collinear, but non-zero Vectors

We consider sub-cases.
Exactly One Vector is Collinear: Collinearity requires two vectors. So, this is not a meaningful case.
If at least two Vectors are Collinear:
If two vectors are collinear, then $\vec{\boldsymbol{b}} = \lambda\vec{\boldsymbol{a}}$ for some scalar λ.
Substitute this in $\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}} + \vec{\boldsymbol{c}} = 0$:

$$\vec{\boldsymbol{a}} + \lambda\vec{\boldsymbol{a}} + \vec{\boldsymbol{c}} = 0 \Rightarrow \vec{\boldsymbol{a}}(1+\lambda) = -\vec{\boldsymbol{c}} \Rightarrow \vec{\boldsymbol{a}} = -\frac{1}{1+\lambda}\vec{\boldsymbol{c}}$$

Hence, we conclude that all three vectors are collinear. The cross product of collinear vectors is zero. Hence:

$$|\vec{\boldsymbol{a}} \times \vec{\boldsymbol{b}}| = |\vec{\boldsymbol{b}} \times \vec{\boldsymbol{c}}| = |\vec{\boldsymbol{c}} \times \vec{\boldsymbol{a}}| = 0$$

Zero Vectors

We consider sub-cases:
A single vector is zero

$$\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}} + \vec{\boldsymbol{c}} = 0 \Rightarrow \vec{\boldsymbol{b}} = -\vec{\boldsymbol{c}}$$

The other vectors must be collinear for their sum to be zero. Hence:

$$|\vec{\boldsymbol{a}} \times \vec{\boldsymbol{b}}| = |\vec{\boldsymbol{b}} \times \vec{\boldsymbol{c}}| = |\vec{\boldsymbol{c}} \times \vec{\boldsymbol{a}}| = 0$$

At least two vectors are zero

$$\vec{\boldsymbol{a}} + \vec{\boldsymbol{b}} + \vec{\boldsymbol{c}} = 0 \Rightarrow \vec{\boldsymbol{c}} = 0$$

If at least two vectors are zero, the third vector must be zero for their sum to be zero. And the cross product of a zero vector is zero:

$$|\vec{\boldsymbol{a}} \times \vec{\boldsymbol{b}}| = |\vec{\boldsymbol{b}} \times \vec{\boldsymbol{c}}| = |\vec{\boldsymbol{c}} \times \vec{\boldsymbol{a}}| = 0$$

3.2 Component Definition

A. Component Definition

The component definition of the cross product is useful when the input vectors are available in component form. This definition is in terms of determinants.[5]

3.36: Component Definition

If $\vec{\boldsymbol{a}} = a_1\hat{\boldsymbol{\imath}} + a_2\hat{\boldsymbol{\jmath}} + a_3\hat{\boldsymbol{k}}$ and $\vec{\boldsymbol{b}} = b_1\hat{\boldsymbol{\imath}} + b_2\hat{\boldsymbol{\jmath}} + b_3\hat{\boldsymbol{k}}$ then their cross product is:

$$\vec{\boldsymbol{a}} \times \vec{\boldsymbol{b}} = \begin{vmatrix} \hat{\boldsymbol{\imath}} & \hat{\boldsymbol{\jmath}} & \hat{\boldsymbol{k}} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

Expanding the determinant along the first column gives us:

$$\vec{\boldsymbol{a}} \times \vec{\boldsymbol{b}} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix}\hat{\boldsymbol{\imath}} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix}\hat{\boldsymbol{\jmath}} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}\hat{\boldsymbol{k}}$$

Expanding each determinant gives us:

[5] You can look up determinants in the note on Determinants and Matrices.

$$= (a_2b_3 - a_3b_2)\hat{\boldsymbol{\imath}} - (a_1b_3 - a_3b_1)\hat{\boldsymbol{\jmath}} + (a_1b_2 - a_2b_1)\hat{\boldsymbol{k}}$$

Example 3.37

A. Find $\vec{\boldsymbol{a}} \times \vec{\boldsymbol{b}}$ if $\vec{\boldsymbol{a}} = 2\hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}} + 3\hat{\boldsymbol{k}}$ and $\vec{\boldsymbol{b}} = 3\hat{\boldsymbol{\imath}} + 5\hat{\boldsymbol{\jmath}} - 2\hat{\boldsymbol{k}}$ (CBSE 2019)

B. Find λ and μ if $(\hat{\boldsymbol{\imath}} + 3\hat{\boldsymbol{\jmath}} + 9\hat{\boldsymbol{k}}) \times (3\hat{\boldsymbol{\imath}} - \lambda\hat{\boldsymbol{\jmath}} + \mu\hat{\boldsymbol{k}}) = \vec{\boldsymbol{0}}$. (CBSE 2016)

C. In Part B, you would have found three equations in two variables. You can find the solution to both the variables from two equations. Is the step where we check in the third equation useful?

Part A

Using the definition, we get $\vec{\boldsymbol{a}} \times \vec{\boldsymbol{b}} = \begin{vmatrix} \hat{\boldsymbol{\imath}} & \hat{\boldsymbol{\jmath}} & \hat{\boldsymbol{k}} \\ 2 & 1 & 3 \\ 3 & 5 & -2 \end{vmatrix}$

$$= \begin{vmatrix} 1 & 3 \\ 5 & -2 \end{vmatrix}\hat{\boldsymbol{\imath}} - \begin{vmatrix} 2 & 3 \\ 3 & -2 \end{vmatrix}\hat{\boldsymbol{\jmath}} + \begin{vmatrix} 2 & 1 \\ 3 & 5 \end{vmatrix}\hat{\boldsymbol{k}}$$

$$= (-2 - 15)\hat{\boldsymbol{\imath}} - (-4 - 9)\hat{\boldsymbol{\jmath}} + (10 - 3)\hat{\boldsymbol{k}}$$

$$= -17\hat{\boldsymbol{\imath}} + 13\hat{\boldsymbol{\jmath}} + 7\hat{\boldsymbol{k}}$$

Part B

Expanding the determinant definition of the cross product $\begin{vmatrix} \hat{\boldsymbol{\imath}} & \hat{\boldsymbol{\jmath}} & \hat{\boldsymbol{k}} \\ 1 & 3 & 9 \\ 3 & -\lambda & \mu \end{vmatrix} = \vec{\boldsymbol{0}}$:

$$(3\mu + 9\lambda)\hat{\boldsymbol{\imath}} - (\mu - 27)\hat{\boldsymbol{\jmath}} + (-\lambda - 9)\hat{\boldsymbol{k}} = \boldsymbol{0}\hat{\boldsymbol{\imath}} + 0\hat{\boldsymbol{\jmath}} + 0\hat{\boldsymbol{k}}$$

Comparing coefficients on both sides:

$$\underbrace{\mu - 27 = 0 \Rightarrow \mu = 27}_{Compare\ \hat{\jmath}}, \qquad \underbrace{3\mu + 9\lambda = 0 \Rightarrow \lambda = -\frac{\mu}{3} = -\frac{27}{3} = -9}_{Compare\ \hat{\imath}}$$

Check in the third equation:

$$Compare\ \hat{\boldsymbol{k}}: -\lambda - 9 = 0 \Rightarrow \lambda = -9$$

Part C

We have two variables and three equations, which means the system is overdetermined. If the values of μ and λ that we find from the first two equations do not satisfy the third equation, then

- The system of equations is inconsistent
- Hence, it has no solutions
- Hence, there would be no values of μ and λ to be found.

Therefore, it is useful/important to check the third equation.

3.38: Cross product of vectors in the xy plane

If $\vec{\boldsymbol{a}} = a_1\hat{\boldsymbol{\imath}} + a_2\hat{\boldsymbol{\jmath}}$ and $\vec{\boldsymbol{b}} = b_1\hat{\boldsymbol{\imath}} + b_2\hat{\boldsymbol{\jmath}}$ then their cross product is:

$$\vec{\boldsymbol{a}} \times \vec{\boldsymbol{b}} = (a_1b_2 - a_2b_1)\hat{\boldsymbol{k}}$$

$$\vec{\boldsymbol{a}} \times \vec{\boldsymbol{b}} = \begin{vmatrix} \hat{\boldsymbol{\imath}} & \hat{\boldsymbol{\jmath}} & \hat{\boldsymbol{k}} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} \hat{\boldsymbol{\imath}} & \hat{\boldsymbol{\jmath}} & \hat{\boldsymbol{k}} \\ a_1 & a_2 & 0 \\ b_1 & b_2 & 0 \end{vmatrix}$$

The determinant is:

$$= (a_2b_3 - a_3b_2)\hat{\boldsymbol{\imath}} - (a_1b_3 - a_3b_1)\hat{\boldsymbol{\jmath}} + (a_1b_2 - a_2b_1)\hat{\boldsymbol{k}}$$

But since $a_3 = b_3 = 0$, the first two terms become zero:

$$= (a_1b_2 - a_2b_1)\hat{\boldsymbol{k}}$$

3.39: Cross product of vectors in a plane
The cross product of two vectors in a plane is perpendicular to the plane.

B. Angle Between Vectors

3.40: Angle Between Vectors

$$\sin\theta = \frac{|\vec{a}\times\vec{b}|}{|\vec{a}||\vec{b}|}$$

- Rearranging the definition of the cross product gives us the property above.

Example 3.41

A. If $|\vec{a}| = 2$, $|\vec{b}| = 7$ and $\vec{a}\times\vec{b} = 3\hat{\imath} + 2\hat{\jmath} + 6\hat{k}$, find the angle between $\vec{a}$ and $\vec{b}$. (CBSE 2019)

B. If θ is the angle between two vectors $\hat{\imath} - 2\hat{\jmath} + 3\hat{k}$ and $3\hat{\imath} - 2\hat{\jmath} + \hat{k}$, find $\sin\theta$. (CBSE 2018)

Part A

$$\sin\theta = \frac{|\vec{a}\times\vec{b}|}{|\vec{a}||\vec{b}|} = \frac{\sqrt{3^2+2^2+6^2}}{2\times 7} = \frac{\sqrt{49}}{2\times 7} = \frac{7}{2\times 7} = \frac{1}{2} \Rightarrow \theta = \sin^{-1}\frac{1}{2} = \frac{\pi}{6}$$

Part B

The cross product of the two vectors is:

$$\begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 1 & -2 & 3 \\ 3 & -2 & 1 \end{vmatrix} = (-2+6)\hat{\imath} - (1-9)\hat{\jmath} + (-2+6)\hat{k} = 4\hat{\imath} + 8\hat{\jmath} + 4\hat{k} = \mathbf{4}(\hat{\imath} + 2\hat{\jmath} + 1\hat{k})$$

Rearranging the definition of the cross product to solve for $\sin\theta$:

$$\sin\theta = \frac{|\vec{a}\times\vec{b}|}{|\vec{a}||\vec{b}|} = \frac{4\sqrt{1^2+2^2+1^2}}{\sqrt{1^2+(-2)^2+3^2}\sqrt{3^2+(-2)^2+1^2}} = \frac{4\sqrt{6}}{\sqrt{14}\sqrt{14}} = \frac{4\sqrt{6}}{14} = \frac{2\sqrt{6}}{7}$$

C. Finding Perpendicular Vectors

Given two vectors on a plane, the vectors perpendicular to them can go "up" or "down" respective to the plane. It is logical that the cross product be used to find these vectors since the output of the cross product is perpendicular to its two input vectors.

- If the length of the vector is determined then we can find two such vectors. For example, the vector may be a unit vector.
- Other conditions can be imposed to uniquely determine the vector. For example, by giving the dot product of the vector with a third vector.

3.42: Unit Perpendicular Vector
The unit vector perpendicular to two vectors is given by:

$$\hat{\boldsymbol{n}} = \pm\frac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}$$

Example 3.43

If $\vec{a} = \hat{\imath} + 2\hat{\jmath} + \hat{k}$, $\boldsymbol{b} = 2\hat{\imath} + \hat{\jmath}$ and $\boldsymbol{c} = 3\hat{\imath} - 4\hat{\jmath} - 5\hat{k}$, then find a unit vector perpendicular to both of the vectors $(\vec{a} - \vec{b})$ and $(\vec{c} - \vec{b})$ (CBSE 2015)

$$\vec{a}-\vec{b}=\hat{\imath}+2\hat{\jmath}+\hat{k}-(2\hat{\imath}+\hat{\jmath})=-\hat{\imath}+\hat{\jmath}+\hat{k}$$
$$\vec{c}-\vec{b}=3\hat{\imath}-4\hat{\jmath}-5\hat{k}-(2\hat{\imath}+\hat{\jmath})=\hat{\imath}-5\hat{\jmath}-5\hat{k}$$

A vector perpendicular to the above two vectors is given by their cross product:

$$\begin{vmatrix}\hat{\imath} & \hat{\jmath} & \hat{k}\\ -1 & 1 & 1\\ 1 & -5 & -5\end{vmatrix}=(-5+5)\hat{\imath}-(5-1)\hat{\jmath}+(5-1)\hat{k}=-4\hat{\jmath}+4\hat{k}=4(-\hat{\jmath}+\hat{k})$$

Convert the above vector into a unit vector by dividing it by its magnitude:

$$\frac{4(-\hat{\jmath}+\hat{k})}{4\sqrt{(-1)^2+1^2}}=\frac{-\hat{\jmath}+\hat{k}}{\sqrt{2}}=-\frac{\sqrt{2}}{2}\hat{\jmath}+\frac{\sqrt{2}}{2}\hat{k}$$

The question asks for only one vector, but we can find one more vector by taking the negative:

$$\frac{\sqrt{2}}{2}\hat{\jmath}-\frac{\sqrt{2}}{2}\hat{k}$$

D. Collinearity

Example 3.44

If $\vec{a}=\alpha\hat{\imath}+3\hat{\jmath}-6\hat{k}$ and $\vec{b}=2\hat{\imath}-\hat{\jmath}+\beta\hat{k}$, then the values of α,β so that $\vec{a}$ and $\vec{b}$ may be collinear are: (AP EAPCET, 17 Sep 2020, Shift-I)

Method I: Determinant Condition

$$\vec{a}\times\vec{b}=\begin{vmatrix}\hat{\imath} & \hat{\jmath} & \hat{k}\\ \alpha & 3 & -6\\ 2 & -1 & \beta\end{vmatrix}=(3\beta-6)\hat{\imath}-(\alpha\beta+12)\hat{\jmath}+(-\alpha-6)\hat{k}=\mathbf{0}$$

Compare coefficients on both sides:

$$3\beta-6=0\Rightarrow\beta=2,\qquad -\alpha-6=0\Rightarrow\alpha=-6$$

Substitute the above into the third equation and check that it works:

$$\alpha\beta+12=(2)(-6)+12=0\Rightarrow Consistent$$

Method II: Scalar Multiple Condition

Recall that two vectors are collinear $if\ and\ only\ if$ one is a scalar multiple of the other.

Compare the $\hat{\jmath}$ coefficients on both that $\vec{a}$ and $\vec{b}$ tells us that

$$3=(-3)(-1)\Rightarrow\vec{a}=(-3)\vec{b}$$

Hence:

$$\alpha=(-3)2=-6,\qquad \beta=\frac{-6}{-3}=2$$

E. Dot Product

3.45: Dot Product

Recall from the previous section that the dot product of two vectors (x_1,y_1,z_1) and (x_2,y_2,z_2) is:

$$(x_1,y_1,z_1)\cdot(x_2,y_2,z_2)=x_1x_2+y_1y_2+z_1z_2$$

Example 3.46

A. Let $\vec{a}=4\hat{\imath}+5\hat{\jmath}-\hat{k},\vec{b}=\hat{\imath}-4\hat{\jmath}+5\hat{k}$ and $\vec{c}=3\hat{\imath}+\hat{\jmath}-\hat{k}$. Find a vector $\vec{d}$ which is perpendicular to both $\vec{c}$ and $\vec{b}$ and $\vec{d}\cdot\vec{a}=21$. (CBSE 2019)

B. If $\vec{a} = \hat{\imath} + \hat{\jmath} + \hat{k}, \vec{b} = \hat{\jmath} - \hat{k}$ then, find a vector $\vec{c}$ such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$. (CBSE 2008, 2013)

Part A

$\vec{c} \times \vec{b}$ is perpendicular to both $\vec{c}$ and $\vec{b}$. Hence, $\vec{d}$ must be a scalar multiple of the cross product of $\vec{c}$ and $\vec{b}$.

$$\vec{d} = \lambda(\vec{c} \times \vec{b}) = \lambda \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 3 & 1 & -1 \\ 1 & -4 & 5 \end{vmatrix} = \lambda[(5-4)\hat{\imath} - (15+1)\hat{\jmath} + (-12-1)\hat{k}] = \lambda(\hat{\imath} - 16\hat{\jmath} - 13\hat{k})$$

Substitute $\vec{d} = \lambda(\hat{\imath} - 16\hat{\jmath} - 13\hat{k})$, and $\vec{a} = 4\hat{\imath} + 5\hat{\jmath} - \hat{k}$ into $\vec{d} \cdot \vec{a} = 21$:

$$\lambda(\hat{\imath} - 16\hat{\jmath} - 13\hat{k}) \cdot (4\hat{\imath} + 5\hat{\jmath} - \hat{k}) = 21 \Rightarrow \lambda(4 - 80 + 13) = 21 \Rightarrow \lambda = -\frac{21}{63} = -\frac{1}{3}$$

Substitute the value of λ into $\vec{d} = \lambda(\hat{\imath} - 16\hat{\jmath} - 13\hat{k})$:

$$\vec{d} = -\frac{1}{3}(\hat{\imath} - 16\hat{\jmath} - 13\hat{k})$$

Part B

Cross Product Condition

$$Let\ \vec{c} = x\hat{\imath} + y\hat{\jmath} + z\hat{k} \Rightarrow (\vec{a} \times \vec{c}) = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 1 & 1 & 1 \\ x & y & z \end{vmatrix} = (z-y)\hat{\imath} - (z-x)\hat{\jmath} + (y-x)\hat{k} = \hat{\jmath} - \hat{k} = \vec{b}$$

Compare coefficients on both sides:

$$\underbrace{z - y = 0 \Rightarrow z = y}_{Compare\ \hat{\imath},\ Equation\ I}, \qquad \underbrace{x - z = 1}_{Compare\ \hat{\jmath},\ Equation\ II}, \qquad \underbrace{y - x = -1}_{Compare\ \hat{k},\ Equation\ III}$$

Dot Product Condition

Substitute values in $\vec{a} \cdot \vec{c} = 3$ to get:

$$(\hat{\imath} + \hat{\jmath} + \hat{k}) \cdot (x\hat{\imath} + y\hat{\jmath} + z\hat{k}) = 3 \Rightarrow \underbrace{x + y + z = 3}_{Equation\ IV}$$

Combining the Conditions

Subtract Equation I from Equation IV to get Equation V, and then add Equations III and V to get Equation VI:

$$\underbrace{x + 2y = 3}_{Equation\ V} \Rightarrow \underbrace{3y = 2}_{Equation\ VI} \Rightarrow z = y = \frac{2}{3}$$

Substitute the value of y and z in Equations II and III and note that both give the same answer:

$$\underbrace{x = 1 + z = 1 + \frac{2}{3} = \frac{5}{3}}_{Equation\ II}, \underbrace{x = 1 + y = 1 + \frac{2}{3} = \frac{5}{3}}_{Equation\ III}$$

$$\vec{c} = x\hat{\imath} + y\hat{\jmath} + z\hat{k} = \frac{5}{3}\hat{\imath} + \frac{2}{3}\hat{\jmath} + \frac{2}{3}\hat{k}$$

F. Area of Parallelogram

3.47: Area of Parallelogram

The area of a parallelogram with $\vec{a}$ and $\vec{b}$ for its sides is:

$$|\vec{a} \times \vec{b}|$$

Draw parallelogram $ABCD$. Let

$$|\overrightarrow{AD}| = |\vec{a}| = s_2, |\overrightarrow{DC}| = |\vec{b}| = s_1$$

Draw the height from vertex $A \perp DC$.

In Right Triangle ΔADP:

$$\sin \angle D = \frac{h}{s_2} \Rightarrow h = \sin \angle D \times s_2$$

Area of Parallelogram

$$= hb = (\sin \angle D \times s_2) \times s_1 = s_1 s_2 \sin \angle D = |\vec{a}||\vec{b}| \sin \angle D = |\vec{a} \times \vec{b}|$$

4

Example 3.48

Find the area of a parallelogram whose adjacent sides are represented by the vectors $2\hat{\imath} - 3\hat{k}$ and $4\hat{\jmath} + 2\hat{k}$. (CBSE 2019)

The cross product of the two vectors is:

$$\begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & 0 & -3 \\ 0 & 4 & 2 \end{vmatrix} = (0+12)\hat{\imath} - (4+0)\hat{\jmath} + (8-0)\hat{k} = 12\hat{\imath} - 4\hat{\jmath} + 8\hat{k}$$

And the magnitude of the cross product is:

$$\sqrt{12^2 + (-4)^2 + 8^2} = \sqrt{144 + 16 + 64} = \sqrt{224} = 4\sqrt{14}\ units^2$$

3.49: Area of Parallelogram using Diagonals

The area of a triangle with $\vec{d_1}$ and $\vec{d_2}$ for its sides is:

$$\frac{1}{2}|\vec{d_1} \times \vec{d_2}|$$

5

Example 3.50

57: The two adjacent sides of a parallelogram are $2\hat{\imath} - 4\hat{\jmath} - 5\hat{k}$ and $2\hat{\imath} + 2\hat{\jmath} + 3\hat{k}$. Find the area of the parallelogram using its diagonal vectors. (CBSE 2016)

$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{AD} = 4\hat{\imath} - 2\hat{\jmath} - 2\hat{k}$$
$$\overrightarrow{BD} = \overrightarrow{AD} - \overrightarrow{AB} = 6\hat{\jmath} + 8\hat{k}$$

The cross product of the two vectors above is:

$$\overrightarrow{AC} \times \overrightarrow{BD} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 4 & -2 & -2 \\ 0 & 6 & 8 \end{vmatrix} = (-16+12)\hat{\imath} - (32+0)\hat{\jmath} + (24-0)\hat{k} = 4(-\hat{\imath} - 8\hat{\jmath} + 6\hat{k})$$

And the area of the parallelogram is half the magnitude of the cross product of the vectors:

$$\frac{1}{2}|\overrightarrow{AC} \times \overrightarrow{BD}| = \frac{1}{2} \times 4\sqrt{(-1)^2 + (-8)^2 + 6^2} = 2\sqrt{101}\ units^2$$

G. Area of Triangle

3.51: Area of Triangle

The area of a triangle with $\vec{a}$ and $\vec{b}$ for its sides is:

$$\frac{1}{2}|\vec{a} \times \vec{b}|$$

- Diagonal of a parallelogram divides it into two congruent triangles
- We want half of the area of the parallelogram.

6

Example 3.52

Two lines with slopes $\frac{1}{2}$ and 2 intersect at (2,2). What is the area of the triangle enclosed by these two lines and the line $x + y = 10$? (AMC 10A 2019/7)

Find the equations of the lines

Use the slope-intercept form of the equation of a line ($y = mx + c$) to find the equations of the two lines.

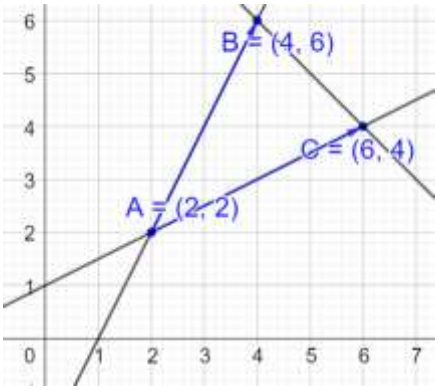
Substitute

$$(x, y) = (2,2), m = slope = 2 \Rightarrow 2 = 2x + c \Rightarrow c = -2$$

$$(x, y) = (2,2), m = slope = \frac{1}{2} \Rightarrow 2 = \frac{2}{2} + c \Rightarrow c = 1$$

Hence, the lines are:

$$y = 2x - 2, y = \frac{x}{2} + 1, y = -x + 10$$



Find the intersections

$$Given\ Intersection: (x, y) = (2,2)$$

$$2x - 2 = -x + 10 \Rightarrow 3x = 12 \Rightarrow x = 4 \Rightarrow y = 2x - 2 = 6 \Rightarrow (x, y) = (4,6)$$

$$\frac{x}{2} + 1 = -x + 10 \Rightarrow \frac{3}{2}x = 9 \Rightarrow x = 6 \Rightarrow y = \frac{x}{2} + 1 = 4 \Rightarrow (x, y) = (6,4)$$

Find the vectors

$$\overrightarrow{\boldsymbol{AB}} = (4 - 2, 6 - 2) = (2,4,0)$$

$$\overrightarrow{\boldsymbol{AC}} = (6 - 2, 4 - 2) = (4,2,0)$$

Find the cross product

The cross product is given by the expression below. Since row 2 and row 3 have zeroes in the last column, the

$$\overrightarrow{\boldsymbol{AB}} \times \overrightarrow{\boldsymbol{AC}} = \begin{vmatrix} \hat{\boldsymbol{\imath}} & \hat{\boldsymbol{\jmath}} & \hat{\boldsymbol{k}} \\ 2 & 4 & 0 \\ 4 & 2 & 0 \end{vmatrix} = (-16 + 12)\hat{\boldsymbol{\imath}} - (32 + 0)\hat{\boldsymbol{\jmath}} + (24 - 0)\hat{\boldsymbol{k}} = 4(-\hat{\boldsymbol{\imath}} - 8\hat{\boldsymbol{\jmath}} + 6\hat{\boldsymbol{k}})$$

3.3 Revision

A. Revision

Example 3.53

44: Let $\boldsymbol{a} = \hat{\boldsymbol{\imath}} + 5\hat{\boldsymbol{\jmath}} + \alpha\hat{\boldsymbol{k}}$, $\boldsymbol{b} = \hat{\boldsymbol{\imath}} + 3\hat{\boldsymbol{\jmath}} + \beta\hat{\boldsymbol{k}}$ and $\boldsymbol{c} = -\hat{\boldsymbol{\imath}} + 2\hat{\boldsymbol{\jmath}} - 3\hat{\boldsymbol{k}}$ be three vectors such that $|\vec{\boldsymbol{b}} \times \vec{\boldsymbol{c}}| = 5\sqrt{3}$ and $\vec{\boldsymbol{a}}$ is perpendicular to $\vec{\boldsymbol{b}}$. Then, the greatest among the values of $|\vec{\boldsymbol{a}}|^2$ is (JEE 2021, 27 Aug, Shift-I)

Since $\vec{\boldsymbol{a}} \perp \vec{\boldsymbol{b}}$, we must have:

$$\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}} = 0$$

$$(1,5,\alpha) \cdot (1,3,\beta) = 0$$

$$1 + 15 + \alpha\beta = 0$$

$$\alpha\beta = -16$$

$$\vec{\boldsymbol{b}} \times \vec{\boldsymbol{c}} = \begin{vmatrix} \hat{\boldsymbol{\imath}} & \hat{\boldsymbol{\jmath}} & \hat{\boldsymbol{k}} \\ 1 & 3 & \beta \\ -1 & 2 & -3 \end{vmatrix} = \hat{\boldsymbol{\imath}}(-9-2\beta) - \hat{\boldsymbol{\jmath}}(-3+\beta) + \hat{\boldsymbol{k}}(2+3) = \hat{\boldsymbol{\imath}}(-1)(9+2\beta) + \hat{\boldsymbol{\jmath}}(3-\beta) + 5\hat{\boldsymbol{k}}$$

$$|\vec{\boldsymbol{b}} \times \vec{\boldsymbol{c}}| = 5\sqrt{3}$$
$$\sqrt{(9+2\beta)^2 + (3-\beta)^2 + 5^2} = 5\sqrt{3}$$

Square both sides:

$$81 + 36\beta + 4\beta^2 + 9 - 6\beta + \beta^2 + 25 = 75$$
$$5\beta^2 + 30\beta + 40 = 0$$
$$\beta^2 + 6\beta + 8 = 0$$
$$(\beta+4)(\beta+2) = 0$$
$$\beta = \{-2, -4\}$$
$$\alpha = -\frac{16}{\beta} \Rightarrow \alpha = \left\{\frac{-16}{-4}, \frac{-16}{-2}\right\} = \{4,8\}$$

$$|\vec{\boldsymbol{a}}|^2 = |(1,5,\alpha)|^2 = 1 + 5^2 + \alpha^2 = 1 + 25 + 64 = 90$$

Example 3.54

46: Let $\vec{\boldsymbol{p}} = 2\hat{\boldsymbol{\imath}} + 3\hat{\boldsymbol{\jmath}} + \hat{\boldsymbol{k}}$ and $\vec{\boldsymbol{q}} = \hat{\boldsymbol{\imath}} + 2\hat{\boldsymbol{\jmath}} + \hat{\boldsymbol{k}}$ be two vectors. If a vector $\vec{\boldsymbol{r}} = \alpha\hat{\boldsymbol{\imath}} + \beta\hat{\boldsymbol{\jmath}} + \gamma\hat{\boldsymbol{k}}$ is perpendicular to each of the vectors $(\vec{\boldsymbol{p}} + \vec{\boldsymbol{q}})$ and $(\vec{\boldsymbol{p}} - \vec{\boldsymbol{q}})$ and $|\vec{\boldsymbol{r}}| = \sqrt{3}$, then $|\alpha| + |\beta| + |\gamma|$ is equal to: (JEE Main 2021, 25 July, Shift-I)

$$\vec{\boldsymbol{r}} \cdot (\vec{\boldsymbol{p}} + \vec{\boldsymbol{q}}) = 0$$
$$(\alpha,\beta,\gamma) \cdot (2+1, 3+2, 1+1) = 0$$
$$(\alpha,\beta,\gamma) \cdot (3,5,2) = 0$$
$$\underbrace{3\alpha + 5\beta + 2\gamma = 0}_{Equation\ I}$$

$$\vec{\boldsymbol{r}} \cdot (\vec{\boldsymbol{p}} - \vec{\boldsymbol{q}}) = 0$$
$$(\alpha,\beta,\gamma) \cdot (2-1, 3-2, 1-1) = 0$$
$$(\alpha,\beta,\gamma) \cdot (1,1,0) = 0$$
$$\alpha + \beta = 0$$
$$\underbrace{\alpha = -\beta}_{Equation\ II}$$

$$3(-\beta) + 5\beta + 2\gamma = 0$$
$$2\beta + 2\gamma = 0$$
$$\gamma = -\beta$$

$$|\vec{\boldsymbol{r}}| = \sqrt{3}$$
$$\sqrt{\alpha^2 + \beta^2 + \gamma^2} = \sqrt{3}$$
$$\alpha^2 + \beta^2 + \gamma^2 = 3$$
$$(-\beta)^2 + \beta^2 + (-\beta)^2 = 3$$
$$3\beta^2 = 3$$
$$\beta^2 = 1$$

$$\alpha^2 = \beta^2 = \gamma^2 = 1$$
$$|\alpha| + |\beta| + |\gamma| = 1 + 1 + 1 = 3$$

Example 3.55

47: If $|\vec{a}| = 2, |\vec{b}| = 5$ and $|\vec{a} \times \vec{b}| = 8$, then $|\vec{a} \cdot \vec{b}|$ is equal to: (JEE Main 2021, 25 July, Shift-II)

$$|\vec{a} \times \vec{b}| = 8$$
$$|\vec{a}||\vec{b}| \sin\theta = \pm 8$$
$$(2)(5)\sin\theta = \pm 8$$
$$\sin\theta = \pm\frac{8}{10} = \pm\frac{4}{5}$$
$$\cos\theta = \pm\frac{3}{5}$$
$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta = (2)(5)\left(\pm\frac{3}{5}\right) = \pm 6$$

$$|\vec{a} \cdot \vec{b}| = 6$$

Example 3.56

48: Let $\vec{a}$ and $\vec{b}$ be two non-zero vectors perpendicular to each other, and $|\vec{a}| = |\vec{b}|$. If $|\vec{a} \times \vec{b}| = |\vec{a}|$, then the angle between $(\vec{a} + \vec{b} + \vec{a} \times \vec{b})$ and $\vec{a}$ is equal to: (JEE Main 2021, 18 March, Shift-II)

$$|\vec{a} \times \vec{b}| = |\vec{a}|$$
$$|\vec{a}||\vec{b}|\sin\theta = |\vec{a}|$$
$$|\vec{b}|\sin 90^\circ = 1$$
$$|\vec{b}| = 1$$
$$|\vec{a}| = |\vec{b}| = 1$$

Without loss of generality, let:

$$\vec{a} = \hat{\imath}, \qquad \vec{b} = \hat{\jmath}$$

$$\vec{x} = \vec{a} + \vec{b} + \vec{a} \times \vec{b} = \hat{\imath} + \hat{\jmath} + \hat{\imath} \times \hat{\jmath} = \hat{\imath} + \hat{\jmath} + \hat{k}$$
$$|\vec{x}| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

$$\vec{x} \cdot \vec{a} = |\vec{x}||\vec{a}|\cos\theta = \sqrt{3} \cdot 1 \cdot \cos\theta = \sqrt{3}\cos\theta$$
$$(\hat{\imath} + \hat{\jmath} + \hat{k}) \cdot (\hat{\imath}) = \sqrt{3}\cos\theta$$
$$1 = \sqrt{3}\cos\theta$$
$$\cos\theta = \frac{1}{\sqrt{3}}$$
$$\theta = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

Example 3.57

Let $\vec{a} = 2\hat{\imath} - 3\hat{\jmath} + 4\hat{k}$ and $\vec{b} = 7\hat{\imath} + \hat{\jmath} - 6\hat{k}$. If $\vec{r} \times \vec{a} = \vec{r} \times \vec{b}$, $\boldsymbol{r} \cdot (\hat{\imath} + 2\hat{\jmath} + \hat{k}) = -3$, then $\vec{r} \cdot (2\hat{\imath} - 3\hat{\jmath} + \hat{k})$ is equal to: (JEE Main 2021, 17 March, Shift-I)

$$\vec{r} \times \vec{a} = \vec{r} \times \vec{b}$$

Collate all terms on the LHS:

$$\vec{r} \times \vec{a} - \vec{r} \times \vec{b} = 0$$

Use the distributive property

$$\vec{r} \times (\vec{a} - \vec{b}) = 0$$

The cross product of two vectors is zero when they are parallel. Hence:

$$\vec{r} = \lambda(\vec{a} - \vec{b}) = \lambda(-5, -4, 10)$$

$$\boldsymbol{r} \cdot (1,2,1) = -3$$
$$\lambda(-5, -4, 10) \cdot (1,2,1) = -3$$
$$\lambda(-5 - 8 + 10) = -3$$
$$\lambda(-3) = -3$$
$$\lambda = 1$$
$$\vec{r} = (-5, -4, 10)$$

$$\vec{r} \cdot (2\hat{\imath} - 3\hat{\jmath} + \hat{k}) = (-5, -4, 10) \cdot (2, -3, 1) = -10 + 12 + 10 = 12$$

Example 3.58

Let $\vec{a} = \hat{\imath} + 2\hat{\jmath} - 3\hat{k}$ and $\vec{b} = 2\hat{\imath} - 3\hat{\jmath} + 5\hat{k}$. If $\vec{r} \times \vec{a} = \vec{b} \times \vec{r}$, $\boldsymbol{r} \cdot (\alpha\hat{\imath} + 2\hat{\jmath} + \hat{k}) = 3$ and $\vec{r} \cdot (2\hat{\imath} + 5\hat{\jmath} - \alpha\hat{k}) = -1$, $\alpha \in \mathbb{R}$, then the value of $\alpha + |\vec{r}|^2$ is equal to: **(JEE Main 2021, 16 March, Shift-II)**

$$\vec{r} \times \vec{a} = \vec{b} \times \vec{r}$$

Use the anti-commutative property

$$\vec{r} \times \vec{a} = -(\vec{r} \times \vec{b})$$
$$\vec{r} \times \vec{a} + \vec{r} \times \vec{b} = 0$$

Use the distributive property:

$$\vec{r} \times (\vec{a} + \vec{b}) = 0$$

$$\vec{r} = \lambda(\vec{a} + \vec{b}) = \lambda(3, -1, 2)$$

$$\boldsymbol{r} \cdot (\alpha\hat{\imath} + 2\hat{\jmath} + \hat{k}) = 3$$
$$\lambda(3, -1, 2) \cdot (\alpha, 2, 1) = 3$$
$$\lambda(3\alpha - 2 + 2) = 3$$
$$\alpha\lambda = 1$$

$$\lambda(3, -1, 2) \cdot (2, 5, -\alpha) = -1$$
$$\lambda(6 - 5 - 2\alpha) = -1$$
$$\lambda - 2\alpha\lambda = -1$$
$$\lambda - 2 = -1$$
$$\lambda - 2 = -1$$
$$\lambda = 1$$
$$\alpha = 1$$

$$|\vec{r}|^2 = 3^2 + 1^2 + 2^2 = 9 + 1 + 4 = 14$$
$$\alpha + |\vec{r}|^2 = 1 + 14 = 15$$

3.4 Triple Products and Further Topics

A. Triple Products

Example 3.59

List the eight possible triple products
Explain which ones which are defined and which ones are not

B. Scalar Triple Product

3.60: Component Definition

If $\vec{\boldsymbol{a}} = a_1\hat{\boldsymbol{\imath}} + a_2\hat{\boldsymbol{\jmath}} + a_3\hat{\boldsymbol{k}}$ and $\vec{\boldsymbol{b}} = b_1\hat{\boldsymbol{\imath}} + b_2\hat{\boldsymbol{\jmath}} + b_3\hat{\boldsymbol{k}}$ then their cross product is:

$$\vec{\boldsymbol{a}} \times \vec{\boldsymbol{b}} = \begin{vmatrix} \hat{\boldsymbol{\imath}} & \hat{\boldsymbol{\jmath}} & \hat{\boldsymbol{k}} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

Expanding the determinant along the first column gives us:

$$\vec{\boldsymbol{a}} \times \vec{\boldsymbol{b}} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \hat{\boldsymbol{\imath}} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \hat{\boldsymbol{\jmath}} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \hat{\boldsymbol{k}}$$

3.61: Scalar Triple Product

If $\vec{\boldsymbol{c}} = c_1\hat{\boldsymbol{\imath}} + c_2\hat{\boldsymbol{\jmath}} + c_3\hat{\boldsymbol{k}}$

$$\vec{\boldsymbol{c}} \cdot \left(\vec{\boldsymbol{a}} \times \vec{\boldsymbol{b}}\right) = \begin{vmatrix} c_1 & c_2 & c_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$\begin{aligned} &\left(c_1\hat{\boldsymbol{\imath}} + c_2\hat{\boldsymbol{\jmath}} + c_3\hat{\boldsymbol{k}}\right) \cdot \left(\begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \hat{\boldsymbol{\imath}} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \hat{\boldsymbol{\jmath}} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \hat{\boldsymbol{k}}\right) \\ &= \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} c_1 - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} c_2 + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} c_3 \\ &= \begin{vmatrix} c_1 & c_2 & c_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \end{aligned}$$

C. Vector Triple Product

3.62: Vector Triple Product

63 Examples